Contents

Lecture 1 – Overview 2

Lecture 3 – Examples & Hodge Theory 5

Lecture 5 – Yang Mills Functional & Equations 15

Lecture 7 – Outline of proof of Donaldson's theorem 25

Lecture 8 – Singularities in M and Holonomy 33

Lecture 9 – Singularities in M, part 2 41

Lecture 11 – Instanton Floer Homology 50

Lecture 12 – The Chern-Simons Functional & Reducibles 58

Lecture 13 – Defining Instanton Floer Homology 65

Lecture 14 – Lifting to Z-grading and the Filtered Chain Complex 72

Lecture 17 – Proof of the NST theorem 80

Lecture 19 – SU(2)-Character Varieties 90

INSTANTONS & 3-MANIFOLDS - Fall 2026 - IRVING DAI

Lecture 1 // Aug 25 // Overview

Thm [Freedman]

Q = unimodular symmetric bilinear form

$\exists!$ $\pi_1 = 1$ Top X^4 s.t. $Q_X = Q$

1) Q even, $\exists!$ SM X^4 / homeo

2) Q odd, $\exists$ at most 2 SM X^4 / homeo

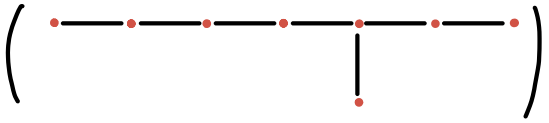
of which at most 1 is smoothable

Thm [Donaldson]

If X^4 is $\pi_1 = 1$, smooth, closed 4-mfd and Q_X is definite,

then Q_X is equivalent to $n(+1)$ or $n(-1)$

Ex E_8 matrix:

adj  $=$

$$\begin{bmatrix} -2 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -2 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & -2 \end{bmatrix}$$

Check: no change of basis to $\pm I_d^8$.

$\Rightarrow$ E_8 realized by non-smoothable top X^4

$\Rightarrow$ Same for nE_8

Over the next few weeks, we'll discuss the pf of Donaldson.

After, we'll discuss IFH and some applications to three manifolds, in particular, the homology cobordism group. The rest of this lecture will be to lay the groundwork for this.

3-Manifolds

Def Homology Cobordism Group

$\Theta^3_{\mathbb{Z}} = \{\text{all } \mathbb{Z}HS^3\text{'s}\}$ / homology cobordism

$Y_1 \sim Y_2$ hx-cob if $\exists\ W^4$ a homology $S^3 \times I$ st $\partial W^4 \cong -Y_1 \cup Y_2$

↳ TOP version is trivial

Thm [Freedman]

If Y is a $\mathbb{Z}HS^3$, then $Y = \partial W$ for <u>some</u> $\mathbb{Z}HB^4$

In fact, $\pi_1 W = 1$

Pf Given unim. sym. bil. pairing Q, can construct sm X^4 with ∂ w/ intersection form Q by attaching 2-h's to B^4 according to Q

E_8:

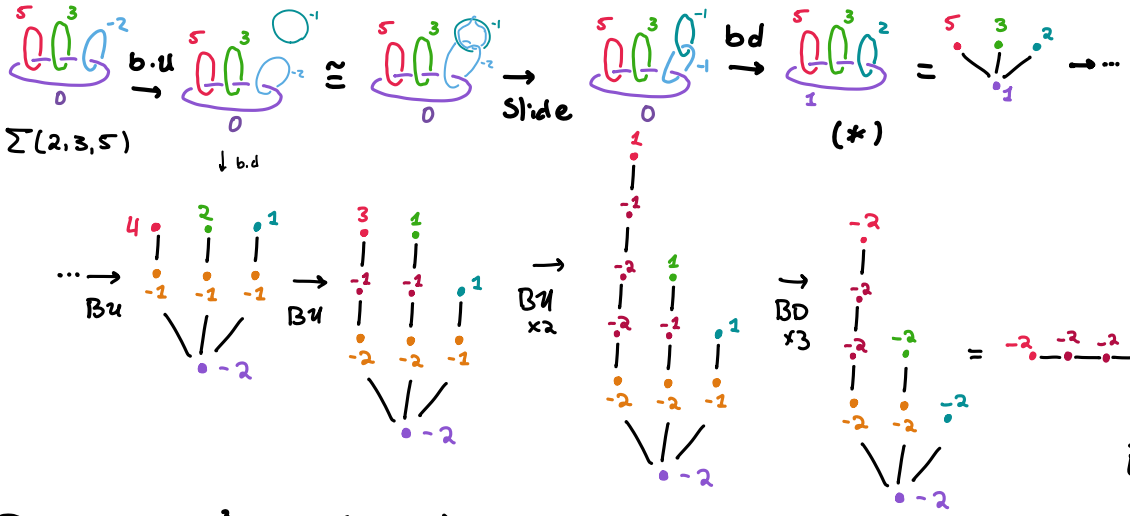
For E_8, $\partial E_8 \cong$ Poincaré hx sphere $\cong \Sigma(2,3,5)$

By Donaldson, $\Sigma(2,3,5)$ does not bound any smooth $\mathbb{Z}HB^4$

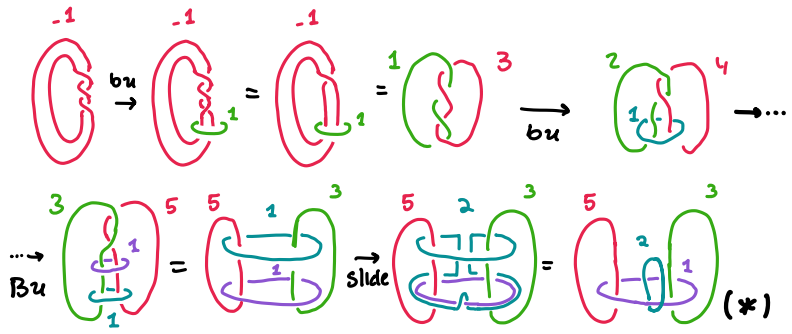
$\Rightarrow [\Sigma(2,3,5)] \neq 1 \in \Theta^3_{\mathbb{Z}}$

Pf of ∂-diffeo:

$\Sigma(2,3,5) \cong \partial E_8$



Poincaré Hx $S^3 \cong \Sigma(2,3,5)$



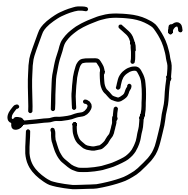
Rmk There are many $\mathbb{Z}HS^3$'s

Ex Brieskorn Spheres: p, q, r pairwise coprime

$\Sigma(p,q,r) := S^5_\varepsilon \cap \{x^p + y^q + z^r = 0\} \subseteq \mathbb{C}^6$

Ex $S^3_{1/n}(k)$ $\forall\, k$

Ex Mazur Mfds: $1\text{-}h^4 \cup 2\text{-}h^4$ s.t. alg linking is ± 1, geom linking $\neq \pm 1$



Q How large is $\Theta^3_{\mathbb{Z}}$?

Thm [Furuta, FS]

$\{\Sigma(2,3,6n-1)\}$ is a $\mathbb{Z}^\infty$ subgp of $\Theta^3_{\mathbb{Z}}$.

Thm [Daiemi-Nozaki-Sato-Taniguchi]

sometimes $\{S^3_{1/n}(k)\}$ are lin indep.

↑ specific condition on IFH_*

Note $\Sigma(2,3,6n-1) = S^3_{1/n}(T_{2,3})$

From concordance to Hx cobordism

$\Sigma_2 : \mathcal{C} \longrightarrow \Theta^3_{\mathbb{Z}/2}$

$K \longmapsto \Sigma_2(K)$ = Double branched cover over K

is a homomorphic

Q Is there a distinction between bounding a $\mathbb{Z}HB^4$ and a $\pi_1 = 0$ $\mathbb{Z}HB^4$?

Ie, is there a $\pi_1 = 1$ hx cob from $\Sigma(2,3,5)$ to itself?

Thm [Taubes]

No.

Thm [Daiemi-Nozaki-Sato-Taniguchi]

often no

cut open cobordism ⇝ get s.c. $\mathbb{Z}HB^4$ bounding $Y^3 \# -Y^3$

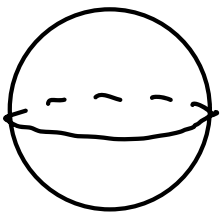
Lecture 3 // Sept 1 // Examples & Hodge Theory

Recall
- A connection is a map $\nabla : \Gamma(E) \to \Gamma(T^*M \otimes E)$
- Locally, connections take the form $\nabla = d + A$ where A is a connection 1-form matrix
- Under change of coordinates, A transforms strangely, not by conjugation: $A \mapsto g^{-1}Ag + g^{-1}dg$ where g is bundle trans. fxn
- The Curvature of a connection is $F_A := dA + A \wedge A$ and by Chern-Weil theory, the chern classes are $\det\left(I - \frac{t}{2\pi} F_A\right)$

Today: We'll explicitly compute one connection & its curvature to illustrate them in a toy example. This example will also generalize to help us understand the Yang-Mills moduli space.

Example: Connections on $\mathbb{CP}^1 = S^2$

Recall that $\mathbb{CP}^1 = \{[z:w] \mid z,w \in \mathbb{C}^2\}$ can be covered with 2-charts, the z chart with all points $[z:w]$ but $[0:1]$, and the w chart. The transition function is $w = \frac{1}{z}$



Now, consider the $\mathbb{C}$-line bundle over $\mathbb{CP}^1$ defined by the bundle transition function $g(z) = z^n$. This is the degree-n $\mathbb{C}$-line bundle where $\mathbb{C}$-fibers get glued by z^n, viewing this as a 1×1 matrix over $\mathbb{C}$.

Fact These are all the $\mathbb{C}$ line bundles over $\mathbb{CP}^1$

What do connections on the degree-n line bundle look like?

To define a connection, we must write down the connection com- 1-form matrix A. Since the bundle is a line bundle, A is a 1×1 matrix over $\mathbb{C}$, ie, a function $\mathbb{C} \to \mathbb{C}$.

Since A is purely local, we must define it in local coordinates. We'll consider the z chart first.

Prop $A := -n \dfrac{\bar{z}\,dz}{1+|z|^2}$ defines a connection on the degree n $\mathbb{C}$-line bundle over $\mathbb{CP}^1$

Pf To check that A actually defines a connection, we must do the following:

① look at A in the other chart

② show that A extends smoothly over the point at ∞.

Each step requires its own coordinate change:

① $A \rightsquigarrow g^{-1}Ag + g^{-1}dg$, the bundle change of coordinates

② $w = \frac{1}{z}$, the manifold change of coordinates

① Recall that the bundle transition function is $g(z) = z^n$.

Thus

$$g^{-1}Ag + g^{-1}dg = z^{-n}\left(-n\,\frac{\bar{z}\,dz}{1+|z|^2}\right)z^n + z^{-n}\,d(z^n)$$

$$= -n\,\frac{\bar{z}\,dz}{1+|z|^2} + \frac{n z^{n-1}}{z^n}$$

$$= -n\,\frac{\bar{z}\,dz}{1+|z|^2} + \frac{n}{z}$$

② We now apply the the manifold change-of-coordinates: $w = \frac{1}{z}$

$$dz = d\left(\tfrac{1}{w}\right) = \frac{-dw}{w^2}$$

So $-n\,\dfrac{\bar{z}\,dz}{1+|z|^2} + \dfrac{n}{z}$ becomes

$$\frac{-n\left(\frac{1}{w}\left(\frac{-1}{w^2}dw\right)\right)}{1+\left|\frac{1}{w}\right|^2} + nw\left(\frac{-1}{w^2}\right)dw = \frac{-n\,dw}{w^3\left(1+\left|\frac{1}{w}\right|^2\right)} - n\frac{dw}{w}$$

$$= n\frac{dw}{w(1+|w|^2)} - n\frac{dw}{w}$$

$$= n\left(\frac{1}{w(1+|w|^2)} - \frac{1}{w}\right)dw$$

$$= n\left(\frac{1}{w(1+|w|^2)} - \frac{(1+|w|^2)}{w(1+|w|^2)}\right)dw$$

$$= n\left(\frac{-|w|^2}{w(1+|w|^2)}\right)dw$$

$$= \frac{-n\,\bar{w}\,dw}{1+|w|^2}$$

this is a valid 1-form on the w-chart, and is defined at $z=\infty \leftrightarrow w=0$

Thus A defines a valid global 1-form. □

Remark A itself does not extend as a smooth 1-form on $\mathbb{CP}^1$:

$n\frac{dw}{w(1+|w|^2)}$ is singular at $w=0 \leftrightarrow z=\infty$, and it is the addition of the $-n\frac{dw}{w}$ term from the $g^{-1}dg$ in the bundle change of coordinates which resolves the issue, making A globally defined.

So A can be defined globally <u>because</u> of the "strange" bundle change-of-coordinates.

We'll now compute the curvature of this 1-form: $F_A = dA + A \wedge A$.

Since $d^2 = 0$, $A \wedge A = 0$, so we need only to compute dA.

$$dA = d\left(-n \frac{\bar{z}\, dz}{1+|z|}\right)$$

$$= -n \left(\frac{(1+|z|^2)\, d\bar{z} - \bar{z}\,(z\, d\bar{z} + \bar{z}\, dz)}{(1+|z|^2)^2} \right) \wedge dz$$

$$= -n \frac{d\hat{z} \wedge dz}{(1+|z|^2)^2} \qquad d\bar{z} = dx - i\,dy$$

$$dz = dx + i\,dy$$

$$= -2ni \frac{dx \wedge dy}{(1+x^2+y^2)^2}$$

Thus $F_A = -2ni \dfrac{dx \wedge dy}{(1+x^2+y^2)^2}$.

Computing the first chern class, we get that

$\Rightarrow$ first chern class: $\det\left(I - \frac{1}{2\pi i} F_A\right) = 1 + c_1$

$= $ Thus $c_1 = \frac{-1}{2\pi i}\left(-2ni \frac{dx \wedge dy}{(1+x^2+y^2)^2}\right) = \frac{n}{\pi} \frac{dx \wedge dy}{(1+x^2+y^2)^2}$

(note that this is the expression of c_1 on the z-chart)

observe: $$\int_{\mathbb{CP}^1} c_1 = n \left(\int_{\mathbb{R}^2} \frac{dx \wedge dy}{(1+x^2+y^2)^2} \right)$$

The previous computation leaves us with some questions.

1) If the final integral was taken over $\mathbb{R}^2$, where did $\mathbb{CP}^1$ come in to play? Why can't we choose another A defined only on $\mathbb{R}^2$?

2) In this example,

$$\int_{\mathbb{CP}^1} c_1 = \int_{\mathbb{CP}^2} F_A = \int_{\mathbb{CP}^2} dA = \int_{\text{large disk}} dA = \int_{\text{large } S^1} A .$$

Could we instead choose A to be compactly supported, so $\int = 0$? Or even choose $A = 0$?

The answer to both is a resounding no. The point here is that not every choice of A defines a connection on a $U(1)$ bundle over $\mathbb{CP}^1$. In this example, even though A is integrated over $\mathbb{R}^2$, we already checked that it extends, so the fact that A is defined globally comes first.

A "knows" what bundle it lives on by "remembering" the singular behavior at $z = \infty$, since it must match the $\frac{n\,dw}{w}$ term to allow for cancellation.

If $A \equiv 0$ or is compactly supported, $\dfrac{dw}{w(1+|w|^2)} = 0$,

so $$n\frac{dw}{w(1+|w|^2)} - n\frac{dw}{w} = -n\frac{dw}{w}.$$

For this to be nonsingular, n must be 0, so the line bundle must be trivial.

Upshot: The curvature integral determines the bundle via the behaviour near ∞.

We'll now review Hodge Theory:

Let (M, g) be an oriented Riemannian manifold.

On each tangent space T_pM, we get an inner product pairing $\langle \cdot, \cdot \rangle_p$ which smoothly varies as p varies in M.

This gives an inner product on each cotangent space T_p^*M, which induces an inner product on $\Lambda^k T^*M$ for each k as follows:

$\{e_1, \dots, e_n\}$ oriented orthonormal basis of T_pM $\leftrightarrow$ Oriented dual basis $\{e_1^*, \dots, e_n^*\}$ which we declare to be orthonormal. This declaration induces the inner product structure $\leftrightarrow$ for each k, we get $\binom{n}{k}$ basis vectors of the form $e_{i_1}^* \wedge \dots \wedge e_{i_k}^*$, where $\{i_1, \dots, i_k\} \subset \{1, \dots, n\}$. We declare this basis to be oriented and orthonormal, which induces the inner product structure

Of course, one must check that the resulting inner product doesn't depend on the original choice of oriented orthonormal basis for T_pM.

By abuse of notation, all of these inner products will be denoted by $\langle -, - \rangle$.

Rmk Let $n = \dim M$

On each $\Lambda^n T_p^* M$, there exists a unique oriented n-wedge with norm 1 via above identification.

This n-wedge is the volume form $e_1^* \wedge \dots \wedge e_n^* := \text{dvol}_g$

Now, note that since $\Lambda^k T^*M$ has $\binom{n}{k}$ basis vectors, $\dim(\Lambda^k T^*M) = \dim(\Lambda^{n-k} T^*M)$; this is also true for differential forms. To exploit this symmetry, we define the Hodge Star.

Def The Hodge Star is a map $* : \Omega^k \to \Omega^{n-k}$ defined pointwise by:

(after fixing an orthonormal oriented basis $\{e_1, \dots, e_n\}$ of T_pM)

For each $I \subset \{1, \dots, n\}$ with $|I| = k$, define $e_I^* := e_{i_1}^* \wedge \cdots \wedge e_{i_k}^*$

For each basis vector e_I^*, the hodge star sends e_I^* to $*e_I^*$ via:

$$*e_I^* := \pm \underbrace{e_{j_1}^* \wedge e_{j_{n-k}}^*}_{e_J^*} \quad \text{where} \quad J = \{1, \dots, n\} \setminus I$$

The $\pm$ sign is chosen such that $e_I^* \wedge e_J^* = e_1^* \wedge \cdots \wedge e_n^*$,

so $*$ gives complementary wedge of forms.

To define $*$ on non-basis vectors, we extend linearly.

Ex $e_I = e_1 \wedge e_3$, $e_J = -e_2 \wedge e_4$

$\Rightarrow e_I \wedge e_J = e_1 \wedge e_3 \wedge -e_2 \wedge e_4$

$= e_1 \wedge e_2 \wedge e_3 \wedge e_4 = dvol$

Rmk 1) This definition again is independent of the choice of orthonormal basis, but it <u>does</u> depend on the choice of metric.

2) $*$ is <u>almost</u> an involution... but not quite:

$$(*)^2 = (-1)^{k(n-k)}$$

$e_I^* \mapsto e_J^*$ s.t. $e_I^* \wedge e_J^* = dvol$

$e_J^* \mapsto e_I^*$ s.t. $e_I^* \wedge e_J^* = dvol$

} total # swaps depends on k.

Defining the Hodge star immediately buys us two things.

First, we can get an inner product structure on Ω^k from that on $\Lambda^k T^*M$, which was induced by the metric.

Def For $\alpha, \beta \in \Omega^k$, define

$$\langle \alpha, \beta \rangle := \int_M \alpha \wedge * \beta \quad \text{since } \alpha \wedge * \beta = n\text{-form.}$$

If α is an orthonormal basis element, $\alpha \wedge * \alpha = d\text{vol}_g$, so

$$\langle \alpha, \alpha \rangle = \int_M |\alpha|^2 d\text{vol}_g$$

Second, we get a formal adjoint of the exterior derivative.

Recall The exterior derivative $d : \Omega^k \to \Omega^{k+1}$

Def The "codifferential" $d^* : \Omega^k \to \Omega^{k-1}$ is defined as the composition $d^* := (-1)^k * d *$

$$d^* : \Omega^k \xrightarrow{*} \Omega^{n-k} \xrightarrow{d} \Omega^{n-k+1} \xrightarrow{*} \Omega^{k-1}$$

Lemma d^* is the formal adjoint of d: $\langle d\alpha, \beta \rangle = \langle \alpha, d^* \beta \rangle$

Pf By Stokes' theorem and the product rule, $*(*d*)\beta$ $\overbrace{\quad\quad}^{\pm 1}$

$$0 = \int_M d(\alpha \wedge * \beta) = \int_M d\alpha \wedge * \beta \pm \int_M \alpha \wedge \overbrace{d(* \beta)}^{\pm 1}$$

$$= \langle d\alpha, \beta \rangle \pm \int_M \alpha \wedge * d^* \beta$$

$$= \langle d\alpha, \beta \rangle \pm \langle \alpha, d^* \beta \rangle$$

$$\Rightarrow \langle d\alpha, \beta \rangle = \mp \langle \alpha, d^* \beta \rangle \quad \square$$

Note $(d^*)^2 = 0$

We now have the tools to discuss the main point of Hodge theory (for our purposes).

Def The Hodge Laplacian is a map $\Delta : \Omega^k \to \Omega^k$ defined as

$$\Delta = (d + d^*)^2 = dd^* + d^*d$$

Def $\alpha \in \Omega^k$ is called harmonic if $\Delta\alpha = 0$

Rmk In local coordinates, $d^*d + dd^*$ is a 2nd order differential operator, since it acts like a 2^{nd} derivative on the coefficients of basis vectors.

Lemma α is harmonic IFF $d\alpha = 0$ (closed) and $d^*\alpha = 0$ (co-closed)

Pf ($\Leftarrow$) obvio

($\Rightarrow$) If $\Delta = 0$, $\langle \Delta\alpha, \alpha \rangle = 0$. Since $\Delta\alpha = dd^*\alpha + d^*d\alpha$,

$$\begin{aligned} 0 = \langle \Delta\alpha, \alpha \rangle &= \langle dd^*\alpha + d^*d\alpha, \alpha \rangle \\ &= \langle dd^*\alpha, \alpha \rangle + \langle d^*d\alpha, \alpha \rangle \\ &= \langle d^*\alpha, d^*\alpha \rangle + \langle d\alpha, d\alpha \rangle \\ &= \langle (d + d^*)\alpha, (d + d^*)\alpha \rangle \end{aligned}$$

$$\Rightarrow (d + d^*)\alpha = 0$$

$$\Rightarrow d\alpha = d^*\alpha = 0 \quad \square$$

Note If α is harmonic, it is closed, so α defines a cohomology class $[\alpha] \in H^*_{dR}$

Lemma α is harmonic IFF α minimizes norm in its de Rham cohomology class

Pf ($\Rightarrow$) Let $\alpha \in H^k_{dR}(M;\mathbb{R})$.

Every other element in this class will differ from α by an exact form, ie, some $d\beta$ for some $\beta \in \Omega^{k-1}$. We will compute the norm of $\alpha + d\beta$.

$$\langle \alpha + d\beta, \alpha + d\beta \rangle = \langle \alpha, \alpha \rangle + \langle d\beta, d\beta \rangle + 2\langle \alpha, d\beta \rangle$$

We know that $\langle d\beta, d\beta \rangle \geq 0$, and $\langle \alpha, d\beta \rangle = \langle d^*\alpha, \beta \rangle = 0$ since α is harmonic.

Thus $\langle \alpha, \alpha \rangle \leq \langle \alpha + d\beta, \alpha + d\beta \rangle$. □

($\Leftarrow$) Assume α is norm minimizing.

For all $\beta \in \Omega^{k-1}$, consider the ray $\langle \alpha + t\,d\beta, \alpha + t\,d\beta \rangle_{t \in [0,1]}$.

Eg, we move α around in its cohomology class along some ray.

Since α is norm minimizing, when $t=0$, the derivative vanishes.

$$0 = \frac{d}{dt}\Big|_{t=0} \langle \alpha + t\,d\beta, \alpha + t\,d\beta \rangle = 2\langle \alpha, d\beta \rangle$$

$$= 2\langle d^*\alpha, \beta \rangle$$

$$\Rightarrow d^*\alpha = 0$$

It remains to show $d\alpha = 0$ and this minimizer is unique; the proof is left as an exercise. □

Thm Hodge Theorem

Define $\mathcal{H}^k$ to be the space of harmonic k-forms.

There exists an isomorphism $\mathcal{H}^k \cong H^k_{dR}$ sending $\alpha \in H^k_{dR}$ to the norm minimizer in $[\alpha]$.

Lecture 5 // Sept 8 // Yang Mills Functional & Equations

Last time: We defined the following objects:

(M^4, g), E an $su(2)$ bundle over M

$\mathcal{A}$:= space of $su(2)$ connections $\{\nabla = d + A\}$ this is an affine space over $T(T^*M \otimes \mathrm{ad}E)$

$\mathcal{G}$ = $SU(2)$ gauge gp $= \Gamma(\mathrm{Aut}E)$

$g \curvearrowright A \quad A \rightsquigarrow g^{-1}Ag + g^{-1}dg$

$\mathcal{M}$ = Moduli space of Yang Mills connections with self dual curvature $= \left\{ \begin{array}{l} SU(2) \text{ connections } \nabla = d + A \\ \text{such that } F_A = *F_A \\ \text{(self dual)} \end{array} \right\} / \mathcal{G}$

This condition of having self-dual curvature is actually a specific instance of a larger phenomena, which we'll discuss today to sit ASD connections in context.

The Yang Mills Functional

Recall that the Lie algebra of $SU(2)$, $su(2)$ is defined as the subset of 2×2 complex matrices which are trace-free and skew-hermitian.

$$su(2) = \left\{ \begin{bmatrix} ix & -y+iz \\ y+iz & -ix \end{bmatrix} \;\middle|\; (x,y,z) \in \mathbb{R}^3 \right\}$$

In particular, this definition clearly shows that $su(2)$ is in bijection with $\mathbb{R}^3$.

$su(2)$ can be endowed with an inner product given by $\langle A, B \rangle = \mathrm{tr}(AB^+) = -\mathrm{tr}(AB)$, which is the same inner product as the one on $\mathbb{R}^3$ (the dot product).

$$\left\langle \begin{bmatrix} ix & -y+iz \\ y+iz & -ix \end{bmatrix}, \begin{bmatrix} ip & -q+ir \\ q+ir & -ip \end{bmatrix} \right\rangle = 2(xp+yq+zr) = 2 \begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} p \\ q \\ r \end{pmatrix}$$

This inner product on $\mathfrak{su}(2)$ can be extended to an L^2 inner product on $\Omega^i(\mathrm{ad}E) = \Gamma(\Lambda^i T^*M \otimes \mathrm{ad}E)$.

Given two sections F and G, their inner product is defined as

$$\langle F, G\rangle_{L^2} = -\int_M \mathrm{tr}(F \wedge * G)$$

where the inner product and trace is taken pointwise for each $p \in M$, giving a 4-form at each point, ie, a function of p which we can integrate over M. By taking $G=F$, the norm of F is thus $\|F\|^2 = \langle F, F\rangle_{L^2}$.

Using this inner product, we may now define the Yang-Mills Functional.

Def [Yang-Mills Functional]

$$YM : \{\mathfrak{su}(2)\text{-connections}\} \to \mathbb{R}$$

$$\nabla \mapsto YM(\nabla) = \langle F_A, F_A\rangle_{L^2} = -\int_M \mathrm{tr}(F_A \wedge * F_A) = -\int_M \|F_A\|^2 \mathrm{dvol}_M$$

Heuristically, $YM(\nabla)$ measures the "strength" of curvature of ∇.

For some connection, what does it mean to be a critical point of the Yang Mills functional? Recall that the space of connections is affine over $\Gamma(T^*M \otimes \mathrm{ad}E)$. A fixed connection ∇ is a critical point of YM if every nearby connection ∇' has larger or smaller curvature. Any such ∇' can be written as $\nabla' = \nabla + tB$, ie, a small perturbation of ∇ by some section $B \in \Gamma(T^*M \otimes \mathrm{ad}E)$. In terms of F_A, the curvature of ∇' becomes

$$\begin{aligned} F_{A+tB} &= d(A+tB) + (A+tB)\wedge(A+tB) \\ &= F_A + t(dB + A\wedge B + B\wedge A) + t^2 B\wedge B \end{aligned}$$

Define $D_A B := dB + A\wedge B + B\wedge A$.

The derivative $\left.\frac{d}{dt}\right|_{t=0} YM(\nabla + tB)$ then becomes

$$\frac{d}{dt}\Big|_{t=0} YM(\nabla + tB) = \frac{d}{dt}\Big|_{t=0} \langle F_{A+tB},\ F_{A+tB} \rangle$$

$$= \frac{d}{dt}\Big|_{t=0} \left\langle F_A + tD_AB + t^2 B\wedge B,\ F_A + tD_AB + t^2 B\wedge B \right\rangle$$

$$\vdots$$

$$= 2\ \langle D_A B, F_A \rangle$$

Therefore ∇ is a critical point of YM IFF this pairing is 0 for all B.

Before we continue, we should discuss the appearance of $D_A B$.

Interlude: More about connections & bundles

Recall the usual deRham complex.

$$\Omega^0 \xrightarrow{d} \Omega^1 \xrightarrow{d} \Omega^2 \xrightarrow{d} \cdots$$

Now, take a vector bundle $E \to B$ and a fixed connection ∇ on E

We have already described ∇ as a map $\nabla : \Gamma(E) \to \Gamma(T^*M \otimes E)$

Can we get an analogous complex to the deRham complex?

To do so, we must generalize the spaces & the maps.

Recall that the Ω^i in the deRham complex are defined to be Sections of the bundle $\Lambda^i T^*E \to M$.

Def E-valued i-forms are Sections of the bundle $\Lambda^i T^*M \otimes E$

$$\Omega^i(E) = \Gamma(\Lambda^i T^*M \otimes E)$$

Note that when $i=0$, we recover $\Omega^0(E) = \Gamma(\Lambda^0 T^*M \otimes E) = \Gamma(E)$ and when $i=1$, we get $\Omega^1(E) = \Gamma(\Lambda^1 T^*M \otimes E)$. So as defined above, a connection gives a map $\nabla : \Omega^0(E) \to \Omega^1(E)$. We extend the definition of ∇ using the Leibniz rule, as follows.

Def $\nabla : \Omega^i(E) \to \Omega^{i+1}(E)$

$$\nabla(\alpha \otimes s) = d\alpha \otimes s + (-1)^i \alpha \wedge \nabla) s$$

where α is an i-form and s = section of E, so $(\alpha \otimes s) \in \Omega^i(E)$

Note that unlike in the deRham complex, $\nabla^2 \neq 0$. However, measuring the failure of this fact is very useful.

Lemma $\nabla^2(s) = F_A s$

Pf Choose basis sections $\{e_1, \ldots, e_n\}$ by choosing a trivialization of the bundle.

Then $s = \sum s_i e_i$, where s_i are functions.

Recall $\nabla = d + A$. Therefore

$$\nabla^2 s = \nabla(\nabla s) = \nabla\left(\sum_{ij} \underbrace{ds_i}_{\text{derivatives of coefcs}} \otimes e_i + s_i A_{ji} e_j\right)$$ (einskim but we have a single story house)

$$= \underbrace{-ds_i \wedge (s_i A_{ji} e_i)}_{\nabla \text{ of 1st term}} + \underbrace{\nabla(s_i A_{ji}) e_j \wedge ds_i e_i}_{\nabla \text{ of 2nd term}}$$

$$= \underbrace{-ds_i \wedge (s_i A_{ji} e_i)}_{\nabla \text{ of 1st term}} + (ds_i \wedge A_{ji} + s_i dA_{ji}) e_j - (s_i A_{ji}) \wedge (A_{kj} e_k)$$

$$= s_i \, dA_{ji} e_j + s_i A_{jl} A_{li} e_j$$

$$= dA + A \wedge A. \square$$

The upshot here is that curvature is the obstruction for the differential on the $\Omega^i(E)$ complex to square to 0.

We'll now consider the bundle $End(E)$. Given a connection $D = d + A$ on E, we can define a connection on $End(E)$ using ∇.

We do so locally

Def A connection D_A on $End(E)$ is locally of the form

$$D_A B := dB + AB - BA$$

where A is a matrix of 1-forms and B is a section of $End(E)$, i.e., a matrix valued function.

This definition can be motivated in two ways.
First, connections on E are identified with matrices, which act on vectors by multiplication and other matrices via the lie bracket. The dB term is the vector part, i.e. the usual entry-wise derivative, and the $AB-BA$ term records the linear action of A on B.

The second way to motivate this is by decomposing $End(E)$ as $End(E) = E \otimes E^*$, and checking that tensoring with the connection on E^* obtained by dualizing gives the same thing.

Now that we have a connection on $End(E)$, we again get a map $D_A: \Omega^0(End\,E) \to \Omega^1(End\,E)$, which we can also extend to a map $D_A: \Omega^i(End\,E) \to \Omega^{i+1}(End\,E)$.

Exercise Use the Leibniz rule to extend D_A to a map $D_A: \Omega^i(End\,E) \to \Omega^{i+1}(End\,E)$.

Further, if A is a $su(2)$ connection, B is $su(2)$-valued, so $[A,B]$ also $su(2)$-valued.
Therefore we can restrict $End(E)$ to $ad(E)$ and get the complex

$$\Omega^0(ad\,E) \xrightarrow{D_A} \Omega^1(ad\,E) \to \dots \quad (*)$$

lemma For $B \in \Omega^1(\mathrm{ad}\, E)$, $D_A B$ as defined above is the $D_A B$ defined

!! Sign issue?

Pf First, any $B \in \Omega^1(\mathrm{ad}E)$ can be expressed as a linear combination of 1-forms ω with coefficients constant $SU(2)$ matrices C. Thus we can assume $B = \omega C$.

Then $D_A B = D_A(\omega C)$

(Leibniz and ω is a 1-form) $= d\omega C - \omega \wedge \nabla C$

(Since C const) $= d\omega C - \omega \wedge (dC + AC - CA)$

(Since ω, A both 1-forms, $\omega \wedge A = -A \wedge \omega$) $= d\omega C - \omega \wedge [A, C]$

$= d\omega C - (\omega \wedge A) C + \omega C \wedge A$

$= d\omega C - (-A \wedge \omega) C + \omega C \wedge A$

$= d(\omega C) + A \wedge (\omega C) + (\omega C) \wedge A$

$= dB + A \wedge B + B \wedge A$. $\square$

Note that this proof, applied to ω = i-form instead, shows in general that the map

$D_A B : \Omega^i(\mathrm{End}\, E) \to \Omega^{i+1}(\mathrm{End}\, E)$ is

$D_A B = dB + A \wedge B - B \wedge A$, for i even,

and $D_A B = dB + A \wedge B + B \wedge A$, for i odd.

Now, seemingly from nowhere, we'll define the Bianci Identity - this will be incredibly useful later.

Thm [Bianchi Identity]

If A is fixed connection, $D_A F_A = 0$

$F_A \in \Omega^2(\mathrm{ad}(E))$

Pf Since $F_A = dA + A \wedge A$,

$$D_A F_A = dF_A + A \wedge F_A - F_A \wedge A$$

$$= d(dA + A\wedge A) + A \wedge (dA + A\wedge A) - (dA + A\wedge A)\wedge A$$

$$= (d^2 A + d(A\wedge A)) + (A\wedge dA + A\wedge A\wedge A)) - (dA\wedge A + A\wedge A\wedge A)$$

$$= d^2 A + d(A\wedge A) + A\wedge dA + A\wedge A\wedge A) - dA\wedge A - A\wedge A\wedge A$$

(Since $d^2 A = 0$ and $dA\wedge A = -A\wedge dA$) $= (dA\wedge A + A\wedge dA) + A\wedge dA - dA\wedge A$

Sign issue?

$$= d^2 A + dA\wedge A + A\wedge dA + A\wedge dA + A\wedge A\wedge A - dA\wedge A - A\wedge A\wedge A$$

$$= 0$$

Now, let $D_A^* = *\, D_A *$. This gives us a codifferential

$$D_A^* : \Omega^2(\mathrm{ad}\, E) \to \Omega^1(\mathrm{ad}\, E)$$

We are now ready to return to discussing the critical points of the Yang-Mills Functional.

Yang-Mills Connections

We have seen that A is a critical point of YM IFF

$$\langle D_A B, F_A \rangle_{L^2} = 0 \quad \text{for all } B \in \Omega^1(\mathrm{ad}\, E)$$

$$\Leftrightarrow \quad \langle B, D_A^* F_A \rangle_{L^2} = 0$$

$$\Leftrightarrow \quad D_A^* F_A = 0$$

Def A connection such that $D_A^* F_A = 0$ is called a Yang Mills connection. These connections are the critical points of the YM functional.

Note the similarity with Hodge theory!

Last week, we saw that the cp of L^2 norm on forms are precisely the harmonic forms: α for which $d\alpha = d^*\alpha = 0$

If a connection A is Yang-Mills, then $D_A^* F_A = 0$. And by the Bianchi identity, we always have that $D_A F_A = 0$.

Therefore YM connections can be thought of as harmonic forms; the slogan here is that "YM theory is the $SU(2)$ version of Hodge theory."

The one caveat is that in general, $D_A * F_A$ is difficult to understand. In Hodge theory, d always exists, but here, both D^*_A and F_A depend on A.

It turns out that some YM connections satisfy a simpler condition, which also guarantees that they are global minima/maxima.

Lemma 1) If A is SD or ASD, then $D_A^* F_A = 0$

2) In fact, (A)SD are the global minima/maxima of the YM functional

Pf 1) This is immediate from Bianchi:

Since $D_A F_A = 0$ and $D_A^* = * D_A *$,

if $F_A = \pm * F_A$, then

$$\begin{aligned} D_A^* F_A &= * D_A * F_A \\ &= * D_A \pm F_A \\ &= \pm * D_A F_A \\ &= 0. \end{aligned}$$

2) For $F_A \in \Omega^2(\mathrm{ad}E) = \Omega^2_+(\mathrm{ad}E) \oplus \Omega^2_-(\mathrm{ad}E)$, it splits as $F_A = F_A^+ + F_A^-$, where $\langle F_A^+, F_A^- \rangle_{L^2} = 0$.

Thus $YM(\nabla) = \langle F_A, F_A \rangle_{L^2}$

$$= -\int_M \mathrm{tr}(F_A \wedge * F_A)$$

$$= \langle F_A^+ + F_A^-, F_A^+ + F_A^- \rangle_{L^2}$$

$$= \langle F_A^+, F_A^+ \rangle_{L^2} + \langle F_A^-, F_A^- \rangle_{L^2} + \langle F_A^+, F_A^- \rangle_{L^2} + \langle F_A^-, F_A^+ \rangle_{L^2}$$

$$= \|F_A^+\|_{L^2} \vee \|F_A^-\|_{L^2}$$

Now, recall the instanton number k defined as

$$k = -\langle c_2, [M] \rangle = \frac{-1}{8\pi^2} \int_M \mathrm{tr}(F_A \wedge F_A).$$

Since $F_A = *(*F_A)$,

$$k = \frac{1}{8\pi^2} \langle F_A, *F_A \rangle_{L^2}$$

$$= \frac{1}{8\pi^2} \langle F_A^+ + F_A^-, *(F_A^+ + F_A^-) \rangle_{L^2}$$

$$= \frac{1}{8\pi^2} \langle F_A^+ + F_A^-, F_A^+ - F_A^- \rangle_{L^2}$$

$$= \frac{1}{8\pi^2} \left(\langle F_A^+, F_A^+ \rangle_{L^2} + \langle F_A^+, -F_A^- \rangle_{L^2} + \langle F_A^-, F_A^+ \rangle_{L^2} + \langle F_A^-, -F_A^- \rangle \right)$$

$$= \frac{1}{8\pi^2} \left(\|F_A^+\|_{L^2} - \|F_A^-\|_{L^2} \right).$$

Therefore $8\pi^2 k = \|F_A^+\|_{L^2} - \|F_A^-\|_{L^2}$.

Combining this with $YM(\nabla) = \|F_A^+\|_{L^2} + \|F_A^-\|_{L^2}$, we see that

If $k>0$, then $YM \geq 8\pi^2 k$, with equality when $F_A^- \equiv 0$, ie, when A is self-dual. On the other hand, if if $k<0$, then $YM \geq 8\pi^2(-k)$ and equality is realized when $F_A^+ \equiv 0$, so A is anti-self-dual. $\square$

Lecture 7 // Sept 15 // Outline of pf of Donaldson's theorem

Last Time: Let (M^4, g) be positive definite and simply connected.

Fix a $k=1$ $SU(2)$ bundle E over M.

Set $\mathcal{M} := \{ \text{SD YM connections on } E \} / \mathcal{G}$

Each splitting $L \oplus \bar{L}$ of E has a unique (up to $SU(2)$ gauge equivalence) split SDYM connection $\nabla \oplus \bar{\nabla}$ for which F_∇ is the unique harmonic representative of $c_1(L)$.

Today, we'll outline a proof of Donaldson's theorem by establishing qualitative features about $\mathcal{M}$.

Thm [Donaldson]

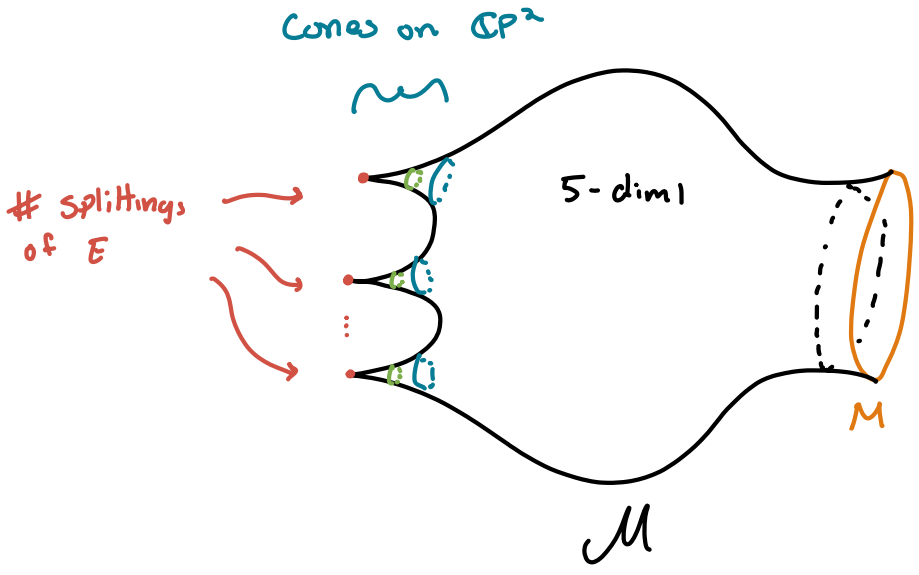
If X^4 is $\pi_1 = 1$, smooth, closed 4-mfd and Q_X is definite, then Q_X is equivalent to $n(+1)$ or $n(-1)$

Claim 1) Away from a finite number of singular points, $\mathcal{M}$ is a smooth 5-dim oriented mfd.

2) The number of these points = the # of splittings of E

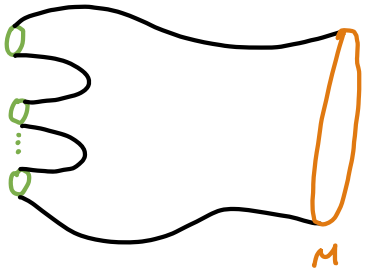
3) Each point has a nbhd $\cong$ homeo to cone on $\mathbb{CP}^2$. Diffeo away from pt.

4) $\mathcal{M}$ is <u>not</u> compact, but it admits a compactification $\overline{\mathcal{M}}$ such that $\partial \overline{\mathcal{M}} \cong M$



We have a cobordism M to $\coprod \mathbb{CP}^2$ (# splittings of E)

obtained by removing the singular points



(*)

It turns out that (*) implies Donaldson's theorem.

Before we discuss this, we'll briefly motivate why bordism is useful here.

Interlude: Bordism in General

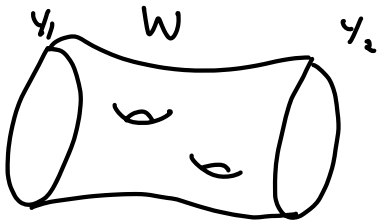
This discussion isn't strictly neccessary, but it places (*) in context.

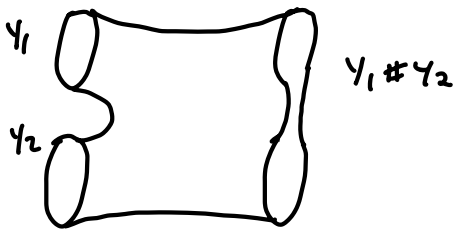
Def The n-th bordism group is $\Omega^n = \{$ Set of all oriented smooth n-mfd. $\}/\sim$

where $Y_1 \sim Y_2$ IFF $\exists W^{n+1}$ s.t. $\partial W = -Y_1 \sqcup Y_2$

If $Y_1 \sim Y_2$, they are called bordant.

Ω^n is a grop under disjoint union and/or connect sum.





This is a generalization of the question: is $M^n \cong \partial W^{n+1}$ for some W^{n+1}?

It turns out that in one of the great alg top triumphs of the 60's, the groups Ω^n are well understood. In particular, Ω^n / torsion is known for all $n \in \mathbb{N}$. Concretely, for $n=3$, $\Omega^3 = 0$ since every 3-manifold bounds a 4-manifold. When $n=4$, $\Omega^4 \cong \mathbb{Z}$, which is determined by signature.

Def The signature of a 4-manifold is $\sigma(M) = \sigma(Q_M) = \#+$ eigenvalues $- \#-$ eigenvalues

Thm In dimension 4, signature is a complete bordism invariant.

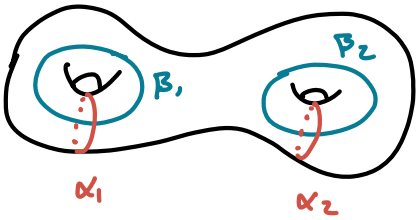
This means that two 4-manifolds are cobordant if and only if they have the same signature. In particular, any M is cobordant to either $\#_{\sigma(M)} \mathbb{CP}^2$, if $\sigma(M) \geq 0$, or $\#_{-\sigma(M)} \overline{\mathbb{CP}}^2$, if $\sigma(M) < 0$.

For our purposes, we'll only need the claim that σ is a bordism invariant.

Prop If M_1 and M_2 are bordant, then $\sigma(M_1) = \sigma(M_2)$.

To prove this proposition, we'll need the following lemma, commonly referred to as the "half lives, half dies" theorem.

To motivate it, consider $\Sigma_g = \partial H_g$. Think of Σ_g as a stand in for M and H_g a stand in for W. The middle dimension of the (co)homology of Σ is H_1, so we look at the map induced by the inclusion $\iota: \Sigma_g \to H_g$.



$$\iota_*: H_1(\Sigma_g) \to H_1(H_g)$$

$$\ker \iota_* \cong \text{Span}\{\alpha_1, \alpha_2\}$$

$$\text{im}\, \iota_* \cong \text{Span}\{\iota_*[B_1], \iota_*[B_2]\}$$

Lemma Half lives, half dies

For W a smooth oriented $2n+1$ dimensional manifold with boundary $\partial W \cong M$, $\ker(\iota_*) \subset H_n(M;\mathbb{R})$ is a half-dimensional Lagrangian subspace with respect to the intersection form $Q: H^n(M;\mathbb{R}) \times H^n(M;\mathbb{R}) \to \mathbb{Z}$, where $\iota_*: H_n(M;\mathbb{R}) \to H_n(W;\mathbb{R})$ is the map on homology induced by the inclusion $\iota: M \cong \partial W \hookrightarrow W$

Proof of Lemma

First recall that by Poincaré duality, $H_n(M;\mathbb{R}) \cong H^n(M;\mathbb{R})$.

Further, from linear algebra, if $T: A \to B$ is a linear map,

the dual map $T^*: B^* \to A^*$ is such that $\dim \operatorname{im} T = \dim \operatorname{im} T^*$:

$$\begin{aligned}\dim \operatorname{im} T &= \dim(A) - \dim(\ker T) \\ &= \dim\left(A/\ker T\right) \\ &= \dim\left(\operatorname{Ann}(\ker T)\right) \quad \left(\begin{array}{l}\operatorname{Ann}(\ker T) = \text{Subsp of } A^* \\ \text{that anihilates } T\end{array}\right) \\ &= \dim \operatorname{im} T^*\end{aligned}$$

Thus the maps $\quad \iota^*: H^n(W;\mathbb{R}) \longrightarrow H^n(M;\mathbb{R})$

$$\iota_*: H_n(M;\mathbb{R}) \longrightarrow H_n(W;\mathbb{R})$$

are dual in the linear algebra sense, so $\dim \operatorname{im} \iota^* = \dim \operatorname{im} \iota_*$.

Again by Poincaré duality, $\dim \operatorname{im} \iota^* = \dim \ker \iota_*$.

Thus $\quad \dim \ker \iota_* + \dim \operatorname{im} \iota_* = \dim H_n(M;\mathbb{R})$

$$\begin{matrix}\| & & \| \\ \dim \operatorname{im} \iota^* & + & \dim \operatorname{im} \iota^*\end{matrix}$$

$$\Rightarrow \dim \operatorname{im} \iota^* = \frac{1}{2} \dim H_n(M;\mathbb{R}) = \frac{1}{2} \dim H^n(M;\mathbb{R})$$

$$\Rightarrow \dim \ker \iota_* = \frac{1}{2} \dim H_n(M;\mathbb{R})$$

Now, take $\alpha, \beta \in H^n(W;\mathbb{R})$, so $\iota^*(\alpha), \iota^*(\beta) \in H^n(M;\mathbb{R})$.

Then $\quad Q_M\left(\iota^*(\alpha), \iota^*(\beta)\right) = \int_M \iota^*(\alpha) \wedge \iota^*(\beta) = \int_M \iota^*(\alpha \wedge \beta) = \int_{\partial W} \iota_*(\alpha \wedge \beta)$

By Stokes' theorem, $\quad \int_{\partial W} \iota_*(\alpha \wedge \beta) = \int_W d(\alpha \wedge \beta) = 0$

Thus $\operatorname{Im}(\iota^*)$ is Lagrangian with respect to Q_M. □

Rmk We may avoid using Stokes' theorem by translating to algebraic topology and cupping against fundamental class. Thus half lives half dies is true when M isn't smoothable.

Having proven half lives half dies, we may now prove the original proposition.

Prop If M_1 and M_2 are bordant, then $\sigma(M_1)=\sigma(M_2)$.

Pf 1) By setting $M=-M_1 \# M_2$, we know that $M \cong \partial(W^5)$.

So it is enough to show that $\sigma(M)=0$.

2) Consider the long exact sequence in homology and the Poincaré-Lefshetz isomorphisms. By naturality, the following diagram commutes:

$$\begin{array}{ccccccc} \cdots \to & H^2(W;\mathbb{R}) & \xrightarrow{\iota^*} & H^2(M;\mathbb{R}) & \to & H^3(W,M;\mathbb{R}) & \to \cdots \\ & \downarrow PD \cong & & \downarrow PD \cong & & \downarrow PD \cong & \\ \cdots \to & H_3(W,M;\mathbb{R}) & \to & H_2(M;\mathbb{R}) & \xrightarrow{\iota_*} & H_2(W;\mathbb{R}) & \to \cdots \end{array}$$

Note that $\operatorname{im}\iota^* \cong \ker \iota_*$ under PD

3) By Half lives, half dies, $\begin{cases} \dim \operatorname{im}\iota^* = \frac{1}{2}\dim H^2(M;\mathbb{R}) \\ \dim \ker \iota_* = \frac{1}{2}\dim H_2(M;\mathbb{R}) \end{cases}$

and Q_M vanishes on $\operatorname{im}\iota^*$ (eg, $\operatorname{im}\iota^* \subset H^2(M;\mathbb{R})$ is Lagrangian)

Thus there is a half-dim subspace on which intersection form vanishes.

4) This implies $\sigma(M)=0$:

Consider the diagonalization of Q_M over $\mathbb{R}$: $Q_M \cong \left[\begin{array}{c|c} +I & 0 \\ \hline 0 & -I \end{array}\right]$.

$$\left[\begin{array}{c|c} +I & \\ \hline & -I \end{array}\right] V = V_+ \oplus V_-$$

This splits V into $V_+ \oplus V_-$, where Q_M is $\pm$-def on $V_\pm$

Let $a=\dim V_+$ and $b=\dim V_-$. Then $a+b=\dim V_+ + \dim V_- = \dim V = 2k$.

Let $U = \operatorname{im}\iota^*$. Since Q_M vanishes on U, $U \cap V_+ = \{0\}$ and $U \cap V_- = \{0\}$.

Thus $\dim(U+V_+) = \dim(U) + \dim(V_+) = k+a \leq 2k$

$\dim(U+V_-) = \dim(U) + \dim(V_-) = k+b \leq 2k$

$\Rightarrow$ $a \leq k$, $b \leq k$. Since $a+b=2k$, $a=b=k \Rightarrow \sigma(M)=a-b=0$. $\square$

Back to Donaldson

Let's return to Donaldson and take stock of what we know:

Thm [Donaldson]

If X^4 is $\pi_1 = 1$, smooth, closed 4-mfd and Q_X is definite,

then Q_X is equivalent to $n(+1)$ or $n(-1)$

The previous discussion of bordism has given us the following three facts, from which Donaldson's theorem will follow.

(i) $\sigma(M) = rk(Q_M)$ (Since M is positive definite)

(ii) $\sigma(M) = \#$ splittings of E

Since 1) σ bordism invariant (Interlude)

2) M bordant to $\bigsqcup_{\#\text{splittings of } E} \mathbb{CP}^2$ $(*)$

(iii) $\#$ splittings of $E = \frac{1}{2} \# x \in H^2$ s.t. $\langle x, x \rangle = 1$

This follows from two claims about vector bundles over X^4:

1) $SU(2)$ bundles are determined by $c_2 \in H^4$ (true only in our setting)

2) $U(1)$ bundles are determined by $c_1 \in H^2$ (true in general)

Via the curvature form or by formal properties of Chern classes, to say $E = L \oplus \bar{L}$ implies that $c_2(E) = c_1(L) \cup c_1(\bar{L}) = -c_1(L) \cup c_1(L)$

Similarly, if $c_2(E) = -x^2$ for $x \in H^2$, the above calculation implies that $E = L \oplus \bar{L}$. Note here that different choices of x give different L's in the splitting.

Therefore, if $k = c_2(E) = 1$, we must search over $x \in H^2$ for which $x^2 = 1$.

Combining (i), (ii), and (iii), we can see that

$$\sigma(M) = \mathrm{rk}(Q_M) = \#\text{ splittings of } E = \frac{1}{2}\#\{x \in H^2 \text{ s.t. } x^2 = 1\}.$$

Heuristically, this equality says that Q_M has "many" x with $\langle x, x\rangle = 1$ where "many" is exactly twice the number of splittings.

To prove Donaldson's theorem, it remains to show that Q_M is equivalent to $n(+1)$ or $n(-1)$, i.e., $\pm n = \mathrm{rk}(Q_M) = \frac{1}{2}\#\{x \in H^2 \text{ s.t. } x^2 = 1\}$.

Lemma Let $Q : V \times V \to \mathbb{Z}$ be a positive-definite symmetric unimodular bilinear pairing. Then $\#\{x \text{ s.t. } Q(x,x) = 1\} \leq 2\,\mathrm{rk}\,Q$ with equality IFF Q is equivalent 1 over $\mathbb{Z}$

Assuming the lemma, since $\mathrm{rk}(Q_M) = \frac{1}{2}\#\{x \in H^2 \text{ s.t. } x^2 = 1\}$, where x^2 is shorthand for $Q(x,x)$, we get that

$$\#\{x \text{ s.t. } Q(x,x) = 1\} \leq 2 \cdot \frac{1}{2}\#\{x \in H^2 \text{ s.t. } Q(x,x) = 1\}$$

$$\#\{x \text{ s.t. } Q(x,x) = 1\} \leq \#\{x \in H^2 \text{ s.t. } Q(x,x) = 1\}$$

So clearly we must have equality, so the lemma implies that Q is is equivalent to 1 over $\mathbb{Z}$, thus $Q \cong \pm(n1)$.

This proves Donaldson's theorem.

We now prove the lemma.

Pf of lemma First note that if $Q = I$, there is clearly $2 \cdot \mathrm{rk}\,Q$ of such elements of H^2, since the standard basis vectors e_i have self pairing 1, as well as $-e_i$.

For a general Q, we'll apply the Gram-Schmidt process over $\mathbb{Z}$. Let x be such that $Q(x,x) = 1$. We claim that $V = \langle x\rangle \oplus \langle x\rangle^\perp$.

Indeed, since any $z \in V$ decomposes as

$$z = \underbrace{Q(x,x)x}_{\in \langle x\rangle} + \underbrace{(z - Q(x,z)x)}_{\in \langle x\rangle^\perp}$$

we know that $V = \langle x\rangle + \langle x\rangle^\perp$; since $\langle x\rangle \cap \langle x\rangle^\perp = \{0\}$,

this decomposition is orthogonal.

Thus $Q: \langle x\rangle \oplus \langle x\rangle^\perp \times \langle x\rangle \oplus \langle x\rangle^\perp \to \mathbb{Z}$, so Q splits as a block matrix $Q = \left[\begin{array}{c|c} 1 & 0 \\ \hline 0 & * \end{array}\right] \begin{matrix} \} x \\ \} x^\perp \end{matrix}$

Now, notice the following:

1) Q is still positive definite on $x^\perp$

2) Any $z \in V$ such that $Q(z,z)=1$ is either $\pm x$ or lies entirely in $\langle x\rangle^\perp$: If $z = cx + w$ for $w \in x^\perp$,

$$Q(z,z) = c^2 + (w,w) \neq 1$$

3) The rank of $Q|_{\langle x\rangle^\perp}$ is $\operatorname{rank}(Q)-1$

Thus we may proceed by induction on $Q|_{\langle x\rangle^\perp}: \langle x\rangle^\perp \times \langle x\rangle^\perp \longrightarrow \mathbb{Z}$.

If $\langle x\rangle^\perp$ contains no elements for which $x^2=1$, the process concludes, and the number of elements for which $x^2=1$ is necessarily less than $2\cdot \operatorname{rk}(Q)$. Otherwise, Q can be completely diagonalized, so Q is equivalent to I. $\square$

Lecture 8 // Sept 17 // Singularities in $\mathcal{M}$ & Holonomy

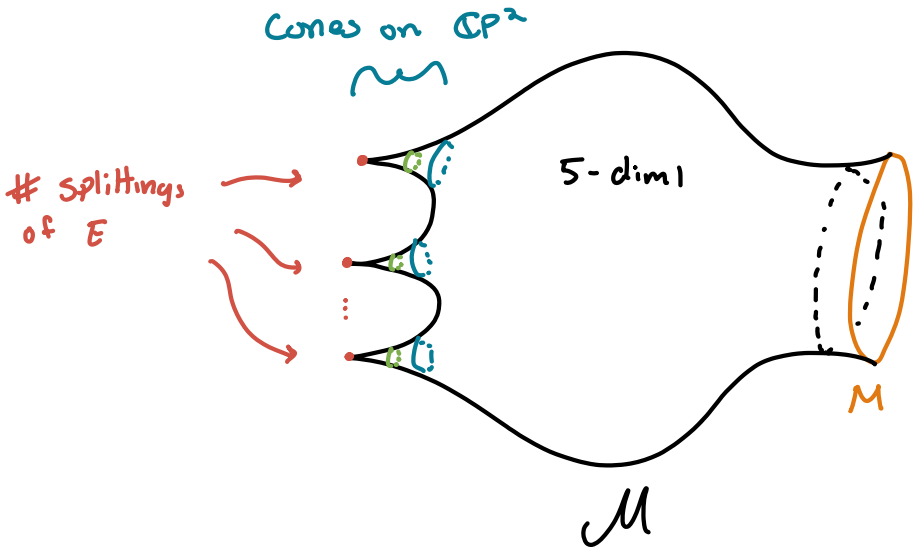
Last Time: Let (M^4, g) be positive definite and simply connected.

Fix a $k=1$ $SU(2)$ bundle E over M.

Consider $\mathcal{M} = \mathcal{A}/\mathcal{G}$ the SDYM moduli space.

We know the following qualitative features of $\mathcal{M}$:

1) Away from a finite number of singular points, $\mathcal{M}$ is a smooth 5-dim oriented manifold.
2) The number of these points = the # of splittings of E
3) Each point has a nbhd $\circ$ homeo to cone on $\mathbb{CP}^2$. Diffeo away from pt
4) $\mathcal{M}$ is <u>not</u> compact, but it admits a compactification $\overline{\mathcal{M}}$ such that $\partial\overline{\mathcal{M}} \cong M$



After some linear algebra, these properties prove Donaldson's theorem:

$$\text{rk } Q_M = \sigma(M) \overset{\text{Since } \sigma \text{ bordism inv.}}{=} \#\text{ Splittings of } E = \frac{1}{2}\#\, x \in H_2 \text{ with } x^2 = 1$$

($\sigma(M) =$ Since definite)

Today, we'll unpack these points: why is $\mathcal{M}$ singular, and why do nbhds look like cone on $\mathbb{CP}^2$

Singularities in $\mathcal{M}$

Recall that $\mathcal{M} = \{\text{SDYM conn.}\}/\mathcal{G}$. $\{\text{SDYM CONN}\} \subseteq \mathcal{A}$, which is a smooth manifold, on which $\mathcal{G}$ acts.

However, $\mathcal{G}$ may have stabilizer subgroups.

Thus the singularities arise from the failure of $\mathcal{G}$ to act freely on $\mathcal{A}$.

Rmk Establishing $\{\text{SDYM conn}\}$ sm mfd is fairly standard:

It is the zero locus of map from oo-dim map to somewhere else,

verify that 0 is regular value, apply Sard-Smale.

Def For $\nabla \in \mathcal{A}$, $\text{Stab}(\nabla) = \{g \in \mathcal{G} \mid g^*(\nabla) = \nabla$

$$A \longrightarrow g^{-1}Ag + g^{-1}dg$$

What can $\text{stab}(\nabla)$ be? First note that for every ∇, $\pm Id \in \text{Stab}(\nabla)$.

Since $\pm Id$ is sent to itself under conjugation, $\pm Id$ is a well-defined section of $\text{Aut}(E)$ These are not interesting gauge transformations, so it's sometimes more convenient to take $\mathcal{G}/\pm Id \curvearrowright \mathcal{A}$ instead.

What else is possible? Let's examine an example.

connections with non trivial stabilizers are related to # splittings.

For a line bundle L, consider the constant gauge transformation given by multiplication by a fixed element of S^1, $e^{i\theta} \in S^1$.

For $E = L \oplus \bar{L}$, consider now $g = \begin{bmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{bmatrix}$

One can check this is a well defined gauge transformation on E.

If ∇ is split, eg, $A = \begin{bmatrix} ia & 0 \\ 0 & -ia \end{bmatrix}$, easy to check that $g^*(\nabla) = \nabla$

So if E is split and ∇ is a split connection, considered as a subgroup of $SU(2)$, $\text{Stab}(\nabla)$ is at least as big as S^1.

Rmk In this observation, we play fast and loose with the difference between a local definition of a gauge transformation and its definition globally. The correct statement is given in the following lemma, and the tension between global & local will be resolved as we introduce the tools to prove it.

Lemma For a $SU(2)$ bundle with $\kappa \neq 0$, a connection has stabilizer bigger than $\pm Id$ IFF it is split, and $Stab(D) \cong S^1$

The proof isn't hard, but requires more technical language.

But this lemma gives why the singularities of $\mathcal{M}$ are what they are. We showed previously that each splitting of E supports a unique (up to $\mathcal{G}$) split SDYM connections.

Each singular point is the unique SDYM connection for a splitting. Thus the # of singular points is the # of splittings. This gives 2)

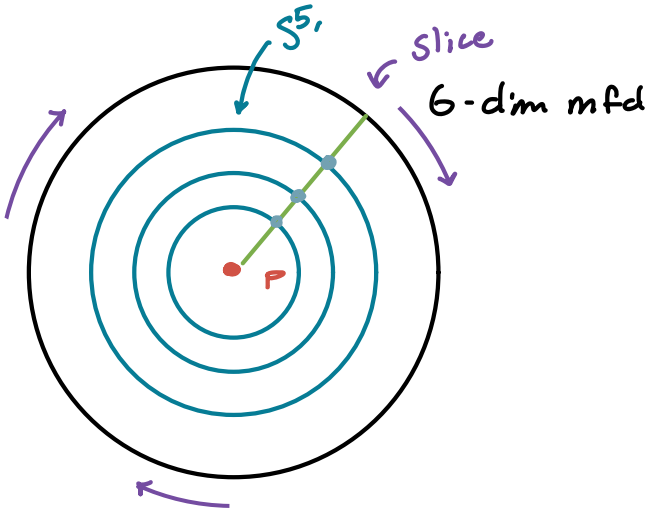
We now ask why local neighborhoods are cones on $\mathbb{CP}^2$.

Near a singular point, $\mathcal{G}/\pm I$ acts freely.

The approximate idea is that we first quotient by $\pm I$ to get a 6-dim mfd with a $S^1/\pm I$ action which acts freely except at one point.

The local model for this is $S^1 \curvearrowright \mathbb{R}^6 \cong \mathbb{C}^3$ via usual S^1 complex multiplication. $\mathbb{C}^3/S^1 \cong$ cone on $\mathbb{CP}^2$ with one singular point, since on each S^5 slice, we have $S^5/S^1 \cong \mathbb{CP}^2$

(via definition $\mathbb{CP}^2 := S^{2n+1}/S^1$)



Note The $\pm I \subset \mathcal{G}$ and S^1 actions are more entertwined in $\mathcal{G}$ than this; we can't actually quotient by gauge in this "partial" way. But it gives an approximate idea.

Q Why is $\mathcal{M}$ 5-dimensional?

This is where we use that $k=1$.

The dimension is from the Atiyah Singer Index formula applied to a fredholm operator whose zero locus are the SDYM connections.

When $k>1$, we get more generally that Sing nbhd $\cong$ cone on $\mathbb{CP}^n$

We now must introduce more differential geometry.

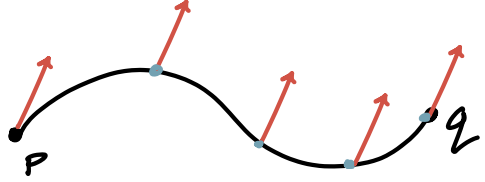
Interlude: Parallel Transport

Let E be a vector bundle over M.

Def Let ∇ be a connection on E, $\gamma : [0,1] \to M$ a path.
A section s defined along γ is parallel transported by ∇ along γ if $\nabla_{\gamma'(t)} s = 0$ for all t.

Lemma Let $\gamma : [0,1] \to M$ be a path from p to q.
If we fix any $v \in E_p$, then there exists a unique section s defined along γ such that s extends v and s is parallel transported by ∇ along γ.

Pf idea



Using the existance of solutions to ODE's, we can construct s and show that $(d+A)s = 0$ ∎

To better see the point of parallel transport, we can observe the effect of parallel transport on a frame of E_p, ie, a basis for E_p, and parallel transport that basis along γ.

Parallel transport preserves orthonormality, so this gives you a a particular trivialization of E along γ for which the basis elements are covariently constant: A vanishes in the path direction γ'.

In essence, the covariant derivative specifies a "preferred frame" along each path, which we can find by choosing a frame at one end and parallel transporting it along the path.

What else can parallel transport do?

For each path γ from p to q, we can build a linear map from E_p to E_q which sends $v \in E_p$ to the endpoint of the parallel transport of v along γ. Call this map $hol_{p,q}(\gamma): E_p \to E_q$. Since $hol_{p,q}(\gamma) \circ hol_{p,q}(\gamma) = id$, this map is an isomorphism.

Now, take $q = p$, so γ is is a based loop in M.

Then $hol_p = hol_{p,p}$ gives a map $E_p \to E_p$, so defines an element $hol_p(\gamma) \in GL(E_p)$.

Def Holonomy is a map $\text{hol}_p : \{ \text{loops based at } p \} \to GL(E_p)$

Properties of Holonomy:

1) If E is a G-bundle and ∇ is a G connection, then $\text{hol}_p(\gamma)$ is $G \leq GL(E_p)$ valued

2) If $F_\nabla = 0$, the holonomy only depends on the based smooth homotopy class of γ.

   Ie, hol_p descends to a map

   $$\text{hol}_p : \pi_1(M) \to GL(E_p) \quad (\text{or } \mathfrak{g} \text{ for } G)$$

   For $G = SU(2)$, $\text{hol}_p : \pi_1(M) \to su(2)$ is a $su(2)$ representation of $\pi_1(M)$.

3) If g is a gauge transformation,

   $$\text{hol}^{g^*(\nabla)}_{p,q}(\gamma) = g_q^{-1} \circ \text{hol}^{\nabla}_{p,q}(\gamma) \circ g_p$$

To illustrate holonomy, consider a trivial line bundle $L = M \times \mathbb{C}$ over M and a $U(1)$ connection on L given by $\nabla = d + ia$, $a \in \Omega^1(M; i\mathbb{R})$.

Thus a section here is just a $\mathbb{C}$-valued function. For a path γ, what does it mean for $\nabla_{\gamma'(t)} s = 0$? Given a based loop γ and a section defined along γ, what does it mean for s to be parallel transported along γ? Since we're working in a line bundle, a section is just a $\mathbb{C}$-valued function, and for $\nabla_{\gamma'(t)} s = 0$, the covariant derivative of this function must vanish along γ.

Explicitly, we get

$$\nabla_{\gamma'(t)} s = 0$$

$$(ds)(\gamma'(t)) + ia(\gamma'(t))s = 0$$

$$\frac{d}{dt}(s \circ \gamma) + ia(\gamma'(t))(s \circ \gamma(t)) = 0$$

But this is just a differential equation, which we can solve as usual:

$$\frac{d}{dt}(s\circ\gamma) = -ia(\gamma'(t))(s\circ\gamma(t))$$

$$\Rightarrow (s\circ\gamma) = \left(e^{-i\int_0^t a(\gamma'(h))dh}\right) s(\gamma(0))$$

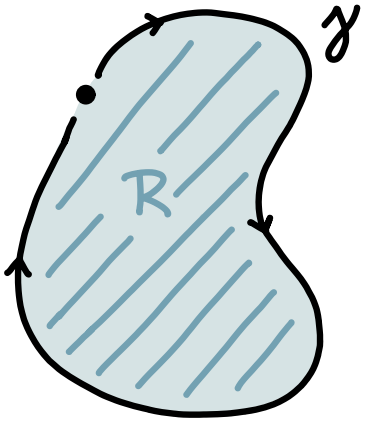
If we conflate s and $s\circ\gamma$, we get that $hol_{\gamma,\nabla} = e^{-i\int_\gamma a} \in S^1$.

Intuitively, we're supposed to think that solving $(d+A)s=0$ gives $s(t) = e^{-\int A} s(0)$, so if A is lie alg. valued, $-\int A$ lie alg. valued, so $e^{-\int A}$ lie gp valued. In reality, this is false – there's an issue with commutivity when $G \neq U(1)$.

However we see directly here that $hol_\nabla(\gamma)$ need not be the identity: if $\int_\gamma a \neq 2\pi n$, $e^{-i\int_\gamma a} \neq 1$.

In fact, we see a nice connection with property #3. Indeed, suppose that γ bounds a region R.

Then Stokes' theorem says that



$$\int_\gamma a = \int_{\partial R} a = \int_R da$$

where da is the curvature of ∇

So the heuristic is that holonomy picks up the curvature of a connection.

This demonstrates, in the $U(1)$ case, assertion #3:

If γ is varied through region where $F_\nabla = 0$, then holonomy is not changed.

However, note that this analysis applies to single coordinate charts; it is purely local. If the bundle is nontriv., it is not true that $F_A = 0$ implies that $\mathrm{hol}(\gamma) = id$. Furthermore, it makes no sense to treat the differential equation like its globally defined, since α isn't. But it is true that holonomy is const. in its homotopy class.

Next time: We'll use parallel transport to discuss the lemma on the exact behaviour of stabilizers of connections.

Lecture 9 // Sept 22 // Singularities in $\mathcal{M}$, part 2

Last time: We saw that connections whose stabilizer is larger than $\pm Id$ lead to singular points of $\mathcal{M}$, and this does happen.

If E is split and ∇ is split, then $Stab(\nabla)$ is at least as large as S^1. Precisely, we had the following lemma:

Lem If E is a $k \neq 0$ $su(2)$ bundle, then $stab(\nabla)$ is larger than $\pm I$ IFF ∇ is split, in which case $Stab(\nabla) \cong S^1$

Towards proving this lemma, we discussed parallel transport and holonomy.

The general question is what are the possibilities of $Stab(\nabla)$?

A priori, we know that $Stab(\nabla) \subseteq G_1$, and this fact actually restricts what $stab(\nabla)$ must be as follows.

Lemma $g \in Stab(\nabla)$ IFF $D_\nabla g = 0$, where $D_\nabla : End(E) \to End(T^*M \otimes E)$

ie, g is parallel transported by D_∇ along every γ.

Pf $D_\nabla g$ is characterized by the Leibniz rule:

$$\nabla(gs) = (D_\nabla g)s + g(\nabla s)$$

After moving terms, we get

$$g^{-1}\nabla(gs) = g^{-1}(D_\nabla g)s + \nabla s$$

By definition, $g^*(\nabla)s = g^{-1}\nabla(gs)$

Thus $g \in Stab(\nabla)$ IFF $g^*(\nabla) = \nabla$

$$\Leftrightarrow \quad \nabla(s) = g^{-1}(D_\nabla g)s + \nabla s \qquad \forall s$$

$$\Leftrightarrow \quad g^{-1}(D_\nabla g)s = 0 \qquad \forall s$$

$$\Leftrightarrow \quad D_\nabla g \equiv 0 \qquad \square$$

Moreover, we know that the parallel transport is unique.

So we can also conclude the following.

Lemma If $g \in \mathrm{Stab}(\nabla)$, then g is determined by its value at a single point.

Rmk Its not true that any g can be transported: g need not be globally defined! But we know $g \in \mathrm{Stab}(\nabla)$, so a priori, it is a global object, so parallel transport makes sense.

Equivantly, we can write this as for any p, the restriction map $\mathcal{G} \to \mathrm{Aut}(E_p)$ is injective on $\mathrm{Stab}(\nabla)$.
$g \mapsto g_p$

Thus (via restriction) $\mathrm{Stab}(\nabla)$ can be considered as a subgroup of $SU(2)$. In fact, $\mathrm{Stab}(\nabla)$ has a nice connection with the holonomy of ∇.

Recall that for a fixed $p \in M$, the holonomy gives a map

$$\mathrm{hol}_p : \{ \text{loops based at } p \} \to \mathrm{Aut}(E_p) = SU(2)$$

$$\gamma \longmapsto \mathrm{hol}_p^{\nabla}(\gamma)$$

Very explicitly, under hol_p, each γ gets sent to a specific linear transformation E_p to E_p, eg, a matrix transformation of the fibre, eg, an element of $SU(2)$. If γ is varied to another γ', we get another matrix transformation.

By considering every γ, the set of corresponding matrices is $\mathrm{im}(\mathrm{hol}_p)$, and since $\mathrm{hol}_p(\gamma_2 \circ \gamma_1) = \mathrm{hol}_p(\gamma_2) \circ \mathrm{hol}_p(\gamma_1)$, $\mathrm{im}(\mathrm{hol}_p)$ forms a subgroup of $SU(2)$.

Lemma The restriction of $\mathrm{Stab}(\nabla)$ to a point p is the centralizer of $\mathrm{im}\,\mathrm{hol}_p$.

Pf ($\subseteq$): Take $g \in \mathrm{Stab}(\nabla)$.

We want to show that g_p commutes with $\mathrm{hol}_p(\gamma)$ for every loop γ.

We know that $\mathrm{hol}_p^{g^*(\nabla)}(\gamma) = g_p^{-1} \circ \mathrm{hol}_p^{\nabla}(\gamma) \circ g_p$.

Since $g \in \mathrm{Stab}(\nabla)$, we know $g^*(\nabla) = \nabla$, so $\mathrm{hol}_p^{g^*(\nabla)}(\gamma) = \mathrm{hol}_p^{\nabla}(\gamma)$.

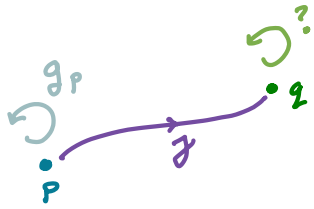
Thus $\mathrm{hol}_p^{\nabla}(\gamma) = g_p^{-1} \circ \mathrm{hol}_p^{\nabla}(\gamma) \circ g_p$ as desired. □

($\supseteq$): Take $g_p \in \mathrm{Aut}(E_p)$. Further assume that $g_p \in Z(\mathrm{im}(\mathrm{hol}_p))$.

We want to show that there exists some $g \in \mathrm{Stab}(\nabla)$ such that the restriction of g to E_p is g_p.

Since $\mathrm{Stab}(\nabla) \equiv \{g \mid D_\nabla g = 0\}$, to define g, we should try to extend g_p via parallel transport by D_∇.

For any other point q, let γ be a path from p to q



Define g_γ such that

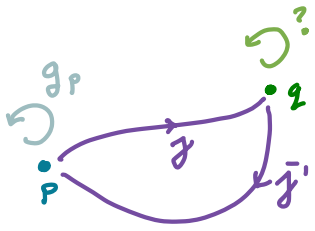
① g_p is extended along γ such that $(D_\nabla)_\gamma g = 0$

② $g_q = \mathrm{hol}_{p,q}(\nabla) \circ g_p \circ \mathrm{hol}_{q,p}(\bar{\gamma})$

We can do this along γ since parallel transport exists.

For any path, we can cook up a g_γ such that $D_\nabla(g_\gamma)$ vanishes along γ; this follows by existance of solutions to ODEs.

However, a priori, we don't know that this extension along γ gives a well-defined global g. Take γ' another path from p to q, and consider g' defined by γ'.



Then g_q and g'_q are defined as

$g_q = \mathrm{hol}(\gamma) \circ g_p \circ \mathrm{hol}(\bar{\gamma})$

$g'_q = \mathrm{hol}(\gamma') \circ g_p \circ \mathrm{hol}(\bar{\gamma}')$.

If g_q is well defined, $g_q^{-1} = (g_q')^{-1}$ for any other g_q', i.e.,
$g_q (g_q')^{-1} = id$.

Indeed, $g_q (g_q')^{-1} = hol(\gamma) \circ g \circ hol(\tilde{\gamma}) \left(hol(\gamma') \circ g \circ hol(\tilde{\gamma}') \right)^{-1} = id$ □

Let's take stock so far: we know that $Stab(\nabla)$ is determined by its restriction to a point which is the centralizer of the holonomy at that point, viewed as a subgroup of $SU(2)$.

So our question now becomes what are the subgroups of $SU(2)$?

Fact They are as follows:

Subgroups of $SU(2)$	Centralizer = $Stab(\nabla)$
I	$SU(2)$
$-I$	$SU(2)$
S^1 or a conjugate of S^1	S^1
Subgroups of S^1 or conjugate	S^1
non-abelian	$\pm I$
$SU(2)$	$\pm I$

This means that the only possibilities for $Stab(\nabla)$ are $\pm Id$, S^1, or $SU(2)$ itself. We are now in the position to prove the lemma.

Lemma If E is a $k \neq 0$ $SU(2)$ bundle, then $Stab(\nabla)$ is larger than $\pm I$ IFF ∇ is split, in which case $Stab(\nabla) \cong S^1$

Proof of lemma

We first show that either $Stab(\nabla) \cong S^1$ or $\pm Id$. From there, we'll show the former happens IFF ∇ is split.

We just saw that there are only three possibilities for $Stab(\nabla)$, so we need only to rule out $SU(2)$.

If ∇ is a connection on an $k \neq 0$ bundle, then F_∇ cannot

be identically 0 (if it was, $c_2 = k = 0$).

Thus holonomy can't be nontrivial, since the holonomy picks up curvature enclosed by a loop. If this path is varied, hol will vary continuously, so $\mathrm{im}(\mathrm{hol}_p)$ can't be contained in a discrete subgroup. Thus the centralizer cannot be $\pm I$, so the subgroup can't be $SU(2)$.

Now, suppose $\mathrm{stab}(\nabla) = S^1$. We just saw that $\mathrm{stab}(\nabla) = S^1$ implies that at a point $p \in M$, the element $g_p \in SU(2)$ must actually be contained in $U(1) \subset SU(2)$.

Therefore $g_p = \begin{bmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{bmatrix}$, which induces a splitting of the fibre over p by considering the eigenspaces.

For any other $q \in M$, g_q is obtained from g_p via conjugation with $\mathrm{hol}_{p,q}(\nabla)$, so the eigenspaces of g_q provide a splitting of E_q compatible with that of E_p. Since $\mathrm{Stab}(\nabla)$ is reconstructed from g_p by parallel transport, the splitting of E_p induces a splitting of E.

To Do: Insert proof of other direction. □

Having proved this lemma, we can conclude our discussion of why the number of singular points in $\mathcal{M}$ is the number of splittings of E.

We'll now begin a discussion of the other major qualitative feature of $\mathcal{M}$: that $\partial\overline{\mathcal{M}} = M$. To do so, we'll first discuss instantons ((A)SD Yang-Mills connections) on S^4, which we'll do by identifying S^4 with $\mathbb{HP}^1$.

Interlude: Review of Quaternions & some useful isomorphisms from algebra

Def A quaternion is an expression of the form $q = x + yi + zj + wk$ where $x, y, z, w \in \mathbb{R}$ and $1, i, j, k$ are basis elements such that

$1 \cdot \alpha = \alpha$ for $\alpha = i, j, k$, $i^2 = j^2 = k^2 = ijk = -1$

$i \cdot j = k$, $j \cdot k = i$, $k \cdot i = j$, and $\alpha \cdot x = x \cdot \alpha$ for $\alpha = i, j, k$ and $x \in \mathbb{R}$

The space of quaternions is denoted $\mathbb{H}$

Remark $\mathbb{H}$ admits the structure of a division ring. However, multiplication fails to be commutative, so $\mathbb{H}$ is not a field: if $q_1 = i$ and $q_2 = j$, $q_1 q_2 = ij \neq ji = q_2 q_1$.

Note that $\mathbb{H} \cong \mathbb{C}^2 \cong \mathbb{R}^4$ by the following isomorphisms.

First rewrite a quaternion as

$$\begin{aligned} x + yi + zj + wk &= (x + yi) + (zj + wk) \\ &= (x + yi) + (zj + wij) \\ &= (x + yi) + (z + wi)j \end{aligned}$$

Then the isomorphisms are given by the maps

$$\begin{array}{ccccc} \mathbb{H} & \longrightarrow & \mathbb{C}^2 & \longrightarrow & \mathbb{R}^4 \\ (x+iy) + (z+iw)j & \mapsto & (x+iy, z+iw) & \mapsto & (x, y, z, w) \end{array}$$

The quaternions can also be identified by a specific subgroup of the space of 2×2 complex matrices.

Def The Pauli Matrices are a set of three 2×2 matrices $\{\sigma_1, \sigma_2, \sigma_3\} \subset Mat_{2\times 2}(\mathbb{C})$ which are traceless, Hermitian, unitary, and involutive:

$$\sigma_1 := \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} \quad \sigma_2 := \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \quad \sigma_3 := \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$$

By sending $i \mapsto \sigma_1$, $j \mapsto \sigma_2$, and $k \mapsto \sigma_3$, any quaternion $q = x + yi + zj + wk$ can be rewritten as the linear combination

$$x\,Id + y\sigma_1 + z\sigma_2 + w\sigma_3 = \begin{bmatrix} x & 0 \\ 0 & x \end{bmatrix} + \begin{bmatrix} yi & 0 \\ 0 & -yi \end{bmatrix} + \begin{bmatrix} 0 & z \\ -z & 0 \end{bmatrix} + \begin{bmatrix} 0 & wi \\ wi & 0 \end{bmatrix}$$

$$= \begin{bmatrix} x+yi & z+wi \\ -z+wi & x-yi \end{bmatrix}$$

Now, notice that the (2,2) entry is the complex conjugate of the (1,1) entry, and the (2,1) entry is the negative conjugate of the (1,2) entry. This relationship is clearly from the chosen basis vectors, so it holds for the image of any quaternion:

$$q \longmapsto \begin{bmatrix} \alpha & \beta \\ -\bar{\beta} & \bar{\alpha} \end{bmatrix} \quad \text{where} \quad q = \alpha + \beta j$$

By now, matrices of this form should be familiar. If we divide by the norm, we recover the $SU(2)$ matrices!

Proposition The Lie group $SU(2) = \left\{ \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix} \mid \alpha, \beta \in \mathbb{C},\ |\alpha|^2 + |\beta|^2 = 1 \right\}$ is isomorphic (as a ring) to the ring of unit quaternions.

Recall that $SU(2)$ is the 2×2 complex matrices of determinant one. By identifying $\mathbb{C}^2$ with $\mathbb{R}^4$, an element of $SU(2)$ determines a determinant one 4×4 matrix, ie, a rotation in $\mathbb{R}^4$. Rotations in $\mathbb{R}^4$ are determined by a rotation of the unit sphere $S^3 \subset \mathbb{R}^4$, which is determined by the image of the $(1,0,0,0)$ vector, which is a point on S^3. This suggests the following claim.

Proposition $SU(2)$ is diffeomorphic to S^3

Exercise Prove this proposition

All together, we can conclude that $SU(2) \cong \{q \in \mathbb{H} \mid |q|^2 = 1\} \cong S^3$

We can now ask: what is $\mathbb{HP}^1$?

We now have the tools to understand why $\mathbb{HP}^2 \cong S^4$.

Recall that one construction of the projective spaces $\mathbb{RP}^1$ and $\mathbb{CP}^1$ are as the one point compactifications of $\mathbb{R}$ and $\mathbb{C}$, respectively; Since $\mathbb{H} \cong \mathbb{R}^4$ and $\mathbb{R}^4 \cup \{\infty\} \cong S^4$, we can conclude that $\mathbb{HP}^2 \cong S^4$!

Alternatively, recall that $\mathbb{RP}^n$ is most commonly constructed as the quotient of S^n by the antipodal map. By viewing the antipodal map as the action of the "unit circle" $\{\pm 1\} \subset \mathbb{R}$, and further identifying $\{\pm 1\} \cong S^0$, $\mathbb{RP}^n$ can be seen as the base space of a fibration $p: S^n \to \mathbb{RP}^n$ with S^0 fibres. This construction is easily generalized to $\mathbb{CP}^n$ and then $\mathbb{HP}^n$: Let V be a vector space over the ring $k = \mathbb{R}, \mathbb{C}$, or $\mathbb{H}$, and let $U(V^d)$ be the unit circle in V^d.

When $k = \mathbb{R}$, V^{n+1} is $n+1$ real dimensions, so $U(V_{\mathbb{R}}^{n+1}) \cong S^{n+1-1} = S^n$.

The unit circle in $\mathbb{R}$ is $\{\pm 1\} \cong S^0$. This recovers the fibration

$$S^0 \to S^n \to \mathbb{RP}^n.$$

When $k = \mathbb{C}$, V^{n+1} is $2(n+1)$ real dimensions, so $U(V_{\mathbb{C}}^{n+1}) \cong S^{2(n+1)-1} = S^{2n+1}$

The unit circle in $\mathbb{C} \cong \mathbb{R}^2$ is $\{e^{i\theta}\} \cong S^1$, so the fibration becomes

$S^1 \to S^{2n+1} \to \mathbb{CP}^n$. Note that when $n = 1$, this recovers the standard Hopf fibration $S^1 \to S^3 \to S^2$.

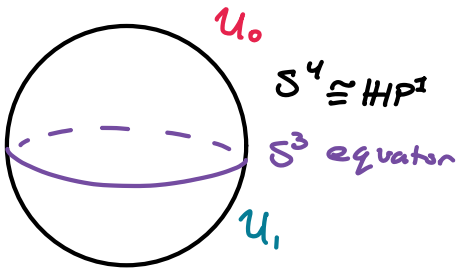
When $k \cong \mathbb{H}$, V^{n+1} is $4(n+1)$ real dimensions, so $U(V_{\mathbb{H}}^{n+1}) \cong S^{4(n+1)-1} = S^{4n+3}$

The unit circle in $\mathbb{H} \cong \mathbb{R}^4$ is S^3, thus the fibration is

$$S^3 \to S^{4n+3} \to \mathbb{HP}^1$$

Instantons on S^4

Mirroring our previous discussion of instantons on $\mathbb{CP}^1$, we'll view $S^4 = \mathbb{HP}^1$. $\mathbb{HP}^1$ admits two charts $U_0, U_1 \cong \mathbb{R}^4 \cong \mathbb{H}$, with the coordinate change function given by $q \longmapsto \frac{1}{q}$



What are $SU(2)$ bundles over S^4?

$SU(2)$ bundles correspond to maps $S^3 \to SU(2)$, where S^3 is the region on which we glue U_0 to U_1. Since $SU(2) \cong S^3$, $SU(2)$ bundles over S^4 are classified by maps $S^3 \to S^3$, of which there are $\mathbb{Z}$-many (degree)

In terms of coordinates, some possible gluing maps are $g(q) = q^n / |q|^n$

The $k=1$ bundle corresponds to $g(q) = q$.

II Instanton Floer Homology

Lecture 11 // Sept 29 // Insanton Floer Homology

The rest of the course will follow this general outline.

1) broad overview of Insanton Floer Homology
2) formal properties & applications
3) more detailed disussion of (1)
4) ?? more recent papers

Instanton Floer Homology is a 3-manifold invariant. Originally, it was defined only for integer homology spheres, and we'll restrict to Y a $\mathbb{Z}HS^3$ for simplicity, but the constructions we'll discuss can be extended.

The general procedure is as follows:

$$Y \rightsquigarrow C(Y) \rightsquigarrow H_*(C(Y))$$

$\mathbb{Z}HS^3$ — Instanton chain complex — Instanton Floer homology

Where does $C(Y)$ come from? Two slogans:

1) ∞-dimensional Morse homology assc. to the CS functional

ie, there is a mysterious S^1-valued functional on an auxilliary space assc. to Y (like $\mathcal{A}/\mathcal{G}$, but for Y^3)

$$CS: \mathcal{A}/\mathcal{G} \longrightarrow \mathbb{R}/\mathbb{Z}$$

Whenever you have a $\mathbb{R}$-valued function, we take the morse hom:

- the cp's of CS become the generators of the chain complex
- the differentials are defined (roughly) by studying the gradient flow of CS.

2) The generators of $C(Y)$ are in correspondence with conjugacy classes of $SU(2)$ representations of $\pi_1(Y)$. So $C(Y)$ actually comes from π_1.

Today, we'll broadly discuss 1&2, and why they're the same.

What is the Chern-Simons Functional

Take Y^3 a $\mathbb{Z}HS^3$ (tho nothing today depends on that)

Fix a $SU(2)$ bundle over Y. Since $c_2(4)=0$, there's only one such bundle, the trivial one $E = Y \times \mathbb{C}^2 \to Y$

Just like before, we study $\mathcal{A} = \{ SU(2) \text{ conn on } E^3 \}$

and $\mathcal{G} \curvearrowright \mathcal{A}$, $\mathcal{G} = \Gamma(\mathrm{Aut}(E))$ Since E is trivial, $\mathcal{G} \cong \mathrm{Maps}(Y, SU(2))$

Note that $\pi_0(\mathcal{G}) \cong \mathbb{Z}$ since $SU(2) \cong S^3$, and maps $Y \to S^3$ are determined by the degree $\deg(g) \in \mathbb{Z}$

$\mathcal{A}/\mathcal{G}$ = configuration space / gauge, the domain of CS

Def The Chern-Simons Functional is a map

$$CS : \mathcal{A}/\mathcal{G} \to S^1$$

$$CS(A) = \frac{1}{8\pi^2} \int_Y \mathrm{tr}\left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right)$$

By this, we mean first: fix a global trivialization of E.

In this global trivialization, ∇ is represented by a matrix of 1-forms A, which is the input of the integral.

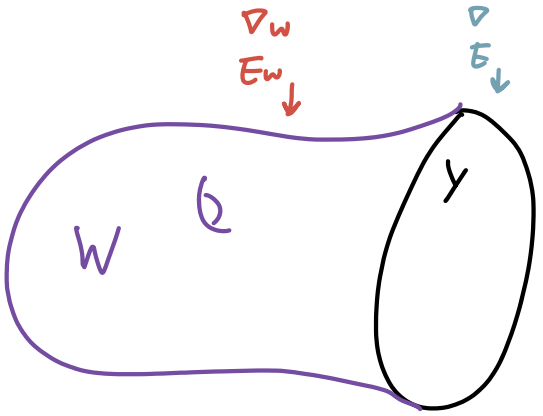
Exercise If $g \in \mathcal{G}$ is a global gauge transformation, $CS(g^* A) = CS(A) + \deg(g)$

This gives another way of saying that $\pi_0(\mathcal{G}) \cong \mathbb{Z}$; each connected component corresponds to a trivialization, so this is saying that, up to homotopy, there are $\mathbb{Z}$-many trivializations of E. The value of CS depends on which trivialization you choose.

This def is super explicit, but isn't as insightful as the following alt. def.

Def Let A be a connection on E. Choose any:

- smooth 4-mfd W^4 such that $\partial W \cong Y$
- $SU(2)$-bundle E_W on W such that $E_W \cong E$
- $SU(2)$-connection ∇_W such that $\nabla_W|_Y = \nabla$



Then $$CS(\nabla) = \frac{1}{8\pi^2}\int_W (F_{\nabla_W} \wedge F_{\nabla_W})$$

This definition follows from the definition of relative characteristic classes.

But as-is, you should be skeptical that this is well defined.

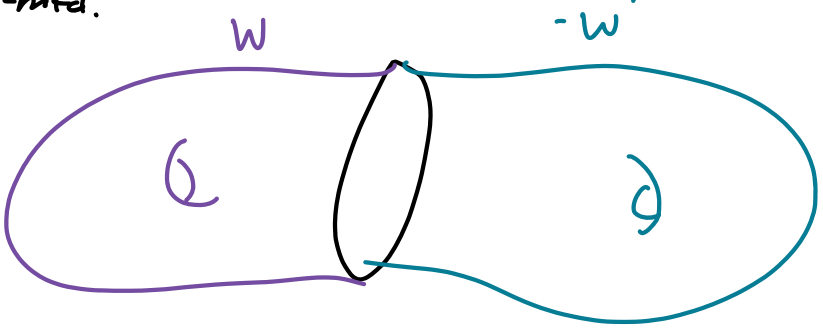
Let's choose a different W', $E_{W'}$, ∇'. Then

$$CS(\nabla) = \frac{1}{8\pi^2}\int_W (F_{\nabla_W} \wedge F_{\nabla_W}) \text{ and}$$

$$CS(\nabla') = \frac{1}{8\pi^2}\int (F_{\nabla_{W'}} \wedge F_{\nabla_{W'}})$$

If this definition is well-defined, then $CS(\nabla) - CS(\nabla') = 0$.

To compute the difference, glue W and $-W'$ along Y to get a $SU(2)$ bundle over a closed 4-mfd.



Then their difference

$$\frac{1}{8\pi^2}\int_W (F_{\nabla_W} \wedge F_{\nabla_W}) - \frac{1}{8\pi^2}\int_W (F_{\nabla_{W'}} \wedge F_{\nabla_{W'}})$$

$$= \langle c_2(E_\Sigma), [W \underset{Y}{\cup} -W'] \rangle \in \mathbb{Z}$$

Thus different W's change $CS(\nabla)$ by an integer.

Thus $CS(\nabla)$ is well-defined in S^1.

Remark The bundles E_W, $E_{W'}$ over W and W' glue together to a a well-defined bundle E_Σ over $W \cup_Y -W'$. The same is true for ∇_W and $\nabla_{W'}$, obtaining ∇_Σ

The Chern-Simons integral of this connection is that of a closed 4-manifold.

$$CS(\nabla_\Sigma) = \frac{1}{8\pi^2}\int_{W\cup_Y -W'} tr\left(F_{\nabla_\Sigma} \wedge F_{\nabla_\Sigma}\right) \in \mathbb{Z}$$

and splits as $\int_W tr\left(F_{\nabla_\Sigma|_W} \wedge F_{\nabla_\Sigma|_W}\right) - \int_{W'} tr\left(F_{\nabla_\Sigma|_{W'}} \wedge F_{\nabla_\Sigma|_{W'}}\right)$.

However, for the connections ∇_W and $\nabla_{W'}$ over 4-manifolds with non-empty boundary, $CS(\nabla_W) \notin \mathbb{Z}$.

In our proof of well-definedness, we relied on some gluing procedure without verifying what that procedure actually is: this was incredibly sloppy. We should also ask whether this def. plays well with G_J.

In order to glue E_W & $E_{W'}$ together, we must fix identifications $E_W|_Y \cong E_{W'}|_Y$. Since there is only one $SU(2)$ bundle over Y, the trivial one, we could argue that E_W, $E_{W'}$ are trivial bundles, thus E_Σ is trivial, so the difference is zero. While the first part of this is true, like we saw with $\mathbb{CP}^1$, two trivial bundles could be glued via some nontrivial boundary maps, like $\mathbb{Z}^n$, which results in a nontrivial bundle. Even simpler, two trivial $\mathbb{R}$-bundles over the interval I could be glued via a twist, resulting in the Möbius band.

The number of these different maps is the number of retrivializations of the bundle on the boundary. So we must ask: How many retrivializations of ∂W are there? The answer is $\mathbb{Z}$, as we saw earlier, and these $\mathbb{Z}$-many gluings give $\mathbb{Z}$-many $SU(2)$ bundles over W!

Claim Re-trivializing a bundle E_W over $\partial W \cong Y$ by composing with some $g \in G_J$ changes $c_2(E_\Sigma)$ by $\deg g \in \mathbb{Z}$.

Lemma The two definitions agree

Pf This follows immediately from Stokes' theorem: $\left(\int_W d\beta = \int_{\partial W} \beta \right)$

$$8\pi^2 \, CS(A) = \int_Y tr\left(A \wedge dA + \frac{2}{3}(A \wedge A \wedge A)\right)$$

$$= \int_W d\left(tr\left(A \wedge dA + \frac{2}{3}(A \wedge A \wedge A)\right)\right)$$

Simplifying, $d\left(tr\left(A \wedge dA + \frac{2}{3}(A \wedge A \wedge A)\right)\right)$

$$= tr\left(d\left(A \wedge dA + \frac{2}{3}(A \wedge A \wedge A)\right)\right)$$

$$= tr\left(d(A \wedge dA) + \frac{2}{3} d(A \wedge A \wedge A)\right)$$

$$= tr\left(dA \wedge dA - \underbrace{A \wedge d^2 A}_{0} + \frac{2}{3}\left(dA \wedge A \wedge A - A \wedge dA \wedge A + A \wedge A \wedge dA\right)\right)$$

$$= tr\left(dA \wedge dA + \frac{2}{3} dA \wedge A \wedge A - \frac{2}{3} A \wedge dA \wedge A + \frac{2}{3} A \wedge A \wedge dA\right)$$

$$= tr\left(dA \wedge dA + 2\, dA \wedge A \wedge A\right)$$

$$= tr\left(dA \wedge dA + dA \wedge A \wedge A + A \wedge A \wedge dA\right)$$

$$= tr\left((dA + A \wedge A) \wedge (dA + A \wedge A)\right) + \underbrace{tr(A \wedge A \wedge A \wedge A)}_{0}$$

$$= tr(F_A \wedge F_A)$$

$$\Rightarrow \int_Y tr\left(A \wedge dA + \frac{2}{3}(A \wedge A \wedge A)\right) = \int_W tr(F_A \wedge F_A)$$

$$= 8\pi^2 \left\langle c_2(E_\Sigma), [W \underset{Y}{\cup} -W'] \right\rangle \quad \square$$

Rmk Timeline: this all happened pre donaldson, post relative characteristic classes.

Just like in Donaldson's theorem, we want to study the critical points of CS, which we'll denote by $crit(cs) \subset \mathcal{A}/\mathcal{G}$ What about interpretation #2?

Thm $\nabla \in \mathcal{A}/\mathcal{G}$ is a critical point of the Chern-Simons functional IFF ∇ is flat, ie, $F_A = 0$

Pf $\nabla \in Crit(CS)$

$$\Leftrightarrow \frac{d}{dt}\Big|_{t=0} CS(\nabla + tb) = 0$$

$$\Leftrightarrow \frac{d}{dt}\Big|_{t=0} 8\pi^2\, CS(\nabla + tb) = 0$$

$$\Leftrightarrow \int_Y tr\left((A+tb)\wedge d(A+tB) + \frac{2}{3}\left((A+tb)\wedge(A+tb)\wedge(A+tb)\right)\right) = 0$$

$$d(A+tB) = dA \wedge B - A \wedge dB$$

$$\Leftrightarrow \int_Y tr\left(B\wedge dA + A\wedge dB + \frac{2}{3} B\wedge A\wedge A + \frac{2}{3} A\wedge B\wedge A + \frac{2}{3} A\wedge A\wedge B\right) = 0$$

$$\Leftrightarrow \int_Y tr\left(2B\wedge dA + 2B\wedge A\wedge A\right) = 0$$

$$= 2\int_Y tr\left(B\wedge F_A\right) = 0 \qquad \forall B$$

$$\Leftrightarrow F_A = 0$$

$$grad_A\, CS = *F_A \qquad \square$$

Thm Critical Points of the Chern-Simons functional modulo gauge are in bijection with $SU(2)$ representations of $\pi_1(Y)$ modulo conjugation.

Def The Character Variety of a 3-mfd is

$$\mathcal{R}(Y) = \left\{ \begin{array}{l} SU(2) \text{ representations} \\ \text{of } \pi_1(Y) \end{array} \right\} / \sim$$

where $\rho_0 \sim \rho_1$ IFF $\rho_1 = g^{-1} \circ \rho_0 \circ g$

In this language, $\mathrm{Crit}(CS) \leftrightarrow \mathcal{R}(Y)$. We now prove the theorem.

Proof Taking the holonomy gives a map

$$\{\text{Flat connections}\} \to \left\{ \begin{array}{l} SU(2) \text{ representations} \\ \text{of } \pi_1(Y) \end{array} \right\}$$

Since $\mathrm{hol}_p^{g^*(\nabla)}(\gamma) = g_p^{-1} \circ \mathrm{hol}_p^{\nabla}(\gamma) \circ g_p$, this holonomy decends to a map on the quotient

$$\{\text{Flat connections}\}/\mathcal{G} \to \left\{ \begin{array}{l} SU(2) \text{ representations} \\ \text{of } \pi_1(Y) \end{array} \right\} / \sim \; = \mathcal{R}$$

It's not obvious this map is surjective.

To show surjectivity, let $\rho : \pi_1(Y) \to SU(2)$ be a rep^n of $\pi_1(Y)$. Construct an $SU(2)$ bundle E and connection ∇ on E as follows:

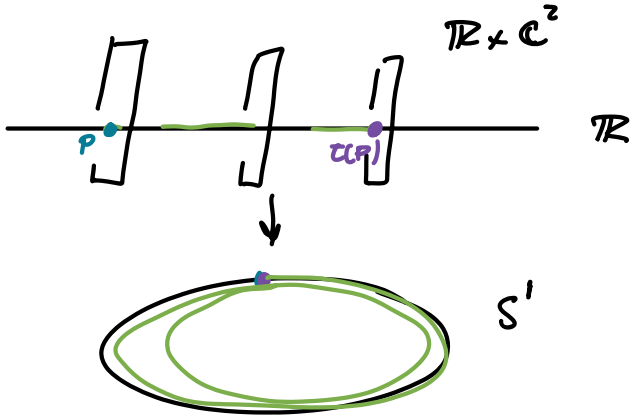
1) Form $\tilde{E} = \tilde{Y} \times \mathbb{C}^2$ trivial bundle over universal cover of Y

2) Quotient out $\tilde{E}$ by identifying the fibres $p \times \mathbb{C}^2$ and $\tau(p) \times \mathbb{C}^2$

for $\tau \in \mathrm{Deck}(\tilde{Y}) \cong \pi_1(Y)$

$\uparrow$ $\gamma \in \pi_1(Y) \leftrightarrow \tilde{\gamma}$ btwn $p, \tau(p)$

via the matrix given by $\rho(\gamma) \in SU(2)$, where γ lifts to $\tilde{\gamma}$



This defines a bundle E over Y, as well as parallel transport on E by pushing down parallel transport on $\widetilde{E}$

By construction, this parallel transport will have holonomy ρ.

This shows surjectivity.

Lecture 12 // Oct. 1 // The Chern-Simons functional & Reducibles

Last time: We defined the Chern-Simons functional

$$CS(A) = \frac{1}{8\pi^2}\int_Y \mathrm{tr}\left(A\wedge dA + \frac{2}{3}A\wedge A\wedge A\right)$$

and saw that $\mathrm{grad}_A CS = *F_A$

We also interpreted $\mathrm{crit}(CS)$ in two ways:

$$\mathrm{Crit}(CS) = \left\{\begin{array}{l}\text{flat connections}\\ \text{modulo gauge}\end{array}\right\} \leftrightarrow \mathcal{R}$$

Today, we'll complete the pf of thm from last time

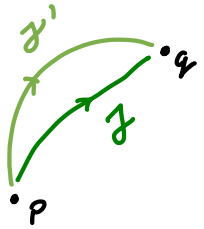
Thm Critical Points of the Chern-Simons functional modulo gauge are in bijection with $SU(2)$ representations of $\pi_1(Y)$ modulo conjugation.

Pf (cont)

We must show injectivity.

Let ∇_0, ∇_1 be two connections ST $\mathrm{hol}_p|^{\nabla_0}$ and $\mathrm{hol}_p^{\nabla_1}$ are related by conjugation via some fixed $g_p \in SU(2)$

We want to show ∇_0 and ∇_1 are gauge equivalent. We construct a global gauge transformation from g_p as follows: For any q, choose a path γ from p to q. Then define



$$g_q = \mathrm{hol}^{\nabla_1}(\gamma)\circ g_p \circ \mathrm{hol}^{\nabla_0}(\bar{\gamma})$$

Is g_q well-defined?

Yes: Choose another γ' from p to q.

Then $g'_q = \mathrm{hol}^{\nabla_1}(\gamma')\circ g_p \circ \mathrm{hol}^{\nabla_0}(\bar{\gamma}')$

and $g_q g_q'^{-1} = \left(\mathrm{hol}^{\nabla_1}(\gamma)\circ g_p\circ \mathrm{hol}^{\nabla_0}(\bar{\gamma})\right)\circ\left(\mathrm{hol}^{\nabla_1}(\gamma')\circ g_p\circ \mathrm{hol}^{\nabla_0}(\bar{\gamma}')\right)^{-1}$

$$= \left(\mathrm{hol}^{\nabla_1}(\gamma)\circ g_p\circ \mathrm{hol}^{\nabla_0}(\bar{\gamma})\right)\circ \mathrm{hol}^{\nabla_0}(\bar{\gamma}')^{-1}\circ g_p^{-1}\circ \mathrm{hol}^{\nabla_1}(\gamma')^{-1}$$

$$= \mathrm{hol}^{\nabla_1}(\gamma)\circ g_p\circ \mathrm{hol}^{\nabla_0}(\bar{\gamma})\circ \mathrm{hol}^{\nabla_0}(\gamma')\circ g_p^{-1}\circ \mathrm{hol}^{\nabla_1}(\bar{\gamma}')$$

Since $\text{hol}^{\nabla_0}(\gamma) \circ \text{hol}^{\nabla_0}(\gamma') \in \text{hol}_p^{\nabla_0}$,

and $\text{hol}_p^{\nabla_0}$ & $\text{hol}_p^{\nabla_1}$ are related by $g_p \in SU(2)$,

$$g_p \circ \text{hol}^{\nabla_0}(\gamma)) \circ \text{hol}^{\nabla_0}(\gamma') \circ g_p^{-1} = \text{hol}^{\nabla_1}(\gamma) \circ \text{hol}^{\nabla_1}(\gamma'),$$

so

$$g_q g_q^{-1} = \text{hol}^{\nabla_1}(\gamma) \circ \text{hol}^{\nabla_1}(\bar{\gamma}) \circ \text{hol}^{\nabla_1}(\gamma') \circ \text{hol}^{\nabla_1}(\bar{\gamma}') = id.$$

By construction, $\text{hol}^{\nabla_0} = g_q^{-1} \circ \text{hol}^{\nabla}(\gamma) \circ g_p$, so certainly the parallel transports of ∇_0, ∇_1 behave as if related by a gauge transformation. □

The Instanton Chain Complex

Recall that the generators of the ICC are the crit pts of the CS functional, and the differential is defined by looking at the grad. flow of CS, $\text{grad}_A(CS) = *F_A$. But how can we do this on a three manifold? The general slogan is that studying the grad. flow of CS on a three manifold Y^3 is the same as studying SD/ASD YM connections on $Y^3 \times \mathbb{R}$. There's a very precise connection btwn the three and four-manifold stories; this slogan is the first piece. Space Instead of studying SD/ASD on closed X^4, we looked at the open 4-mfd $\mathbb{R} \times Y$. On $\mathbb{R} \times Y$, consider the triv. $SU(2)$-bundle. $\mathbb{R} \times Y \times \mathbb{C}^2$. Since $c_2(\mathbb{R} \times Y) = 0$. By the Künneth formula, $c_2(\mathbb{R} \times Y) = \sum_{i+j=2} c_i(\mathbb{R}) \otimes c_j(Y) = 0$, this is only one.

Now, lets fix a metric g_Y on Y, taking the product metric $dt^2 + g_Y$ on $\mathbb{R} \times Y$. We may use this to orient $\mathbb{R} \times Y$ by declaring a positively-oriented basis to be $\{\underbrace{dt}_{\mathbb{R}}, \underbrace{dx, dy, dz}_{Y \text{ directions}}\}$.

Let's think about the bundle as a 1P1 of objects over Y

A GT g over over $\mathbb{R}\times Y$ is 1PF of GT's g_t:

eg, a GT on Y sm. varying with t

A cxn (rather, a cxn 1-form) on $\mathbb{R}\times Y$ is, in general, of the form $A = A_t + B_t\,dt$. This is achieved by separating the dt from dx, dy, dz in 1-form so

A_t consists of one forms in Y and B_t is a matrix of 0-forms.

Note that A_t, B_t still vary in t, since the x, y, z coordinates are themselves fxns of t: $x = x(t)$, $y = y(t)$, $z = z(t)$. Also note that B_t is not a cxn on Y.

For each t, define the inclusion map $\iota_t : Y \to \{t\}\times Y \subset \mathbb{R}\times Y$; then A_t is $\iota_t^*(A)$. It will be convenient for us to define the following.

Def A is in temporal gauge if $B_t = 0$.

Lemma Up to GT on $\mathbb{R}\times Y$, every A can be put into temporal gauge.

Pf We want to find a 1PF of GTs $\{g_t\}$ which kills the dt component.

Applying the transformation law, g_t sends B_t to $g_t^{-1} B_t g_t^{-1} - g_t^{-1} dg_t$. Therefore g_t acts on A via

$$A \xmapsto{\curvearrowright g_t} g_t^{-1} A g_t + g_t^{-1} dg_t$$

Simplifying,

$$g_t^{-1} A g_t + g_t^{-1} dg_t = g_t^{-1}(A_t + B_t\,dt) g_t + g_t^{-1} dg_t$$
$$= g_t^{-1} A_t g_t + g_t^{-1} B_t g_t\,dt + g_t^{-1} dg_t$$

The dt comp. of $g_t^{-1} dg_t$ is $g_t^{-1} \frac{\partial}{\partial t} g_t\,dt$, so the dt comp. of $g_t^{-1} A g_t + g_t^{-1} dg_t$ is

$$g_t^{-1} B_t g_t\,dt + g_t^{-1} \frac{\partial}{\partial t} g_t\,dt.$$

Therefore we would like to show that for every t, there is

a g_t such that

$$g_t^{-1} B_t g_t \, dt + g_t^{-1} \frac{\partial}{\partial t} g_t \, dt = 0$$

In other words, we want to solve the (IPF) of ODEs:

$$g_t^{-1} B_t g_t \, dt + g_t^{-1} \frac{\partial}{\partial t} g_t \, dt = 0$$

$$\Leftrightarrow \quad B_t g_t \, dt + \frac{\partial}{\partial t} g_t \, dt = 0$$

$$\Leftrightarrow \quad B_t g_t + \frac{\partial}{\partial t} g_t = 0$$

$$\Leftrightarrow \quad \frac{\partial}{\partial t} g_t = -B_t g_t$$

For each fixed $y \in Y$, we want that

$$\frac{\partial}{\partial t} g_t(y) = -B_t(y) \, g_t(y).$$

Let's interpret the right hand side.

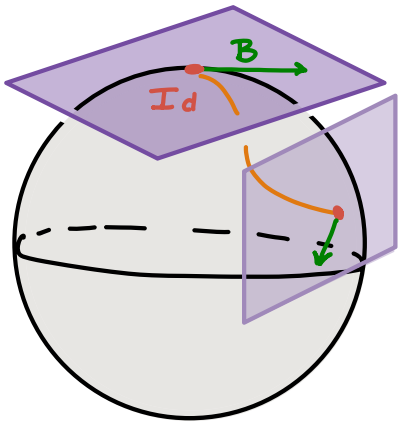
Note that $g_t(y)$ is a path in $SU(2)$, and $B_t(y)$ is a time-varying element of the lie algebra of $SU(2)$, $\mathfrak{su}(2)$, which is the tangent space at the identity $T_{Id} SU(2)$.

Explicitly, for a fixed t and y, $B_t(y)$ is a 2×2 matrix in $\mathfrak{su}(2)$, and $g_t(y)$ is a 2×2 matrix in $SU(2)$.

We can get a new 2×2 matrix in $SU(2)$ by right-multiplying $B_t(y)$ by $g_t(y)$:

$$B_t(y) \cdot g_t(y) = \begin{bmatrix} b_{11}(t,y) & b_{21}(t,y) \\ b_{12}(t,y) & b_{22}(t,y) \end{bmatrix} \cdot \begin{bmatrix} g_{11}(t,y) & g_{21}(t,y) \\ g_{12}(t,y) & g_{22}(t,y) \end{bmatrix}$$

$T_{Id} SU(2) \cong \mathfrak{su}(2)$



Doing so at each t, still fixing y, gives us the pushfoward of $B_t(y)$ under right mult. by $g_t(y)$, defining a v.f. over all of $SU(2)$.

Combining this with the LHS, if this vector field equals $\frac{d}{dt} g_t(y)$, the path must be an integral curve (eg, a flowline) for this VF. By ODE theory, this always exists; if the initial conditions are fixed, it is unique. □

Up to GT, we can think of D on $\mathbb{R}_{xy}$ as $\{\nabla_t\}$ on Y

We're now able to ask: what do the SDYM equations look like?

First, the self-duality equation is $F_A = *F_A$.

Let $A = A_t$ be in temporal gauge. Calculating $F_A = dA_t + A \wedge A$, note that it's not true that the 4-dimensional curvature is a 1PF 3-dimensional curvatures. Even though the connection itself has no dt component, the entries of this matrix are functions in all four variables, ie, these entries are allowed to vary in t. This means that dA will have a $\frac{\partial}{\partial t} A_t \, dt$ component, which encodes how the curvature of Y changes in the time direction.

$$dA = dt \wedge \frac{\partial}{\partial t} A_t + \left(\sum_{i=1}^{3} dx_i \wedge \frac{\partial}{\partial x_i} A_t \right)$$

$$= \frac{-\partial}{\partial t} A_t \wedge dt + \underbrace{\left(\sum_{i=1}^{3} dx_i \wedge \frac{\partial}{\partial x_i} A_t \right)}_{d^{(3)} A_t}$$

$$\Rightarrow F_A = \frac{-\partial}{\partial t} A_t \wedge dt + d^{(3)} A_t + A_t \wedge A_t$$

For notational convenience, define $F_{A_t} := d^{(3)} A_t + A_t \wedge A_t$

Then $F_A = -\frac{\partial}{\partial t} A_t \wedge dt + F_{A_t}$.

To calculate $*F_A$, lets find the relationship between the three and four-dimensional Hodge Stars. There are two main cases: if a 2-form has a time component, and if it doesn't.

1) $*^{(4)}(dx_1 \wedge dt) = -dx_2 \wedge dx_3 = -*^{(3)} dx_1$

2) $*^{(4)}(dx_1 \wedge dx_2) = -dx_3 \wedge dt = -*^{(3)}(dx_1 \wedge dx_2) \wedge dt$

This generalizes to the following relationship: let $i,j,k \in \{1,2,3\}$.

1) $*^{(4)}(dx_i \wedge dt) = -dx_j \wedge dx_k = -*^{(3)} dx_i$

2) $*^{(4)}(dx_i \wedge dx_j) = -dx_k \wedge dt = -*^{(3)}(dx_i \wedge dx_j) \wedge dt$

Thus $*^{(4)} F_A$ can be expressed using $*^{(3)}$ as

$$*^{(4)} F_A = *^{(4)}\left(\frac{-\partial}{\partial t} A_t \wedge dt + F_{A_t}\right)$$

$$= *^{(4)}\left(\frac{-\partial}{\partial t} A_t \wedge dt\right) + *^{(4)} F_{A_t}$$

$$= *^{(3)} \frac{\partial}{\partial t} A_t - *^{(3)} F_{A_t} \wedge dt$$

In full, the SDYM equation becomes:

$$F_A = *F_A.$$

$$-\frac{\partial}{\partial t} A_t \wedge dt + F_{A_t} = *^{(3)} \frac{\partial}{\partial t} A_t - *^{(3)} F_{A_t} \wedge dt$$

$$\frac{\partial}{\partial t} A_t = (*)^3 F_{A_t}$$

For $A = A_t$ in temporal gauge, A is a SD YM connection IFF A_t follows $\text{grad}_B CS = *F_B$

Similarly, A is ASD iff A_t is a flow of $-\text{grad}_B CS$.

Now that we've made sense of what it means to be a generator of the ICC, there's one technical point we must discuss.

Just like in the proof of Donaldson's theorem, the space $\mathcal{A}/\mathcal{G}$ in the 3-dimensional setting is not a manifold, since there can be connections with nontrivial stabilizer. Recall that we proved that the stabilizer of a connection is the centralizer of the connection.

This holds in the 3-d case too.

Subgroups of $SU(2)$	Centralizer
I	$SU(2)$
$-I$	$SU(2)$
S^1 or a conjugate of S^1	S^1
Subgroups of S^1 or conjugate	S^1
non-abelian	$\pm I$
$SU(2)$	$\pm I$

Lemma Flat connections with stabilizer larger than $\pm I$ correspond bijectively to abelian representations of $\pi_1(Y)$, ie, those which factor through $H_1(Y;\mathbb{Z})$.

Def ∇ is called reducible if $\mathrm{stab}(\nabla)$ is larger than $\pm I$

This is the first place we'll use that Y^3 is a $\mathbb{Z}HS^3$.

If Y is a homology sphere, there is only one reducible flat connection which corresponds to the trivial representation into $SU(2)$

Rmk if Y is a $\mathbb{Q}HS^3$, could have torsion

Computability of I breaks down based on how computable the character variety is, but also on how computible the differential is

The sing points aren't necc. isolated, but they can be post perturbation.

Lecture 13 // Oct. 6 // Defining Instanton Floer Homology

Last time: We restricted to Y^3 a $\mathbb{Z}HS^3$, and fixed the (trivial) $SU(2)$ bundle E over Y. Let $\mathcal{A}^{irr} \subset \mathcal{A}$ be the subset of connections with stabilizer $\pm I$.

Define $\mathcal{B} = \underbrace{\mathcal{A}}_{SU(2)\text{ connections}} / \underbrace{\mathcal{G}}_{\text{gauge group}}$

$\mathcal{B}^* = \mathcal{A}^{irred}/\mathcal{G}$

Then the Chern Simons Functional is a map $CS: \mathcal{B} \to \mathbb{R}/\mathbb{Z}$

Def (IFH, irreducible complex)

$C(Y) = \mathbb{F}\langle \text{crit points of } CS \text{ on } \mathcal{B}^* \rangle$

Rmk We have to perturb CS to ensure that the cp's are isolated. This perturbation happens in the same way one perturbs a function $f: M \to \mathbb{R}$ to be Morse.

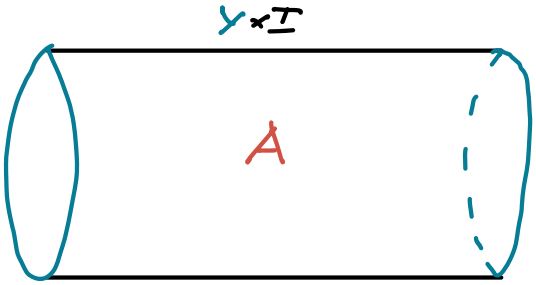
Since $\mathcal{G}$ has $\mathbb{Z}$-many path components, $\pi_0(\mathcal{G}) = \mathbb{Z}$, so $\pi_1(\mathcal{B}^*) = \mathbb{Z}$.

Now, take $x, y \in Crit(CS)$ and let γ be a path from x to y in $\mathcal{B}^*$.

The set of homotopy classes of such paths is $\mathbb{Z}$.

Def The moduli space of γ-flow lines from x to y is

$$M_\gamma(x,y) = \left\{ \begin{array}{l} \text{ASD connections on } Y \times I \\ \text{which limit to } x \text{ and } y \text{ on each} \\ \text{end and lie in htpy class of } \gamma \end{array} \right\} \Big/ \begin{array}{l} \text{gauge trans.} \\ \text{on } Y \times I \end{array}$$



Note that there is a free $\mathbb{R}$ action on $M_\gamma(x,y)$ by translation. If you've seen Morse theory, you know where this is going: we'll first quotient by $\mathbb{R}$, then count flowlines.

Def $\check{\mathcal{M}}_{\mathfrak{z}}(x,y) = \mathcal{M}_{\mathfrak{z}}(x,y)/\mathbb{R}$

$$\partial x = \sum_{y} \sum_{\mathfrak{z} \in S} |\check{\mathcal{M}}_{\mathfrak{z}}(x,y)|\, y$$

where S is the set of homotopy classes of paths from x to y for which $\dim(\check{\mathcal{M}}_{\mathfrak{z}}(x,y)) = 0$.

It turns out that, in contrast to fin. dim Morse theory, the $\dim(\check{\mathcal{M}}_{\mathfrak{z}}(x,y))$ depends on $\mathfrak{z}$: different homotopy classes of path $\mathfrak{z}$ from x to y give different dimensional moduli spaces. In particular, the expected dimensions of the ASD moduli spaces differ by 8. Thus $C(Y)$ is a relatively $\mathbb{Z}/8$-graded complex. The relative grading difference between x and y is $gr(x) - gr(y) = \dim(\mathcal{M}(x,y)) \bmod 8$.

To get an absolute $\mathbb{Z}/8$ grading on $CS(Y)$, recall that for Y a $\mathbb{Z}HS^3$, there is a unique (up to gauge) reducible flat conn. on Y. Denote this connection by Θ, and define absolute gradings by counting flows into Θ.

Def The absolute grading of $x \in Crit(CS)$ is

$$gr(x) := \dim(\mathcal{M}(x,\Theta)) \bmod 8.$$

However, something goes wrong when we try to include Θ as a generator of our chain complex. To see this, let's add Θ. Since the boundary map ∂ is only defined for irreducible critical points, we must define new boundary maps for flows into and out of Θ. These are Donaldson's D_1 and D_2 maps.

Def $D_1 : C(Y) \to \mathbb{F}$ (think of as $\mathbb{F}\langle\Theta\rangle$)

$$D_1(x) = \sum_{\mathfrak{z} \text{ from } x \text{ to } \Theta \text{ s.t. } \dim(\mathcal{M}(x,\Theta))=1} |\mathcal{M}(x,\Theta)/\mathbb{R}|$$

Def $D_2 : \mathbb{F} \to C(Y)$

$$D_2(1) = \sum_y \left(\sum_{\phi \text{ from } \Theta \text{ to } y \text{ s.t. } \dim(\mathcal{M}(\Theta, y)) = 1} |\mathcal{M}(\Theta, y)/\mathbb{R}| \, y \right)$$

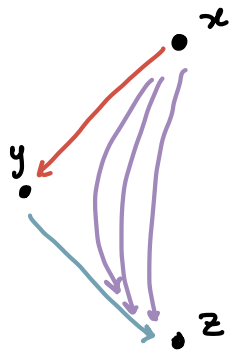
Notice that D_1 is supported on

$C_1(Y) = \{x \in C(Y) \text{ s.t. } gr(x) = 1\} \subset C(Y)$, since the only points which could flow into Θ must have grading 1. However, it turns out that D_2 has image in $C_{1-4}(Y)$. This is strange.

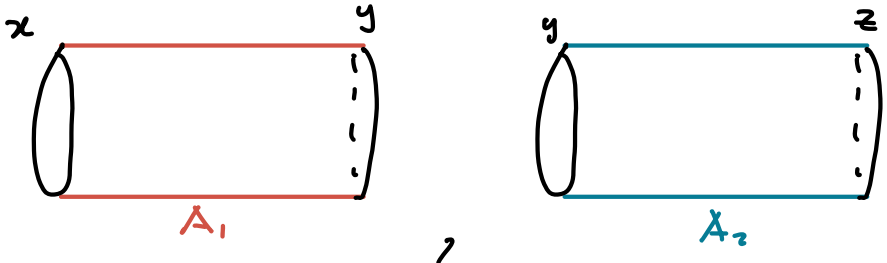
Let's go back to our def. of relative gradings. Why were these well-defined? The implict claim here was that for $x, y, z \in Crit(CS)$,

$$\dim(\mathcal{M}(x, y)) + \dim(\mathcal{M}(y, z)) = \dim(\mathcal{M}(x, z))$$

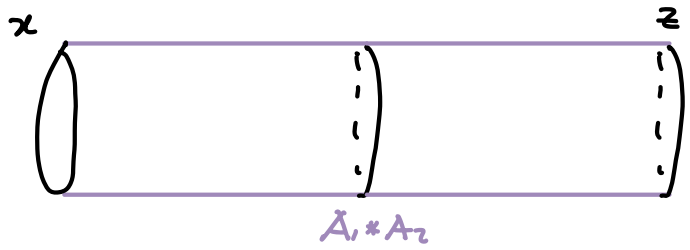
eg, we can construct a flow from x to z by gluing flows from $x \to y$ to flows from $y \to z$.



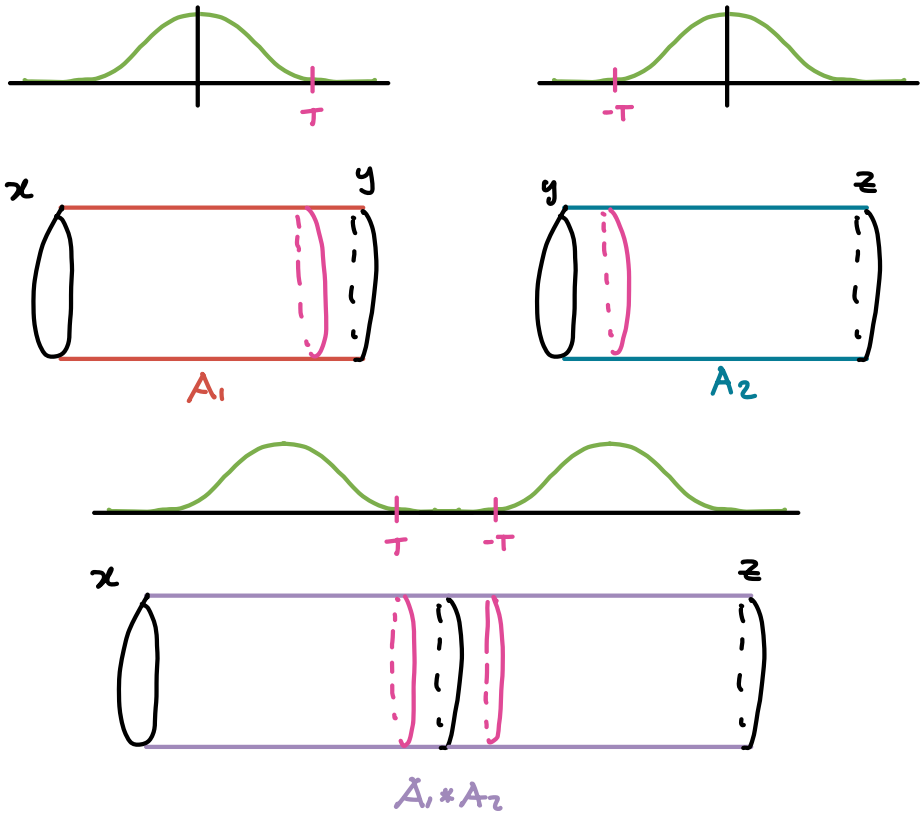
But something goes wrong in the Instanton Floer Complex when $y = \Theta$. Say we believe in some gluing procedure. Then, given an ASD connection A_1 on $Y \times \mathbb{R}$ which limits to x as $t \to -\infty$ and y as $t \to \infty$, and similarly an A_2 from y to z, we'd want a connection $A_1 * A_2$ from x to z on $Y \times \mathbb{R}$ built from A_1 and A_2.



$\wr$

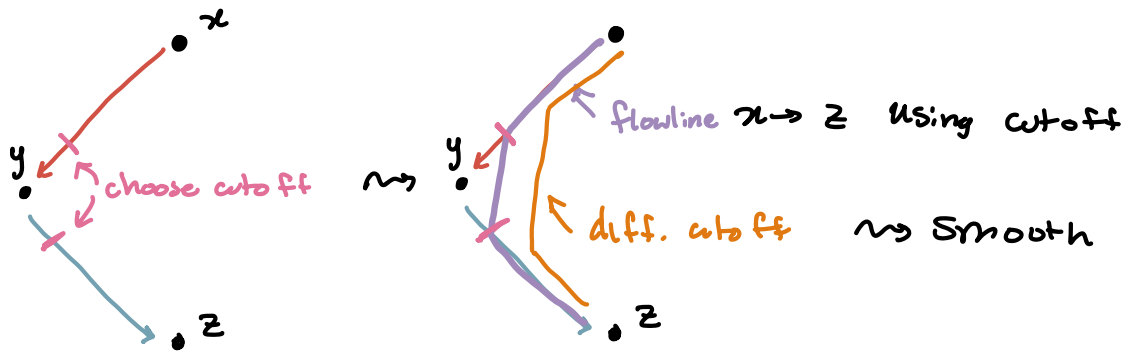


To "glue" A_1 and A_2 together at Y to get something smooth, we'll need to add a decay function that ensures that, after a certain time T, $f_0 A_1|_{[T,\infty)} = f_0 A_2|_{(-\infty,T]}$.



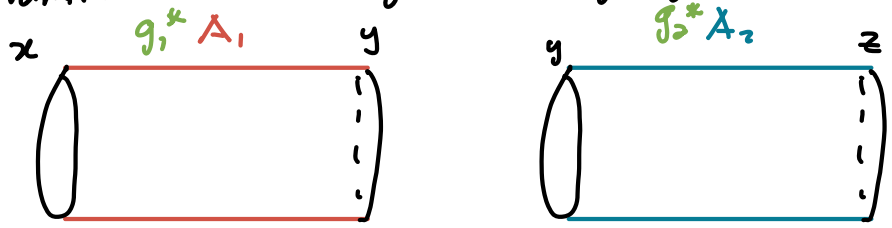
This gives us a 1-dimensional gluing parameter T.

In the flowline picture, the analog is building a flowline from x to z by choosing a cutoff parameter, following $x \to y$ until then,



What goes wrong when y has large stabilizer?

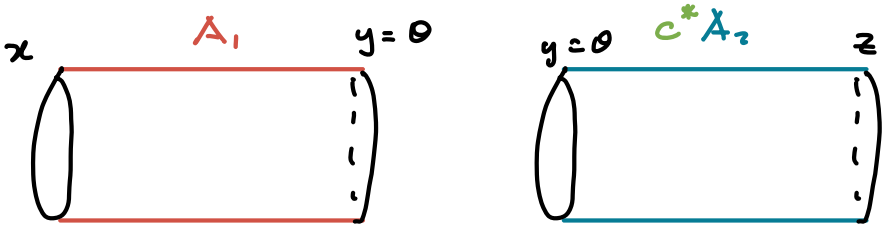
To glue, we can pick representatives on each end which are literally the same. But we can also compose with gauge transformations to get new gluings.



If $g_1^* A_1|_y = g_2^* A_2|_y$, then the connection on the composite cylinder is g-equivalent to the original connection glued via the identity.

But if Stab(y) big, we get many more Cxn. on long than would expect!

C = const, non-id gauge trans on A_2



Since y has big Stab,

$C^*(A_2)|_y = A_2|_y$, so we can still glue to get a connection on the entire composite cylinder.

However, $A_1 * C^* A_2$ is not gauge equiv. to $A_1 * A_2$

Since any gauge trans. $C^* A_2 \to A_2$ must change A_1 as well.

Therefore, we get a Stab(y)'s worth of extra distinct connections on $A_1 * A_2$.

Since $Y = \Theta$, and Stab(Θ) $\cong SU(2)$,

$$\dim(\mathcal{M}(x,z)) = \dim(\mathcal{M}(x,y)) + \dim(\mathcal{M}(y,z)) + \dim(SU(2))$$
$$= \dim(\mathcal{M}(x,y)) + \dim(\mathcal{M}(y,z)) + 3$$

This exactly matches what was claimed earlier:

For $D_1: x \to y$, $D_2: y \to z$, we expect a grading difference of 2 for $x \to z$, but $1 - (-4) = 5 = 2+3$

Another important fact is that in general, $D_2 \circ D_1 \neq 0$.

This is in contrast with fin-dim morse theory, where $\partial^2(x) = 0$ for all x. Let's recall why this is true.

$$\partial^2(x) = \partial(\partial(x)) = \partial\left(\sum_{y \in C_{k-1}} n(x,y)\, x \right)$$

($n(x,y)$ = # flows $x \to y$)

$$= \sum_{z \in C_{k-2}} \left(\sum_{y \in C_{k-1}} n(x,y)\, x \right) n(y,z)\, y$$

$\partial^2(x)$ breaks up as a sum of terms cz, where c is the number of broken flowlines from x to z.

$$\partial^2(x) = \sum \cdots + cz + \cdots,$$

The claim is that the coefficient c is always 0.

Indeed, the moduli space $\mathcal{M}(x,z)/\mathbb{R}$ is a 2-dim mfd $/\mathbb{R}$, so it is an open 1-manifold.

It admits a compactification, where the endpoints are broken flowlines. Thus the number of broken flowlines is even, thus zero mod 2, and with slightly more work, one can show the signed count is also zero.

However, this relies on the fact that $\mathcal{M}(x,z)$ is a 2-dimensional manifold, which comes from the index difference of 2. Therefore this argument fails for $\mathcal{O}$ in IFH, so we

Can't add in the reducible and get a chain complex, But $D_1 \circ \partial = 0$ and $\partial \circ D_2 = 0$, so there's some hope of working around this. One such way is by augmenting.

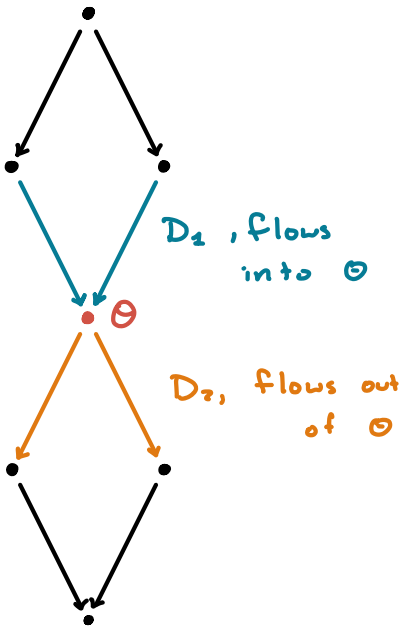
Note that $D_2(\theta)$ is a cycle in $C(Y)$, and thus defines a hx class $[D_2(\theta)] \in H_{-4}(C)$

D_1 defines a cohx element since it vanishes on $\mathrm{im}(\partial)$, so $[D_1] \in H^1(C)$

Given the irreducible complex C with D_1 and D_2, we can define the reducible complex by either precomposing D_2 or postcomposing D_1 with ∂, ie, we only allow either flows into or out of θ.

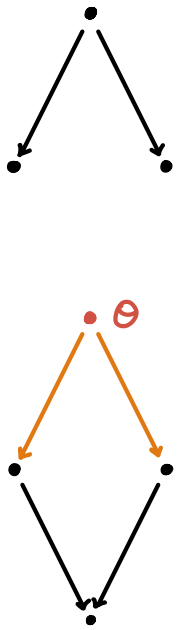
Def The "upper bar complex" is $\overline{C} = (C \oplus \mathbb{F}_{[0]}, \overline{\partial} = \partial + D_1)$

The "lower bar complex" is $\underline{C} = (C \oplus \mathbb{F}_{(-3)}, \underline{\partial} = \partial + D_2)$



$\overline{C}$

Cuts out flows out of θ



$\underline{C}$

Cuts out flows into θ

Lecture 14 // Oct. 8 // Lifting to $\mathbb{Z}$-grading and the Filtered Chain Complex

Last time: we defined the Irred Inst. Chain Cpx $C(4)$, which is a $\mathbb{Z}/8$ graded complex based at the reducible Θ. The chain cpx acts strangely at Θ: the grading drops by 5 for flows passing through it, so there's an issue with broken flowlines and ∂^2 isn't 0.

We can get a chain cpx. with the reducible sol'n by encoding the behavior at Θ into two maps.

$D_1 : C_1 \to \mathbb{F}$, which encodes flows into Θ

$D_2 \mathbb{F} \to C_{(-4)}$, which encodes flows out of Θ

From these we can define two chain cpx's for which $\partial^2 = 0$ by either forgetting flows into / out Θ

$$\overline{C} = C \oplus \mathbb{F}_0, \quad \overline{\partial} = \partial + D_1$$

$$\underline{C} = C \oplus \mathbb{F}_{(-3)} \quad \underline{\partial} = \partial + D_2$$

Q Why do we care about the reducible(s)?

A The cobordism maps usually come from the D_1, D_2 maps

$\mathbb{Z}$-Lifted version of IFH

We can lift the CS functional to be $\mathbb{R}$ valued (instead of $\mathbb{Z}$) by choosing a lift of $\mathcal{B}^*$ to the universal cover, $\tilde{\mathcal{B}}^*$

$$\pi_1(\mathcal{B}^*) \cong \mathbb{Z}, \quad \pi_1(\tilde{\mathcal{B}}^*) = 1$$

Recall that $\pi_1(\mathcal{B}^*) \cong \mathbb{Z}$ since we are quotienting the space of connections, which is contractible, by $\mathcal{G}$ which has $\mathbb{Z}$-many path components, one in each degree.

To get $\tilde{\mathcal{B}}^*$, we must instead quotient by by a subgroup of $\mathcal{G}$ which is $\mathbb{Z}$-times smaller, ie, choosing one path components as a "basepoint". The most obvious one to choose is

the degree 0 one.

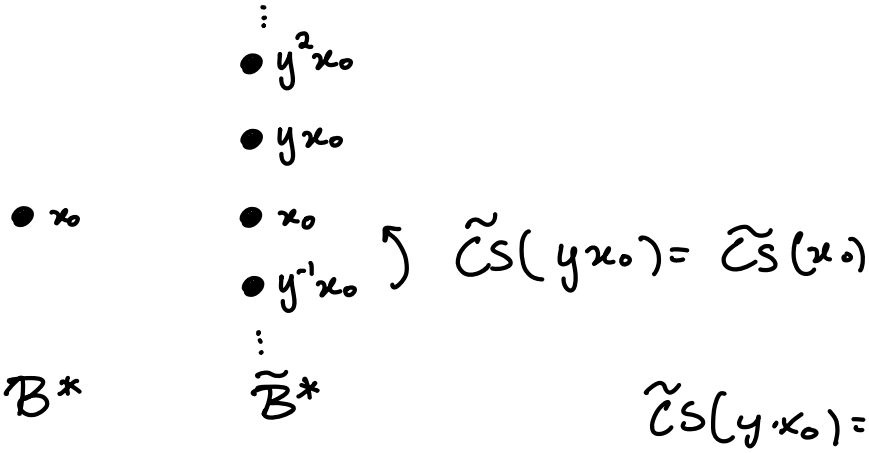
Def The based gauge group is the subgroup $\mathcal{G}_{\mathcal{J}0} \leq \mathcal{G}_{\mathcal{J}}$ of degree zero gauge transformations

$$\mathcal{G}_{\mathcal{J}0} := \{ g \in \mathcal{G}_{\mathcal{J}} \mid deg(g^{-1}) = 0 \}$$

By construction, $\pi_0(\mathcal{G}_{\mathcal{J}0}) = 1$. Like before, we define the configuration spaces $\widetilde{\mathcal{B}}$ and $\widetilde{\mathcal{B}}^*$ by

$$\widetilde{\mathcal{B}} = \mathcal{A} / \mathcal{G}_{\mathcal{J}0}$$
$$\widetilde{\mathcal{B}}^* = \mathcal{A}^{irred} / \mathcal{G}_{\mathcal{J}0}$$

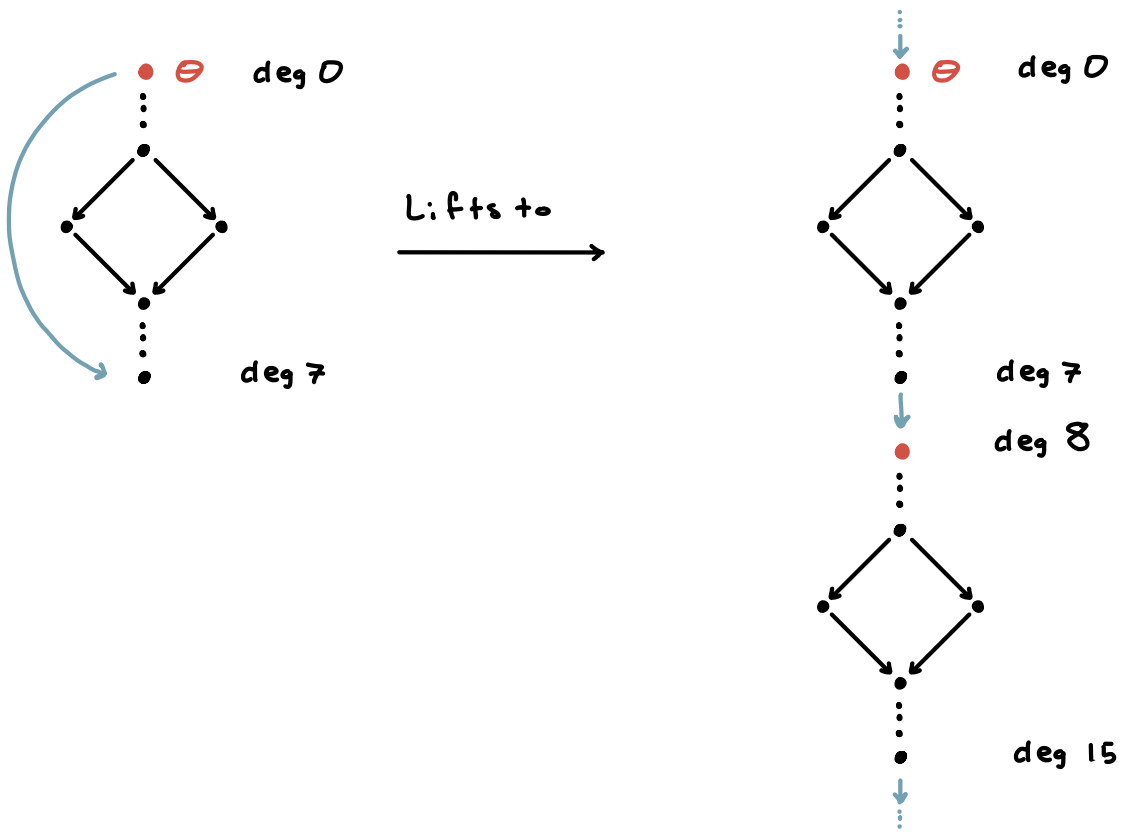
Note that $\widetilde{\mathcal{B}}^*$ is a $\mathbb{Z}:1$ cover of $\mathcal{B}^*$, so the Chern-Simons functional $CS: \mathcal{B}^* \to \mathbb{R}/\mathbb{Z}$ lifts to a $\mathbb{R}$-valued map $\widetilde{CS}: \widetilde{\mathcal{B}}^* \to \mathbb{R}$. Repeating the IFH construction, we obtain a $\mathbb{Z}$-graded complex over $\mathbb{F}[y^{\pm 1}]$, where y is in degree 8 and multiplying by y encodes switching between lifts, so increases the value of the Chern-Simons functional by one.



We will also refer to the $\mathbb{Z}$-lifted version as $C(y)$.

$$\overline{C} = C \otimes \mathbb{F}[y^{\pm 1}]_{(0)}$$

$$\underline{C} = C \otimes \mathbb{F}[y^{\pm 1}]_{(-3)}$$



The Chern-Simons Filtration

Def A $\mathbb{R}$-valued filtration on C is a collection of subcomplexes $\{C_s\}_{s\in\mathbb{R}}$ s.t. 1) $C_r \subseteq C_s$ for $r \le s$

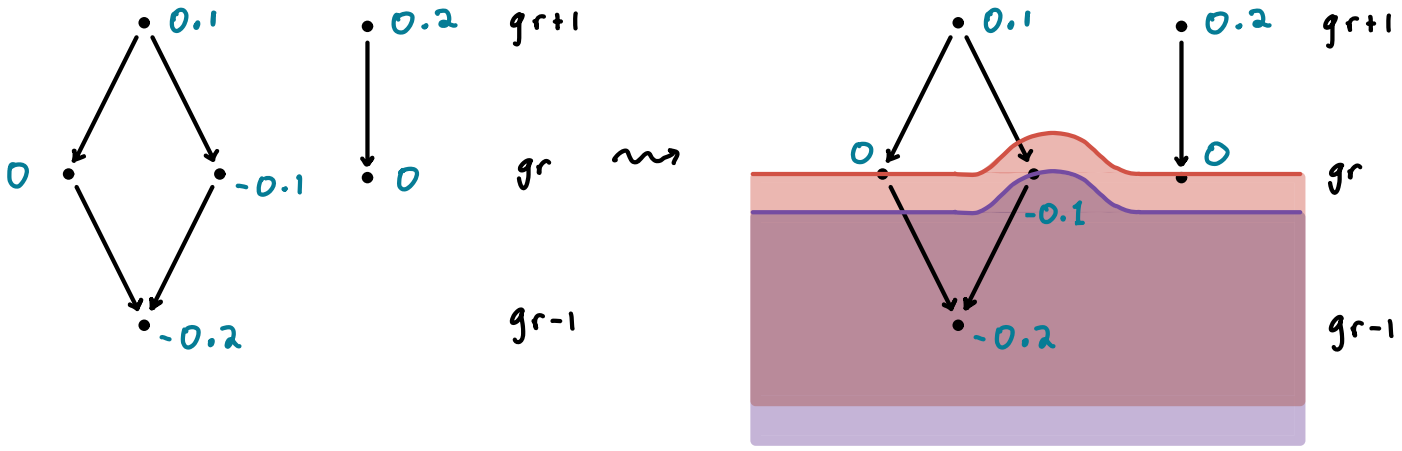
2) $\bigcup_s C_s = C$

Given a filtered cpx, we can get filtered subquotients $C^{[r,s]} = C^s/C^r$ for each pair $r<s$. Note that $[r,s]$ actually means $(r,s]$, so the generators $x \in C^{[r,s]}$ are the generators $x \in C$ with Chern-Simons value $r < \widetilde{CS}(x) \le s$.

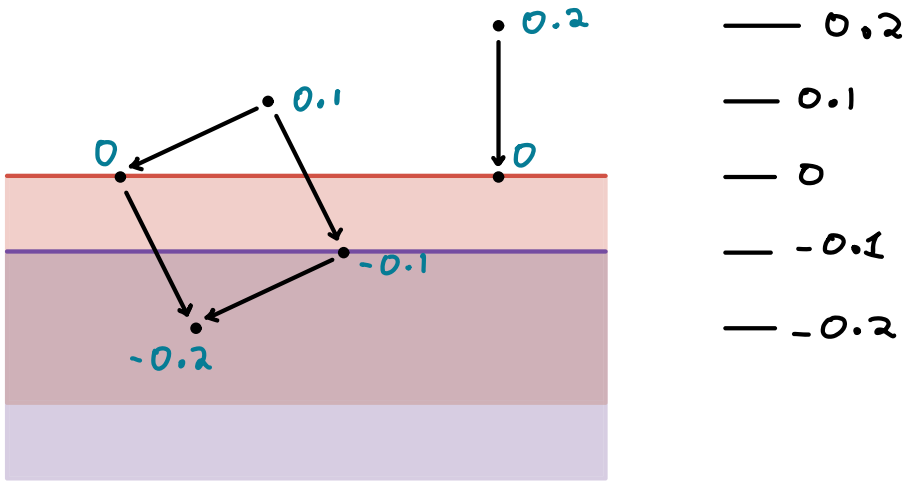
Observe that $\widetilde{CS}$ gives C the structure of a filtered complex. For each generator (critical point), we can look at the value of the (perturbed) CS-functional at each x.

Diagrammatically, the filtration looks like the following. The values of CS for each generator are written in blue, and the complex is arranged

vertically by $\mathbb{Z}$-grading.



If this complex was instead arranged vertically by Chern-Simons value, the filtration would look like horizontal lines:



Rmk The $\mathbb{Z}$-grading is indep from the cs value, but they both both go down under the differential

Rmk If B^* has nontrivial π_1, then the grading shift for flows which use this π_1 is larger than one, since the expected dim for the flowline is higher. Lifting to $\tilde{B}^*$ "unwinds" this π_1, so gradings always drop by 1.

By trivializing the bundle, we get a connection for which $\widetilde{CS}$ vanishes; we take this to be the lift of Θ.

This filtration also descends to one on C and $\overline{C}$.

Def A filtered chain map $f: C_1 \to C_2$ is a chain map which is non-increasing with respect to the filtration, ie, $f(C_1^s) \subseteq C_2^s \quad \forall s$

Def A filtered chain homotopy H between filtered chain maps f and g is a chain homotopy for which $H(C_1^s) < C_2^s \quad \forall s$

If H exists for f, g, f and g are filtered chain homotopic.

We claim that filtered htpy equiv class of C is an invariant of Y^3. To show this, we ask: what must we keep track of? What aux data was used in our def of $CC(Y)$?

Well, the first thing we had to do for any bundle stuff was choosing a metric on Y.

The crit pts of $CS(Y)$ don't depend on g: the point is that they're inh. topological, eg, counting $SU(2)$ repn's of π_1

However, the differential does depend on g, since to even define a grad-like vf, we need a metric. Moreover, the definition uses the Hodge star! $\text{grad}_A CS = *_g F_A$

So strictly speaking, the CPX does depend on g a priori. Thus we must check that changing g doesn't change the FHEC.

There is a second thing. If CS doesn't have discrete CPs, we must perturb the CS funct to get discrete CPs.

So far, we've brushed this fact under the rug; it's actually super annoying to deal with.

A random perturbation of CS funct (or even a general Morse function) will certainly change value of $\widetilde{CS}$ at CP's. So we have no reason to think that any perturbation should play well with the CS filtration.

In reality, our defn of the IFC is incorrect. It's actually defined as a "limiting object." The idea is that we can find an arbitrarily small sequence of perturbations $\{\pi_i\}_{i=1}^{\infty}$ for which $\widetilde{CS}$ has discrete CP's.

For each perturb., we get a filtered CPX. $C^{\pi_i}(Y)$.

If we do this with enough regularity, as $\|\pi_i\| \to 0$, these complexes converge to some limiting object which we define to be $C(Y)$: $C^{\pi_i}(Y) \xrightarrow{\|\pi_i\| \to 0} C(Y)$.

However, there are many cases for which a perturb. is not needed, in which case the defn. is true as given.

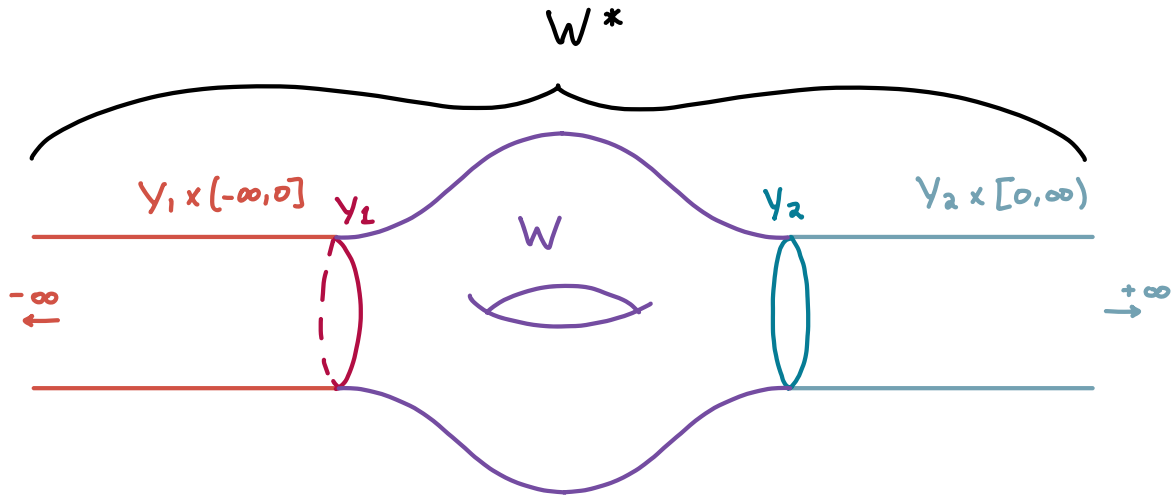
For example, the Brieskorn Hz spheres $\Sigma(p,q,r)$ with three singular fibres have discrete critical points without a perturbation. Conceptually, we'll discuss the complex as if no perturbation is needed.

Cobordism Maps

If you've seen Floer theory before, you'll have heard of the so-called TQFT structure, eg, "bordisms induce maps". This is formalized as follows

Def A Floer Homology theory F is said to have the structure of a TQFT if, for W a cob. from Y_1 to Y_2, there is a cobordism map $F_W : C(Y_1) \to C(Y_2)$

In our setting, let W be a neg-def cob. with $b_1(W) = 0$. From W, we construct the "cylindrical end mfd" W^* by attaching $Y_1 \times (-\infty, 0]$ and $-Y_2 \times [0, \infty)$ to their resp ends.



Consider the trivial bundle $W^* \times \mathbb{C}^2$.

Given $x \in C(Y_1)$ and $y \in C(Y_2)$, look at ASD connections on W^* which limit to x and y (on the left and right, respectively), modulo gauge trans on W^*.

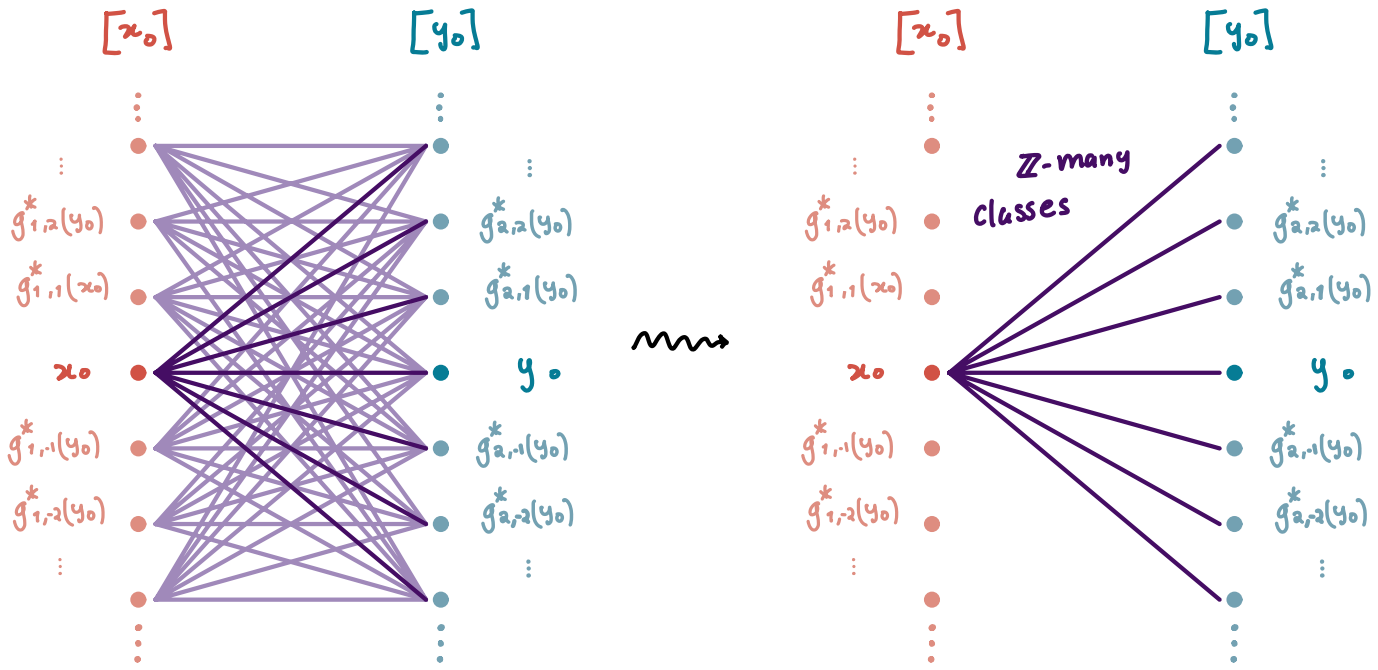
As with ∂, space of cxns on W^* mod gauge on W^* has many diffs. top. comp's. We can think about this as follows. On Y_1 and Y_2, x and y are really just representatives of gauge equiv. classes $[x_0]$ and $[y_0]$ of cxns on Y_1 and Y_2.

If $g^* y_0 \in [y_0]$, then we also want to count connections limiting to x_0 and $g^* y_0$. As with before, we claim there is no gauge trans. on W^* sending a connection from $x_0 \to g^* y_0$ to one from $x_0 \to y_0$. In particular, if g is a gt on W^*, then $\deg(g|_{Y_1}) = \deg(g|_{Y_2})$.

Exercise Prove the claim

This shows that there are $\mathbb{Z}$-many top. classes of cxns interpolating between x_0 and y_0, determined by the difference of degree $\deg(g_1) - \deg(g_2)$ for gauge trans. g_1 on Y_1 and g_2 on Y_2.

Diagramatically, this can be seen by quotienting by a vertical $\mathbb{Z}$-shift:



Lecture 17 // Oct. 20 // Proof of the NST theorem

Last time: We defined the r_s-invariant for $s \in [-\infty, 0]$

$$r_s(Y) = -\inf_{r<0} \{ r \mid \exists\ \Theta\text{-supported } [r, -s] \text{ cycle} \}$$

which asks the question "In which filtered subquotients do we have a cycle of the form $\Theta^t \dots$"

1) r_s-invt. is a piecewise const. non-inc. fxn

2) If $\exists$ negative-def. cob. W (with $H_1(W)=0$) from Y_1 to Y_2, then
$r_s(Y_1) \leq r_s(Y_2)$
In Part., r_s is a hom. cob. inv.: If Y bounds a negative-def. 4-mfd W (with $H_1(W)=0$), then $r_s(Y) = \infty$

3) In addl.,
If, in addl., $\pi_1(W) = 1$, then the inequality is strict (allowing $\infty < \infty$)

Today, we'll prove one of the main thms of the NST paper.

Thm [NST] Let $K \subset S^3$. Denote $Y_n = S^3_{1/n}(K)$, $n > 0$.
If $[D_2^{Y_1}] \neq 0$, then none of the Y_n are hx-cob to eachother.

If $r_s(Y_1) < \infty$ for any s, then $\{Y_n\}_{n=1}^{\infty}$ is lin. indep. in $\Theta^3_{\mathbb{Z}}$

Rmk A-priori, it's not obvious that $\Theta^3_{\mathbb{Z}}$ is at all interesting. For many years after def, not known whether $\Theta^3_{\mathbb{Z}}$ is 0, or ∞-gen. It is: In particular, it contains a $\mathbb{Z}^\infty$ subgroup. ($\bigoplus_{i=1}^{\infty} \mathbb{Z}_i$)
The o.g. proof was due to Furuta and FS in the 80's.
In some sense, this thm is a generalization of theirs.
FFS had same family, and also used Yang-Mills theory.
(Modern language (Floer) allows them to generalize it)

Rmk It's actually very difficult to study $1/n$ surgeries using HF. There's no known inv. coming from HF which distinguish ∞-ly many $1/n$ surgeries on a fixed knot.

To study lin. combos of Y_n, we must first understand the behav. of r_s under #.

Thm For any $s \in [-\infty, 0]$ and $s_1, s_2 \in [-\infty, 0]$ such that $s_1 + s_2 = s$,
$$r_s(Y_1 \# Y_2) \geq \min\{ r_{s_1}(Y_1) + s_2,\ r_{s_2}(Y_2) + s_1 \}$$
Perhaps the form of this theorem – ie, it is an inequality, and takes a minimum over a sum of r_s and s's – is strange. We'll first give a pseudo-pf of the inequality, since the reason why why it's of such a strange form will become apparent during the proof

Pf (sketch)
We'll pretend that $\underline{C}(Y_1 \# Y_2) = \underline{C}(Y_1) \otimes \underline{C}(Y_2)$
This (# corresponds to $\otimes$) is the behav. one usually expects for complexes under connect sum.

The reducible in $\underline{C}(Y_1 \# Y_2)$ is $\Theta_1 \otimes \Theta_2$, and
$\widetilde{CS}(\alpha \otimes \beta) = \widetilde{CS}(\alpha) + \widetilde{CS}(\beta)$.

Now, fix s_1 and s_2. Spse that we have a ΘSC α for $\underline{C}(Y_1)$ in $[r_1, -s_1]$, and a ΘSC β for $\underline{C}(Y_2)$ in $[r_2, -s_2]$.

What "Θ-supported" means is that in $\underline{C}(Y_1)$, $\alpha = \Theta + \dots$.
To say that α is a cycle in this subqu. is to say
that a) α is in the subquotient, ie, $\widetilde{CS}(\alpha) \leq -s$
and b) $\widetilde{CS}(\underline{\partial}\alpha) < r_1$. In particular, $\partial\alpha$ need not be 0,
but if it isn't, it's outside subquotient.
The same is true for β: $\widetilde{CS}(\beta) \leq -s_2$ and $\widetilde{CS}(\underline{\partial}\beta) < r_2$.

Claim: $\alpha \otimes \beta$ is a $\Theta_1 \otimes \Theta_2$-supported
(a.k.a., Θ-supp in the $\otimes$)
we know that $\widetilde{CS}(\alpha \otimes \beta) \leq (-s_1) + (-s_2) = -s$.
So $\alpha \otimes \beta$ is in the subquotient.
By the Leibniz rule,
$$\underline{\partial}(\alpha \otimes \beta) = (\underline{\partial}\alpha) \otimes \beta + \alpha \otimes (\underline{\partial}\beta).$$
Individually, $(\underline{\partial}\alpha) \otimes \beta < r_1 + |-s_2|$ and
$\alpha \otimes (\underline{\partial}\beta) < (-s_1) + r_2$
Therefore $\alpha \otimes \beta$ is a cycle in the
subquotient $\left[\max\{r_1 + (-s_2), (-s_1) + r_2, -s\}\right]$.

This is where the formula appears.

Now, note that this is true $\forall$ ΘSC in these subqt's.
When does this occur? By def^n, $-r_3(Y_1)$ and

$-r_{s_2}(Y_2)$ are the borders of when $\exists$ ΘSC in each resp. subqt. So the claim is that you can take any r_1 up to $-r_{s_1}(Y_1)$ and any r_2 up to $r_2 \longrightarrow -r_{s_2}(Y_2)$.

Thus we get a ΘSC in the subq's

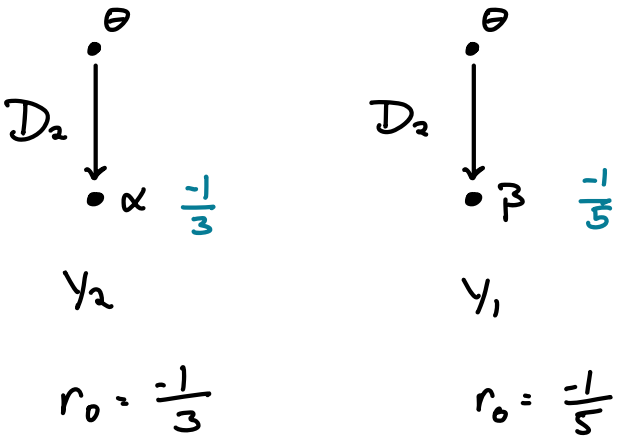
$$\left[\max\left\{-r_{s_1}(Y_1)+(-s_2),\ -r_{s_2}(Y_2)+(-s_1)\right\},\ -s\right]$$

$$\Rightarrow\ r_s(Y_1\#Y_2) \geq \max\left\{r_{s_1}(Y_1)+s_2,\ r_{s_2}(Y_2)+s_1\right\}\ \square$$

Cor $\quad r_0(Y_1\#Y_2) \geq \min\{r_0(Y_1), r_0(Y_2)\}$

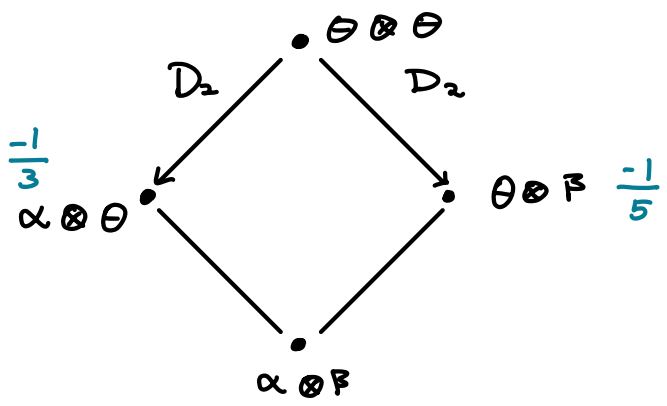
To see why this behavior — taking a minimum instead of adding — is more natural in this setting, lets do an example.

Example Let the following be the u-bar cpxes for Y_1 and Y_2.



Here, r_0 is measuring the cs displacement of the D_2 arrow.

The point is that if we pretend $\#$ is $\otimes$, the Complex for $Y_1\#Y_2$ looks like:



Notice that $D_2(\theta)$ has 2 arrows, corr to the 2 summands in the Leibniz rule.

Here, it's clear that to kill all the D_2's, you must take the minima of all of the summands.

Unfortunately, it's not acc. true, in general, that

$$\underline{C}(Y_1 \# Y_2) = \underline{C}(Y_1) \otimes \underline{C}(Y_2).$$

The rough reason is that it turns out that $\underline{C}(Y_1 \# Y_2)$ cannot be determined from $\underline{C}(Y_1)$ and $\underline{C}(Y_2)$ alone. To see why, lets look at the potential types of generators of $\underline{C}(Y_1 \# Y_2)$:

- $\theta \otimes \theta$ (red ⊗ red) } This is red
- $\alpha \otimes \beta$ (irred ⊗ irred) } These are irred in the tensor product
- $\theta \otimes \beta$ (red ⊗ irred) } These are irred in the tensor product
- $\alpha \otimes \theta$ (irred ⊗ red) } These are irred in the tensor product

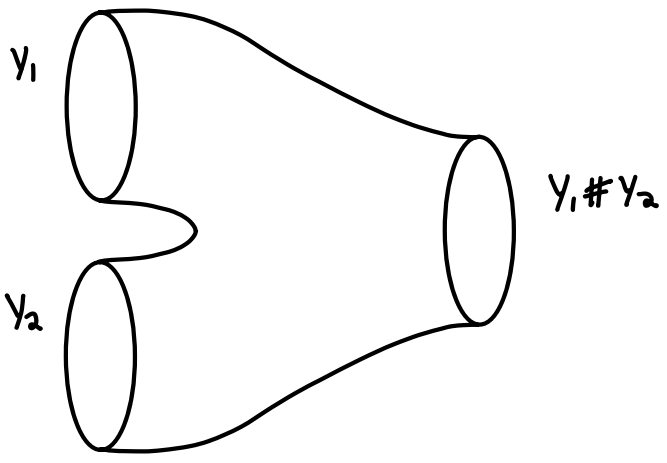
The reason why the last two, which are combinations of reducible and irreducible generators, are irreducible in the product is as follows. Recall that generators correspond to $SU(2)$ representations, and that $\pi_1(Y_1 \# Y_2) \cong \pi_1(Y_1) * \pi_1(Y_2)$. If we have $SU(2)$ reps. ρ_1 and ρ_2 of $\pi_1(Y_1)$ and $\pi_1(Y_2)$, and ρ_1 has trivial centralizer but ρ_2 doesn't, then $\rho_1 * \rho_2$ also has trivial centralizer. The intuition here is the same as for gauge transformations on the composite cylinder: even if some element commutes with ρ_2,

It doesn't with ρ_1, so it doesn't commute with $\rho_1 * \rho_2$ either.

The issue arises because $\alpha \otimes \Theta$ and $\Theta \otimes \beta$ are irreducible: there is no reason to expect that there are no flows into/between them! In particular, we should be able to have maps from $\alpha \otimes \beta$ to $\alpha \otimes \Theta$. However, even if there is a D_1 map where $D_1(\beta) = \Theta$, in choosing the u-bar cpx, we threw these out! So you actually need both the D_1 and D_2 maps of each factor to calculate the u-bar cpx of the connect sum.

Rmk Being able to write down an explicit formula for $\underline{C}(Y_1 \# Y_2)$ as some tensor product is one motivation for why one may want to try harder to incorporate both D_1 and D_2 into the diff'l.

What salvages this is the fact that there is a cobordism map $\underline{C}(Y_1) \otimes \underline{C}(Y_2) \to \underline{C}(Y_1 \# Y_2)$. This cobordism is literally the pair of pants, from $Y_1 \sqcup Y_2$ to $Y_1 \# Y_2$.



Thus far, we've assumed that the Y's are connected which seems silly to note, but the distinction btwn conn and disconn. boundary really matters to the gauge gp: they're qualitatively different. It turns out that we get get bordism maps from disconn → conn, but not the other way. But this means that the rs inv. of the algebraic cpx $\underline{C}(Y_1) \otimes \underline{C}(Y_2)$ bounds the rs inv. of the complex we actually care about, $\underline{C}(Y_1 \# Y_2)$.

Q Is the inequality always strict?

A No! For example, take $Y_1 = -\Sigma(2,3,5)$ and $Y_2 = \Sigma(2,3,5)$. Then $Y_1 \# Y_2$ is homology cobordant to S^3. The U-bar complex for each summand looks like

$r_0(Y_1) = \frac{-1}{120}$ $\quad$ $r_0(Y_2) = \infty$

Since rs is a hx cob inv.,

$$\infty = r_0(S^3) = r_0(Y \# -Y) \geq \min\{r_0(Y), r_0(-Y)\}$$

$$\geq \min\left\{\frac{1}{120}, \infty\right\}$$

$$\geq \frac{1}{120}$$

We are now prepared to prove the thm. We'll prove a more general algebraic statement.

Thm Let $\{Y_n\}_{n=1}^{\infty}$ be a sequence of $\mathbb{Z}HS^3$'s such that

1) $\infty > r_0(Y_1) > r_0(Y_2) > \dots$

2) $\infty = r_0(-Y_1) = r_0(-Y_2) = \dots$

Then the $\{Y_n\}_{n=1}^{\infty}$ are lin. indep. in $\Theta^3_{\mathbb{Z}}$.

Pf Assume not: spse $\sum_{i\le m} n_i Y_i = 0$ in $\Theta^3_{\mathbb{Z}}$.

Then $n_m Y_m = \sum_{i<m} n_i Y_i$ (assume $n_m > 0$),

so $r_0(n_m Y_m) = r_0\left(\sum_{i<m} n_i Y_i\right)$. Note that these n_i can be positive or negative

Iterating this inequality, we get that

$r_0(n_m Y_m) = \dots \ge \min\{ r_0(\pm Y_1), r_0(\pm Y_2), \dots \}$

where the $+$ or $-$ in each $\pm$ is chosen based on $\operatorname{sign}(n_i)$.

OTOH, Since $Y_m = n_m Y_m \# (-n_{m-1}) Y_m$,

$$r_0(Y_m) = r_0\left(n_m Y_m \# [-n_{m-1}) Y_m\right)$$

$$\ge \min\{ r_0(n_m Y_m), r_0(-Y_m) \}$$
$$\ge \min\{ r_0(n_m Y_m), \infty \}$$
$$= r_0(n_m Y_m)$$

since $r_0(-Y_m) = 0$ by hypothesis.

Thus $r_0(Y_m) \ge r_0(n_m Y_m)$, so

$$r_0(Y_m) \ge \min\{ r_0(\pm Y_1), r_0(\pm Y_2), \dots \}$$

which contradicts the initial hypotheses. □

The approx. intuition for this result is that small and nonzero CS displacement of D_2 cannot lie in the span of large CS displacemnt.

Does this apply to our situation?
Yes! Heres a slick topological argument, which concludes the proof of [NST].

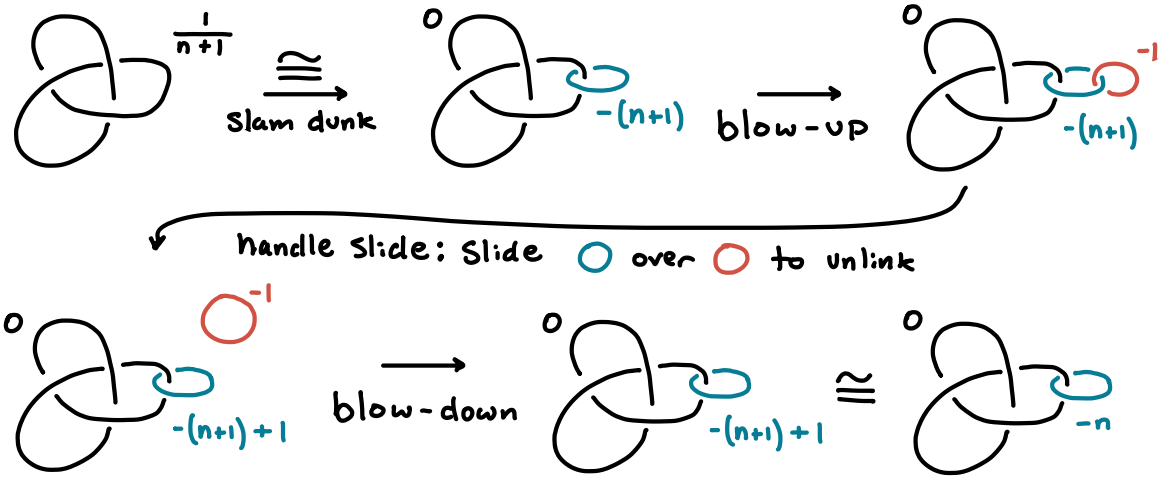
Thm [NST] Let $K \subset S^3$. Denote $Y_n = S^3_{1/n}(K)$, $n > 0$.
If $[D_2^{Y_1}] \neq 0$, then none of the Y_n are hx-cob to eachother.

Pf We only need to show that the conditions of the prev. thm are satisfied.
Eg, we need to verify that

1) $\infty > r_0(Y_1) > r_0(Y_2) > \dots$
2) $\infty = r_0(-Y_1) = r_0(-Y_2) = \dots$

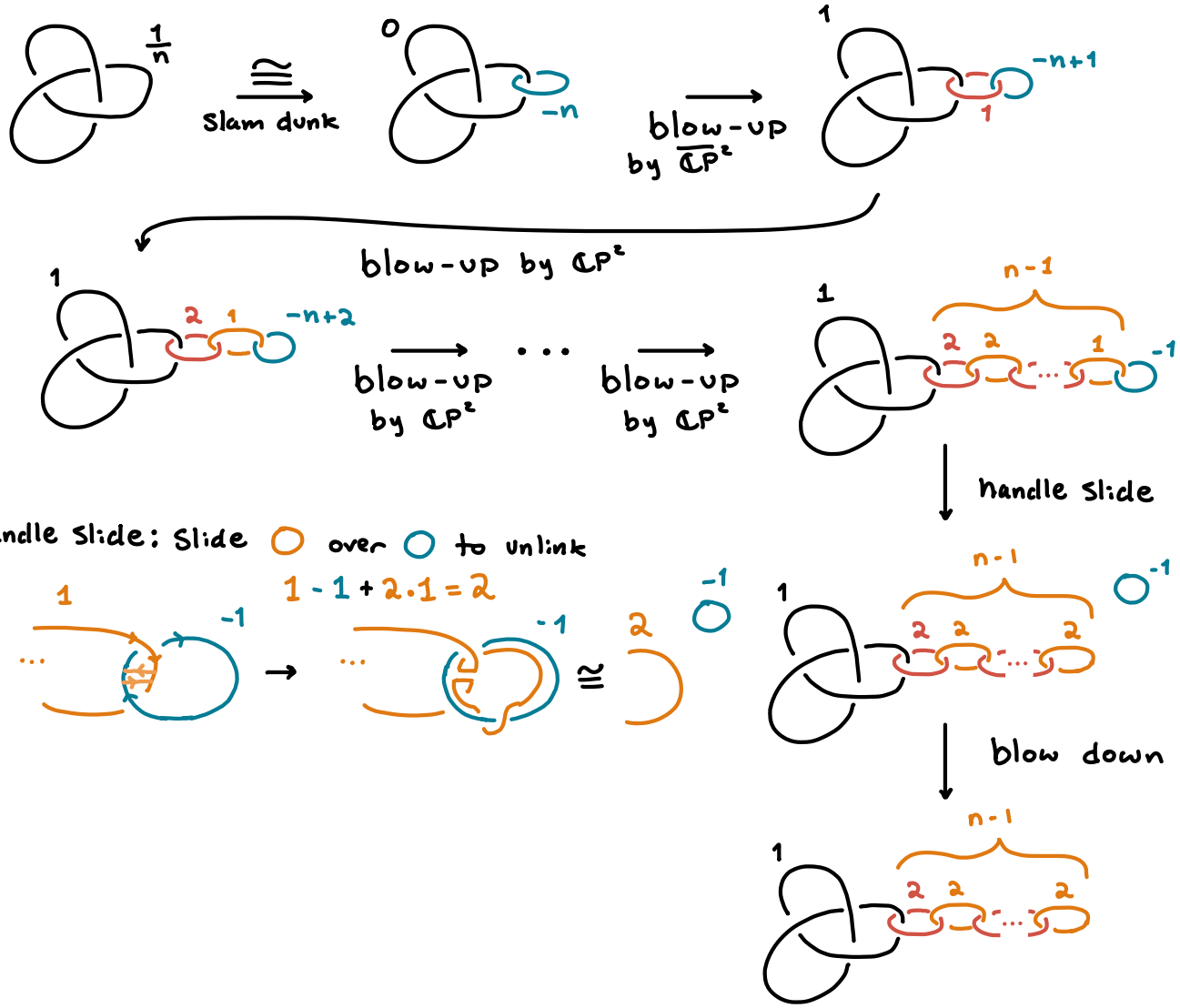
1) holds from last time:

We constructed a $\pi_1 = 1$ neg.-def. cob from Y_{n+1} to Y_n:



2) Follows from the fact that $S^3_{1/n}(K)$ bounds a +-def mfd if $n>0$ and a --def mfd if $n<0$.

The +-def mfd can be constructed as follows ($n>0$):



The intersection form is $\begin{matrix} 1\{ \\ n-1\{ \\ 1\{ \end{matrix}\begin{bmatrix} 1 & 1 & & & \\ 1 & 2 & 1 & & 0 \\ & 1 & 2 & 1 & \\ & & 1 & \ddots & \\ & 0 & & & \begin{matrix} 2 & 1 \\ 1 & 2 \end{matrix} \end{bmatrix}$, which we leave to the reader to verify has Smith normal form Id_4.

The same process can be applied when $n<0$ by blowing up with $\overline{\mathbb{C}P^2}$'s.

Since this process gives a neg.-def. cob from $-Y_n$ to S^3, we can conclude that $r_0(-Y_n)=\infty$.

Lecture 19 // Oct. 27 // SU(2) - Character Varieties

This week, we'll return to our discussion of the SU(2) character variety. We briefly mentioned this perspective when we discussed the critical points of the CS functional, but we haven't done any examples nor explicit calculations yet. Recall that we can study the fund. gp of a 3-mfd by considering instead the image of $\pi_1(Y)$ under some homomorphism to a simpler group, SU(2).

Def The Character Variety of a 3-mfd is

$$R(Y) = \left\{ \begin{array}{c} SU(2) \text{ representations} \\ \text{of } \pi_1(Y) \end{array} \right\} / \sim$$

where $\rho_0 \sim \rho_1$ IFF $\rho_1 = g^{-1} \cdot \rho_0 \cdot g$

Today, we'll discuss F-S's computation of IFH of Brieskorn spheres $\Sigma(a_1, \dots, a_n)$. Since we mentioned that

may need to perturb, will do

Recall that if the set of cp's of $CS(Y)$ is not discrete, we need to perturb the functional to get a discrete set.

With this in mind, we'll do ex's of both situations: when when R is discrete and when R has higher-dim components.

On Thursday, we'll discuss the common idea of viewing one's favorite gauge theory as the categorification of something. For IFHs, this "something" is the Casson invariant:

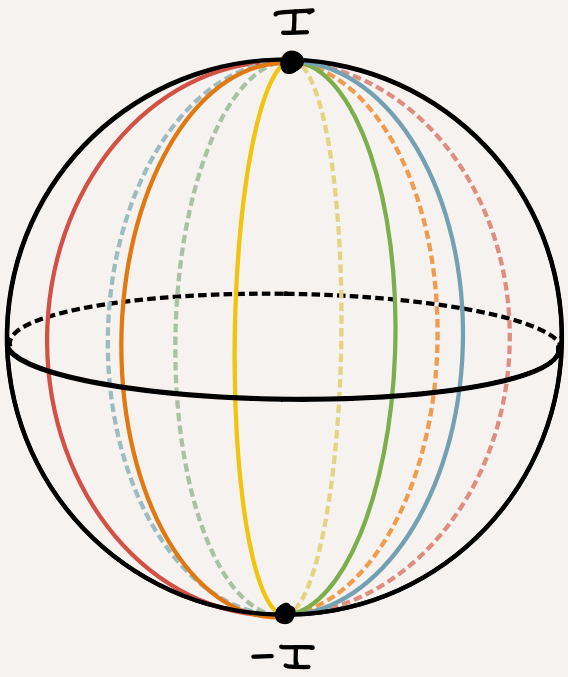
$$\frac{1}{2}\chi(H_*(CC(Y))) = \lambda(Y)$$

Examples of SU-2 char. varieties

First, lets recall some facts about $SU(2)$.

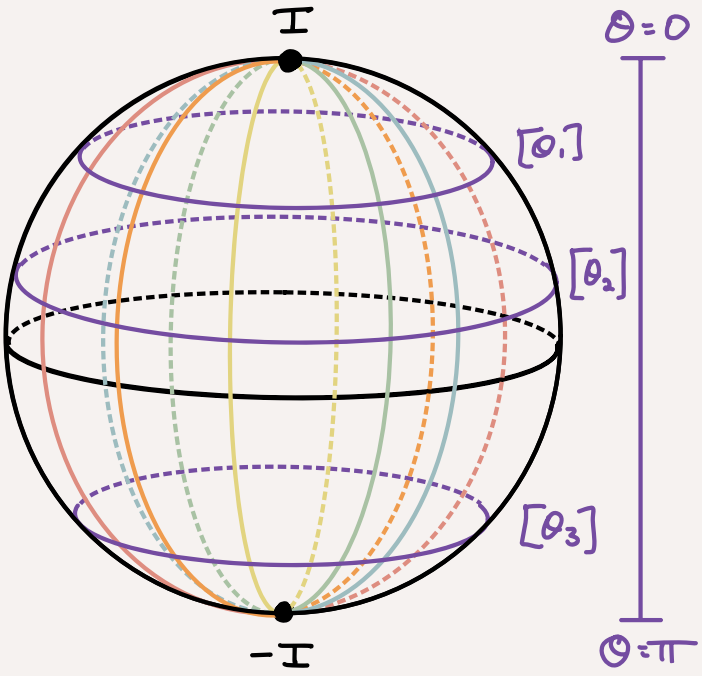
1) $SU(2) = \left\{ \begin{pmatrix} \alpha & -\overline{\beta} \\ \beta & \overline{\alpha} \end{pmatrix} \mid \alpha, \beta \in \mathbb{C},\ |\alpha|^2 + |\beta|^2 = 1 \right\}$ is diffeomorphic to S^3.

Going forward, we'll think of $SU(2)$ as S^3, which we'll draw 1-dimension down as S^2.



2) $SU(2)$ has many S^1 subgps, which are all conj to each other & sweep out $SU(2)$. They are all disjoint except at $\pm I$. Ie, every element of $SU(2)$ is conjugate to some diagonal matrix $\begin{bmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{bmatrix}$ which is unique up to the sign of θ.

3) For $A \in SU(2)$, $Tr(A)$ is a complete invariant up to conjugation: If A and B are conjugate, $Tr(A) = Tr(B)$.



Each conj class is a 2-sphere S^2 parameterized by the value of tr. For $\theta \in [0, \pi]$, we will speak of the conj. class of $[\theta]$ to mean the conj class of $\begin{bmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{bmatrix}$, where

$$\theta = \cos^{-1}\left(\frac{1}{2} Tr\left(\begin{bmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{bmatrix} \right) \right)$$

We can now compute the representation variety of a Brieskorn sphere $\Sigma(a_1, \dots, a_n)$, where the a_i are coprime.

Def The Brieskorn spheres $\Sigma(a_1, \dots, a_n)$ are $\mathbb{Z}HS^3$'s constructed as follows:

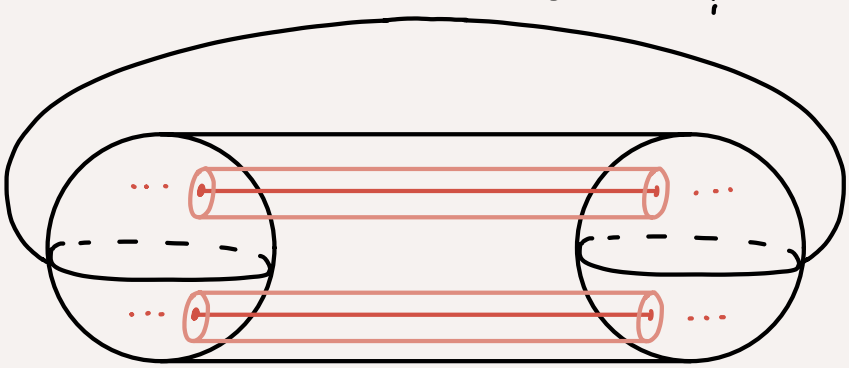
1) Solve the linear eq^n $\left(\prod_{i=1}^{n} a_i\right)\left(\sum_{i=1}^{n} \frac{b_i}{a_i}\right) = 1$

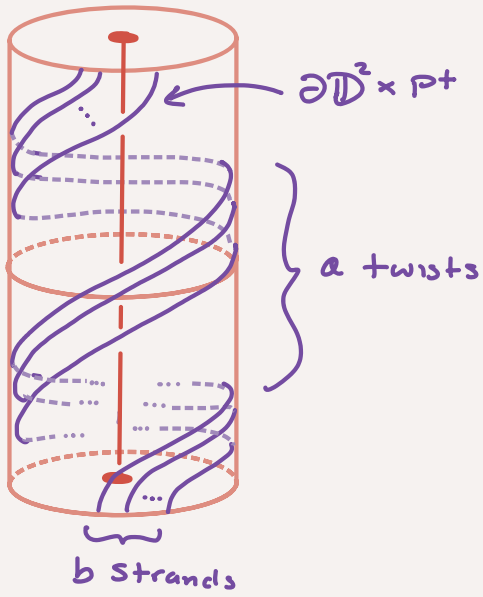
for a set $\{b_1, \dots, b_n\} \in \mathbb{Z}^n$

2) Surger $S^2 \times S^1$ along n longitudinal fibres with surgery coefficients $\left\{\frac{a_i}{b_i}\right\}_{i=1}^{n}$.

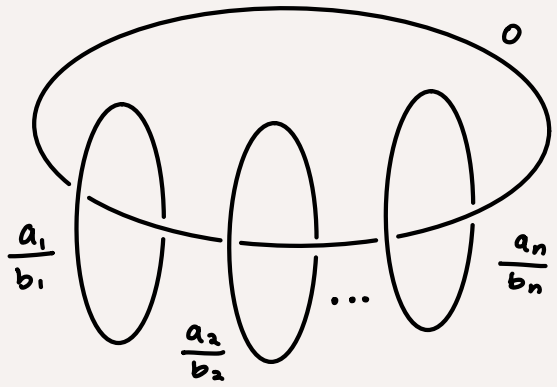
Eg. punch out n longitudinal S^1 fibres to get n $S^1 \times S^1$ boundary components. Let $pt \times S^1$ in $S^1 \times S^2$ be homologous to $pt \times S^1$ in $S^2 \times S^1$, and a/b be the class $a[S^1 \times pt] + b[pt \times S^1]$. To each component, reglue $S^1 \times D^2$ so that $[\partial D^2] \mapsto a[S^1 \times pt] + b[pt \times S^1]$.

$S^2 \times S^1 \setminus \bigsqcup_i D^2 \times S^1$

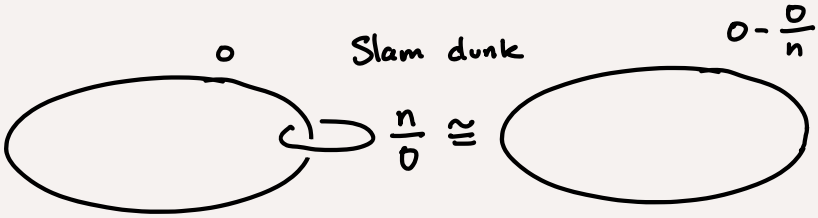




This gives the following surgery description



Rmk Sanity Check for coeffs.

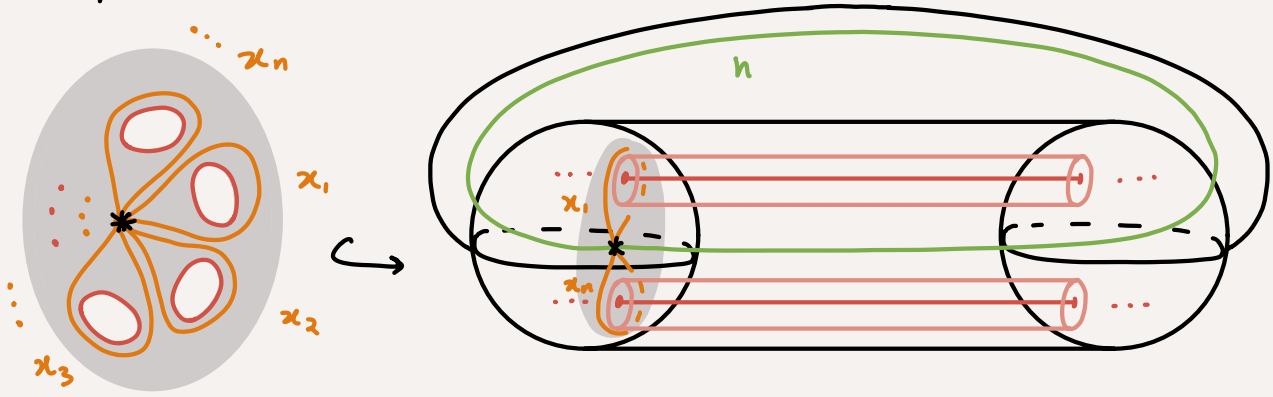


$n \to S^1 \times pt \subset S^2 \times pt$

$0 \to pt \times S^1 \subset pt \times S^1$

Of course, the $\{a_1, \dots, a_n\}$ do not uniquely determine a 3-mfld. However, if we demand that the 3-mfld is a $\mathbb{Z}HS^3$, solving the equⁿ above gives a set $\{b_i\}_{i=1}^n$, which is not necc. unique, but one can show that they define the same 3-mfld. (This notion of equivalency can be ported over to equivalency of the sets $\{b_i\}$.

To get the Char var, we first must compute the fund gp. The generators are $x_1, \dots, x_n, h$, where the $x_1, \dots, x_n$ are loops about the meridional gen of each fibre and h is the longitudinal direction.



To get the relations, check that

- the longitude commutes with w/ each meridional gen, so $[x_i, h] = 1$ for all $1 \leq i \leq n$
- For each solid torus $S^1 \times D^2$ we glue back in, $pt \times \partial D^2$ gets attached along the (p,q) torus knot, so
$$[\partial D^2] = x_i^{a_i} h^{b_i} = 1$$
- the product of all x_i's is nullhomotopic, so $\prod_{i=1}^{n} x_i = 1$

This gives the presentation for π_1:

$$\pi_1\left(\Sigma(a_1, \dots, a_n)\right) \cong \left\langle x_1, \dots, x_n, h \mid [x_i, h] = 1,\ x_i^{a_i} h^{b_i} = 1, \prod_{i=1}^{n} x_i = 1 \right\rangle$$

Now, lets think about the char var of this fund q_p.

To each gen, we need to assign an element of $SL(2)$ and this assignment must preserve the relations

We want to find irred rep^{n}s, eg, rep^{n}s with the smallest possible Stab under Conj action.

If a rep has a large Stab, then its trivial, since for ρ reducible, it factors thru the abelianization which is zero for homology spheres.

Lemma If ρ is irred then $\rho(h) = \pm Id$

Pf Assume not: $\rho(h) \neq \pm id$.

Then up to conjugation, $\rho(h)$ is diag matrix:

$$\rho(h) = \begin{bmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{bmatrix}, \quad \theta \neq 0, \pi$$

One can check that the only elements in $SU(2)$ which commute w/ a diag matrix are the elem. in that S^1 subgp:

if $$\begin{bmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{bmatrix} \begin{bmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{bmatrix} = \begin{bmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{bmatrix} \begin{bmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{bmatrix},$$

$$\begin{bmatrix} \alpha e^{i\theta} & -\bar{\beta} e^{-i\theta} \\ \beta e^{i\theta} & \bar{\alpha} e^{-i\theta} \end{bmatrix} = \begin{bmatrix} \alpha e^{i\theta} & -\bar{\beta} e^{i\theta} \\ \beta e^{-i\theta} & \bar{\alpha} e^{-i\theta} \end{bmatrix}$$

so $\beta e^{i\theta} = \beta e^{-i\theta}$ and $-\bar{\beta} e^{i\theta} = -\bar{\beta} e^{-i\theta}$,

so $\beta = 0$. Thus $\begin{bmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{bmatrix} = \begin{bmatrix} \alpha & 0 \\ 0 & \bar{\alpha} \end{bmatrix}$

Since x_i commutes w/ h, this implies that, $\rho(x_i)$ is in the same subgp as $\rho(h)$, impling that ρ is reducible. □

So we may as well assume that $\rho(h) = \pm I$.

Now, lets ask ourselves what $\rho(x_i)$ can be for each i.

For each of the 2 choices, $\rho(h) = I$ or $\rho(h) = -I$.

using the third rel'n, we have that

$$\rho(x_i^{a_i} h^{b_i}) = \rho(1)$$

$$\rho(x_i)^{a_i} \rho(h)^{b_i} = 1$$

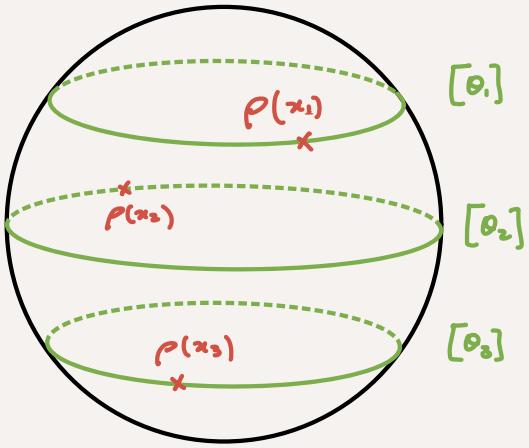
$$\rho(x_i)^{a_i} (\pm 1)^{b_i} = 1$$

$$\rho(x_i)^{a_i} = (\pm 1)^{-b_i}$$

Therefore $\rho(x_i)$ is a root of unity, so there is a discrete set of choices for the conj class of $\rho(x_i)$. In particular, $\rho(x_i)$ is in the conj class of a $\begin{cases} a_i^{th} \text{ root of unity, if } (\pm 1)^{-b_i} = id \\ 2a_i^{th} \text{ root of unity, if } (\pm 1)^{-b_i} = -id. \end{cases}$

Having fixed the (a_i, b_i)'s, for each tuple of choices for the conj. classes of the $\rho(x_i)$, we only need to enforce the last relation: $\pi \rho(x_i) = Id$

Another way of phrasing this constraint is: Fix conj classes $[\theta_i]$ for each x_i. Can we find $\rho(x_i) \in [\theta_i]$ such that $\rho(x_1) \cdot \ldots \cdot \rho(x_n) = Id$?

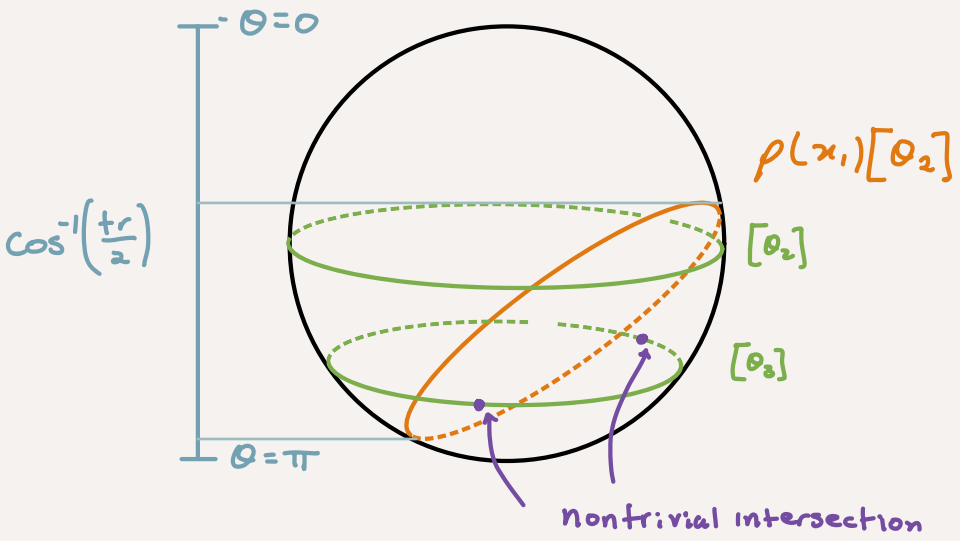


Lets now specialize to the case of $n=3$ singular fibres, $R(\Sigma(a_1, a_2, a_3))$. The last rel^n becomes $x_1 x_2 = x_3^{-1}$.

Since $\rho(x_3^{-1})$ is in the same cc as $\rho(x_3)$, $\rho(x_3)$ is in the cc as $\rho(x_1 x_2)$.

Up to conj, we can assume that $\rho(x_1)$ is in the std S^1 subgp, eg, is diag.

Then, we can ask if $\rho(x_1)\cdot[\theta_2] \cap [\theta_3] \neq \emptyset$?



Since left mult. by a fixed element of $SU(2)$ is an isometry of $SU(2)$, $\rho(x_1)[\theta_2]$ is another S^2 but it is slanted.

To check whether it intersects with $[\theta_3]$, we can look at the range of values of $\cos^{-1}(\frac{1}{2}tr)$ on the S^2 given by $\rho(x_1)[\theta_2]$.

Since the norm of matrices in $SU(2)$ is 1, the trace is is minimized/maximized when the matrix is diagonal, so the endpoints must be at $|\theta_1-\theta_2|$ and $\theta_1+\theta_2$ (or, if $\theta_1+\theta_2>\pi$, $2\pi-(\theta_1+\theta_2)$).

$$\begin{bmatrix} e^{i\theta_1} & 0 \\ 0 & e^{-i\theta_1} \end{bmatrix} \cdot \begin{bmatrix} e^{i\theta_2} & 0 \\ 0 & e^{i\theta_2} \end{bmatrix} = \begin{bmatrix} e^{i(\theta_1-\theta_2)} & 0 \\ 0 & e^{i(\theta_1-\theta_2)} \end{bmatrix}$$

$$\begin{bmatrix} e^{i\theta_1} & 0 \\ 0 & e^{-i\theta_1} \end{bmatrix} \quad \begin{bmatrix} e^{i\theta_2} & 0 \\ 0 & e^{-i\theta_2} \end{bmatrix} \quad \begin{bmatrix} e^{i(\theta_1+\theta_2)} & 0 \\ 0 & e^{-i(\theta_1+\theta_2)} \end{bmatrix}$$

Therefore $\cos^{-1}\left(\frac{tr(A)}{2}\right)$ when $A \in \rho(x_1)[\theta_2]$. lies in the interval

$$\left[|\theta_1 - \theta_2| \,,\; \begin{cases} \theta_1 + \theta_2 & \text{if } \theta_1 + \theta_2 \in \pi \\ 2\pi - (\theta_1 + \theta_2) & \theta_1 + \theta_2 > \pi \end{cases} \right]$$

If θ_3 is not in this interval, then we don't get a rep^n for that choice of conj classes. If it does, then in general, the intersection $\rho(x_1)\cdot[\theta_2] \cap [\theta_3]$ will be a copy of S^1, since generically $S^1 = S^2 \cap S^1$ in S^3.

But remember that we're quotienting by conjugation! We alr. quot. a bit by choosing θ_1 in the skel S^1, but we can still conjugate within this S^1 & preserve the intersection. This S^1 is the "leftover" conj by the S^1 subgp which contains $\rho(x_1)$. After doing so, for each $\{\theta_1, \theta_2, \theta_3\}$, we get one pt in the char var.

Rmk If the int is in a pt (instead of an S^1), this says that the leftover action fixes a pt, which implies that the x_i all commute, so π_1 is abelian and hence reducible.

Rmk Since the a_i are coprime, θ_3 can never be at an end of the interval.

Lets do an example for our favorite 3-mfd so far: $\Sigma(2,3,5)$,

Step 1: Solve the eqn $\left(\prod_{i=1}^{3} a_i\right)\left(\sum_{i=1}^{3} \frac{b_i}{a_i}\right) = 1$:

$$(2\cdot 3\cdot 5)\left(\frac{b_1}{2} + \frac{b_2}{3} + \frac{b_3}{5}\right) = 1$$

$$15b_1 + 10b_2 + 6b_3 = 1$$

Notice: $15+6=21$, $10\cdot 2=20$, $21-20=1$

$\Rightarrow$ $b_1=b_3=-1$, $b_2=2$ works

Step 2: Compute π_1:

$$\pi_1\left[\Sigma(2,3,5)\right] = \langle x_1, x_2, x_3, h \mid [x_i,h]=1,\ x_1^2=h,\ x_2^3=h^{-1},\ x_3^5=h^{-1},\ x_1x_2x_3=1\rangle$$

Step 3: Find rep's which satisfy the first two rel'ns:

If $\rho(h)=I$:

$\rho(x_1)^2 = Id \Rightarrow \rho(x_1) = \pm Id$

$\Rightarrow \rho(x_2x_3)=1 \Rightarrow \rho(x_2)\rho(x_3)=\pm Id$

$\Rightarrow$ rep^n is reducible

If $\rho(h)=-Id$:

$\rho(x_1)^2 = -Id \Rightarrow \rho(x_1)^4 = Id$

$\Rightarrow \rho(x_1)$ is a 4^{th} root of unity whose square is -1

$\Rightarrow \rho(x_1) \in [\pi/2]$

$\rho(x_2)^3 = -(-Id) = Id \Rightarrow \rho(x_2) \in [\pi/3]$ or $[2\pi/3]$

$\rho(x_3)^5 = -(-Id) = Id \Rightarrow \rho(x_3) \in [n\pi/5]$, $n=1,2,3,4$

Step 4: Use the range of $\rho(x_1)[\theta_2]$ to verify the last relation:

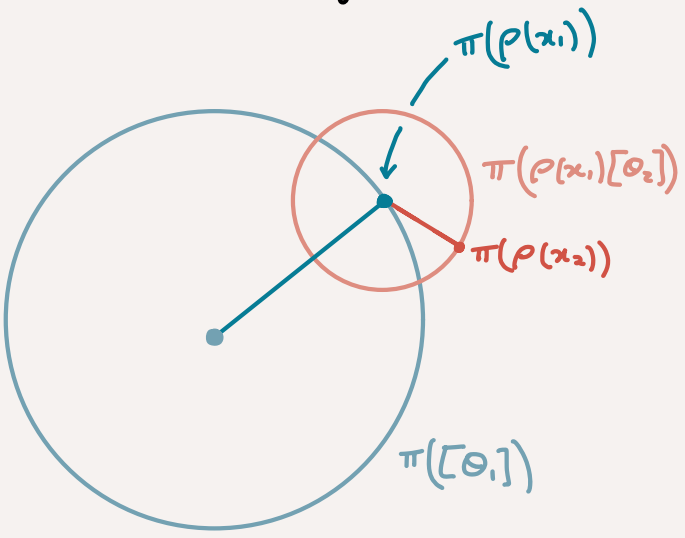
INSERT THIS STEP

It turns out that in general, the Char-Var for $n=3$ is a discrete set. Strictly speaking, this is still not enough to prove that no perturbation is needed (we still need to verify transversality, like in normal morse theory) but in this case, everything works out. But it is true that if the Char-Var is not a discrete set, we definitely must perturb. Lets now take $n=4$, and show how $\mathcal{R}(\Sigma(a_1,a_2,a_3,a_4))$ need not be discrete.

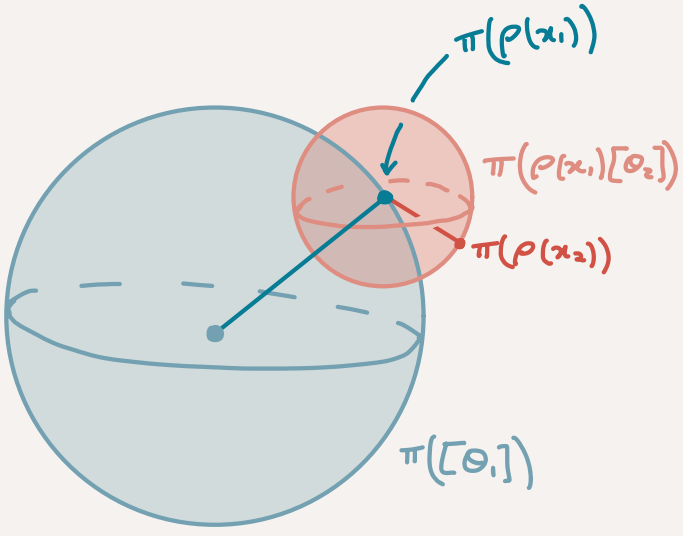
Here is the same story up until we get to the third condition. For each 4-tuple of conj classes $\{[\theta_1],[\theta_2],[\theta_3],[\theta_4]\}$, we ask if we can choose representatives in each class satisfying $x_1x_2x_3x_4=1 \iff x_1x_2 = x_4^{-1}x_3^{-1}$.

As before, we formulate this as finding $\rho(x_1) \in [\theta_1]$ and $\rho(x_4^{-1}) \in [\theta_4]$ such that $\rho(x_1)[\theta_2] \cap \rho(x_4^{-1})[\theta_3] \neq \emptyset$.

To better visualize this condition, lets look at the image of Stereographic proj $\pi: S^3 \to \mathbb{R}^3$. Since π is conformal, spheres get mapped to spheres. The size of each $[\theta_i]$ sphere is inversely proportional to θ_i, and rotating $[\theta_2]$ by $\rho(x_1)$ corresponds to translating the center of $\pi([\theta_2])$ to lie at $\rho(x) \in \mathbb{R}^3$

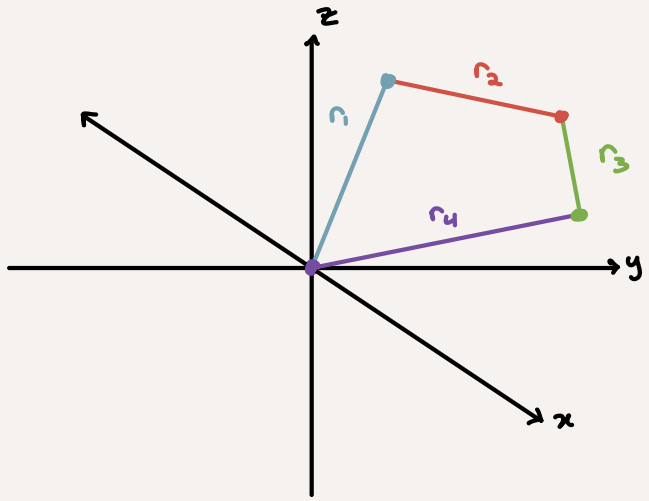


2-d picture



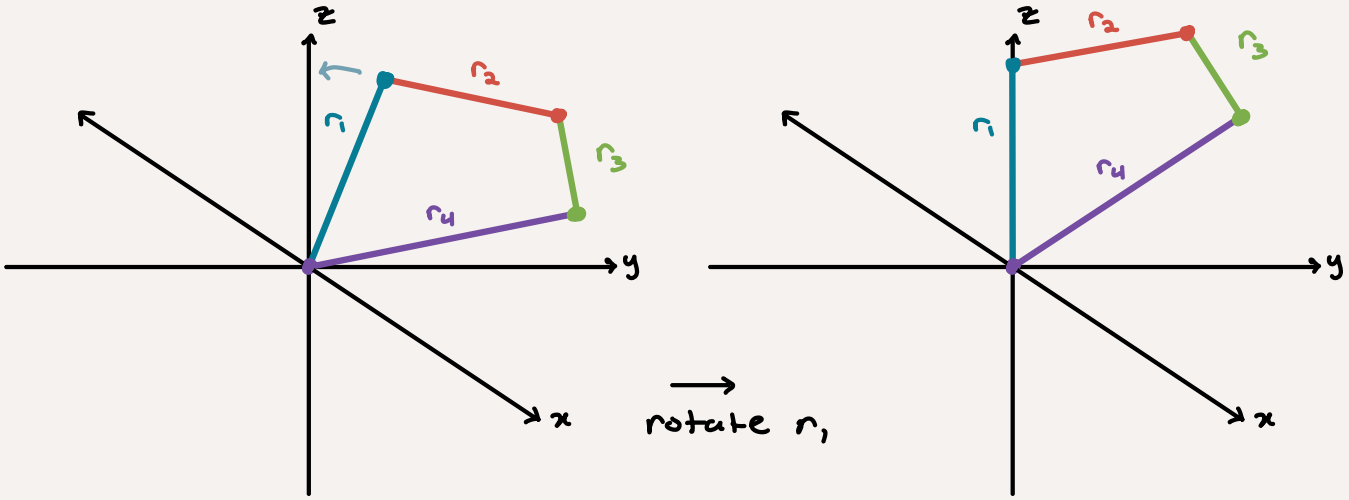
3-d picture

However, trying to include the other two spheres makes this picture very busy! The relevant data is the radius of each sphere, and the location of the center-point of the subsequent sphere. So an even better way to visualize the condition that $x_1x_2x_3x_4 = 1$ is by a collection of four rigid rods whose lengths depend on θ_i, where the start of rod i is attached to the end of rod $i-1$. Then $x_1x_2x_3x_4 = 1$ exactly when the rods form a closed polygon in $\mathbb{R}^3$.

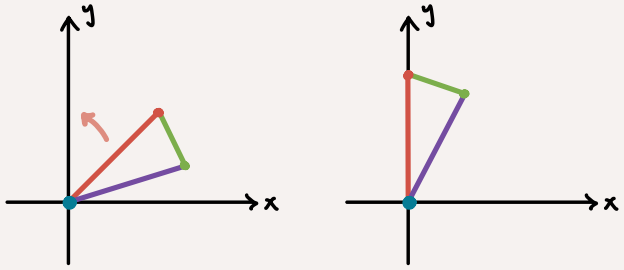


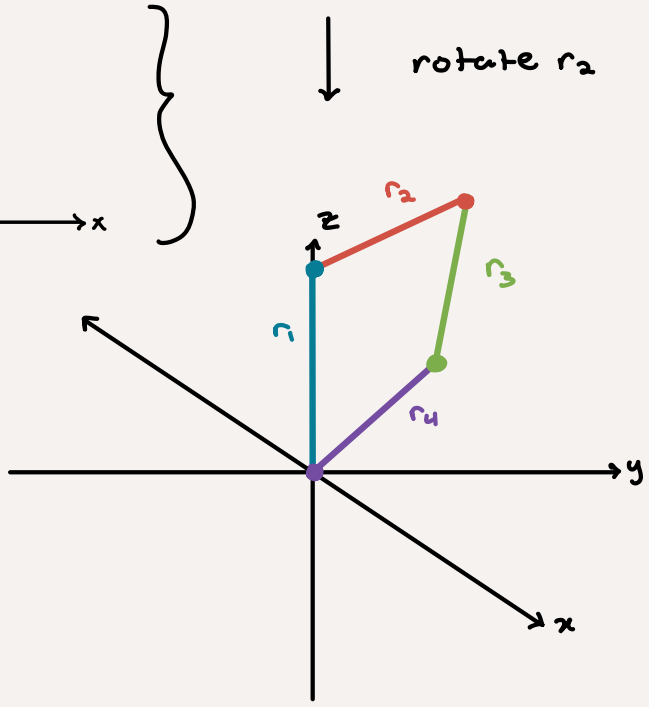
In general, such a configuration of rods for a given set of side lengths does exist. The question then becomes how many are there, up to conjugation. The conjugation action is that of $SU(2)$ acting on $\mathbb{R}^3$, so by rotations.

We may first move r_1 to align w/ the z axis. This is the same thing as conjugating to turn $\rho(x_1)$ into a diagonal matrix. The leftover action corr. to rotating about the z-axis, since that fixes $\rho(x_1)$. Therefore we can rotate such that r_2 lies in the RHS of the y-z plane. However, there's still some deg's of freedom! So the moduli space is still 1-dimensional.



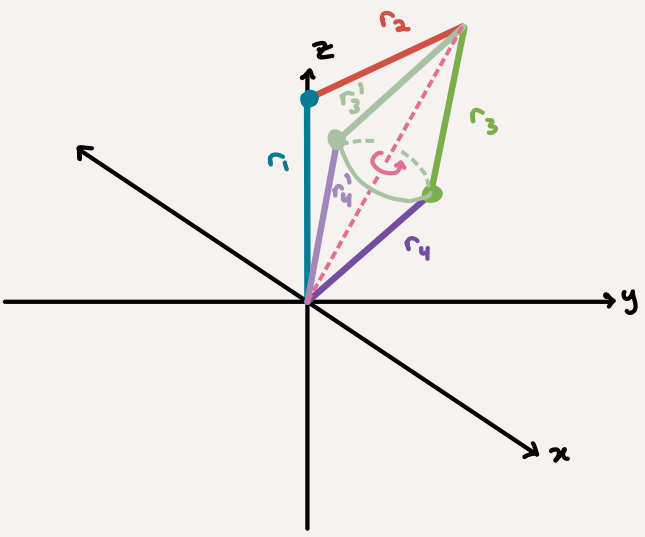
Top down r_2 rotation:



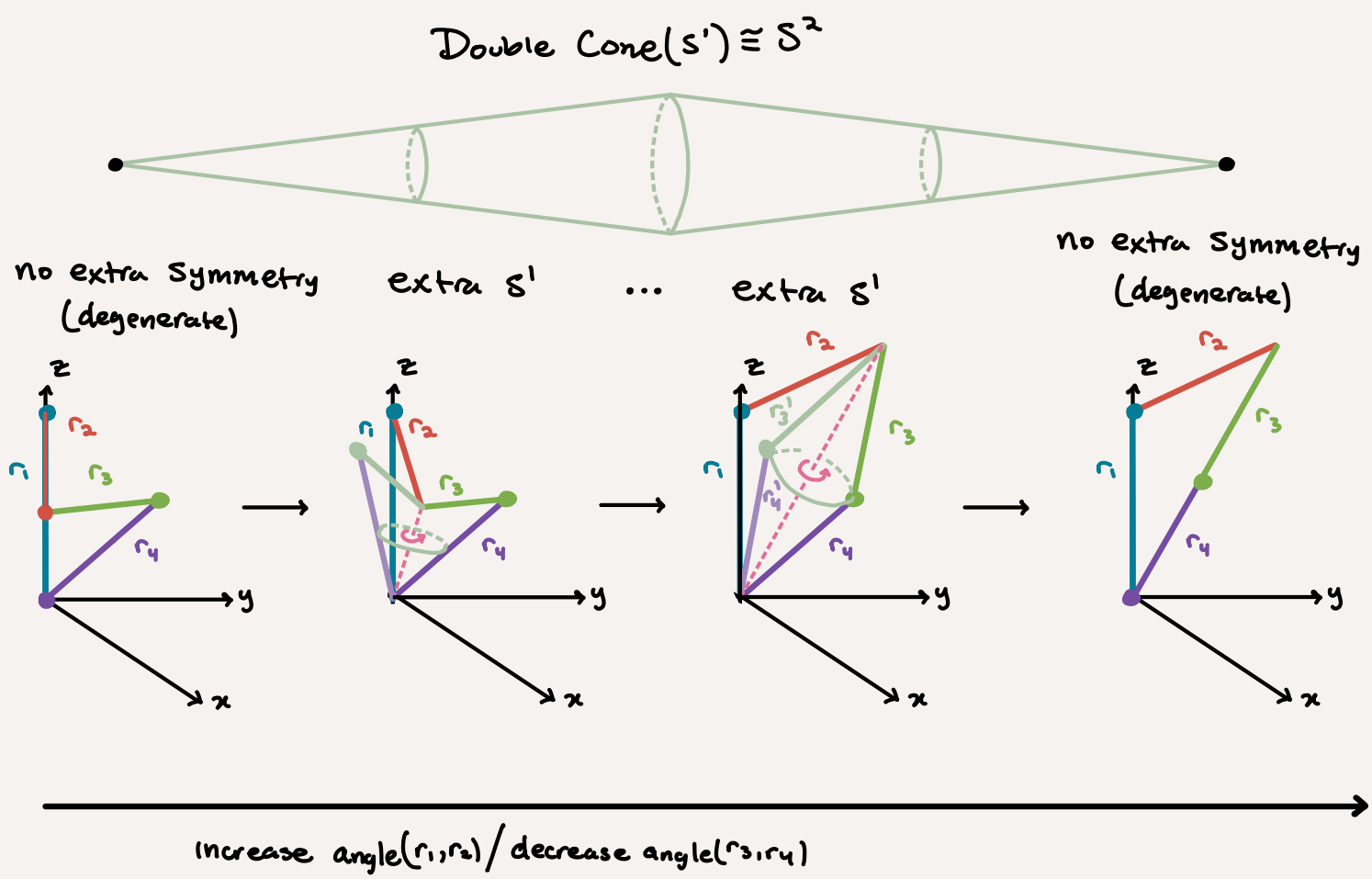


This is the final polytope →

However, there is still an S^1 action on this polytope which is not given by conjugation! In fact this S^1 action is a slice of a residual S^2's worth of action in $\mathbb{R}$.



This S^2 can be thought of as the double cone on S^1, where the cone points are the degeneration of the symmetry due to the angle between r_1 and r_2, and r_3 and r_4.


Double Cone(S¹) ≅ S²
no extra Symmetry
(degenerate)
extra S¹
...
extra S¹
no extra Symmetry
(degenerate)
z
y
x
r1
r2
r3
r4
r3'
r4'
Increase angle(r1,r2) / decrease angle(r3,r4)